

**DEPLOYMENT
GUIDE**

Securing Applications in Azure with VM-Series Firewalls and Panorama

DECEMBER 2024

Table of Contents

Preface.....	3
Purpose of This Guide.....	5
Objectives.....	5
Audience.....	5
Related Documentation.....	6
Deployment Overview.....	7
Choosing a Design Model Option.....	7
Design Model.....	8
Outbound Traffic.....	10
East-West Traffic.....	11
Inbound Traffic.....	11
Enterprise Network and Management Traffic.....	14
Assumptions and Prerequisites.....	16
Deploying Azure Infrastructure.....	17
Configuring the Application Resource Groups, VNet, Subnets, and Services.....	17
Deploying Outbound and East-West Security.....	21
Configuring the Transit Resource Group, VNet, Subnets, and Services.....	21
Configuring Panorama Templates, Template Stacks, and Device Groups.....	29
Deploying the VM-Series Virtual Machines.....	47
Configuring the VM-Series After Deployment.....	54
Configuring Azure After Deployment.....	58
Deploying Inbound Security.....	66
Configuring Panorama Templates, Template Stacks, and Device Groups.....	67
Deploying the VM-Series Virtual Machines.....	72
Configuring the VM-Series After Deployment.....	78
Deploying the Public Load-Balancer Option.....	82
Deploying the Application Gateway Option.....	91
Deploying VPN to Prisma Access.....	114
Creating a Virtual Network Gateway and VPN Attachments.....	115
Configuring Prisma Access Service Connection VPN Tunnels and Routing.....	120
Configuring Azure VPN Tunnels and Routing.....	124
Configuring Prisma Access Security Policy.....	127
Configuring Azure VM-Series Security Policy.....	130
Feedback.....	132

Preface

GUIDE TYPES



Overview guides provide high-level introductions to technologies or concepts.

Design guides provide an architectural overview for using Palo Alto Networks® technologies to provide visibility, control, and protection to applications built in a specific environment. These guides are required reading prior to using their companion deployment guides.

Deployment guides provide decision criteria for deployment scenarios, as well as procedures for combining Palo Alto Networks technologies with third-party technologies in an integrated design.

Solution guides provide add-on solutions for post-deployment use cases.

DOCUMENT CONVENTIONS



Notes provide additional information.



Cautions warn about possible data loss, hardware damage, or compromise of security.

Blue text indicates a configuration variable for which you need to substitute the correct value for your environment.

In the **IP** box, enter **10.5.0.4/24**, and then click **OK**.

Bold text denotes:

- Command-line commands.

show device-group branch-offices

- User-interface elements.

In the **Interface Type** list, choose **Layer 3**.

- Navigational paths.

Navigate to **Network > Virtual Routers**.

- A value to be entered.

Enter the password **admin**.

Italic text denotes the introduction of important terminology.

An *external dynamic* list is a file hosted on an external web server so that the firewall can import objects.

Highlighted text denotes emphasis.

Total valid entries: 755

ABOUT PROCEDURES

These guides sometimes describe other companies' products. Although steps and screen-shots were up-to-date at the time of publication, those companies might have since changed their user interface, processes, or requirements.

GETTING THE LATEST VERSIONS OF GUIDES

We continually update reference architecture guides. You can access the latest version of this and all guides at this location:

<https://www.paloaltonetworks.com/referencearchitectures>

WHAT'S NEW IN THIS RELEASE

Since the last version of this guide, Palo Alto Networks made the following changes:

- Improved guidance for securing the management interface
- Updated the health-check method from the external load-balancer and application gateway
- Updated version of PAN-OS® to 10.2.10-h9
- Changed phrasing, terminology, and diagrams for technical accuracy

Purpose of This Guide

This guide provides design and deployment details for VM-Series firewalls on Microsoft Azure. This deployment guide focuses specifically on the Transit Virtual Network (VNet) design model.

This deployment guide:

- Provides architectural guidance and deployment details for using Palo Alto Networks next-generation firewalls in order to provide visibility, control, and protection to your applications built on Microsoft Azure.
- Requires that you first read the **Securing Applications in Azure: Design Guide**. The reference architecture guide provides architectural insight and guidance for your organization to plan linkage of pertinent features with the next-generation firewall in a scalable and highly available design.
- Provides decision criteria for deployment scenarios, as well as procedures for programming features of Microsoft Azure and the Palo Alto Networks VM-Series next-generation firewall in order to achieve an integrated design.

OBJECTIVES

Completing the procedures in this guide, you can successfully deploy a Palo Alto Networks VM-Series next-generation firewall in the Azure environment. The main objectives are to enable the following functionality:

- Protection and inspection of flows outbound and east-west from private networks and for secure communication with enterprise devices
- Application layer visibility and control for all flows
- Preparing the firewalls for WildFire® analytics, URL filtering, identity-based services, and the full Threat Prevention services
- Resilient and scalable operation through integration with Azure Load Balancer
- Panorama® centralized management using templates and device groups
- Centralized reporting with Palo Alto Networks cloud-delivered Strata™ Logging Service (formerly *Cortex Data Lake*)

AUDIENCE

This deployment guide is written for technical readers, including system architects and design engineers, who want to deploy VM-Series firewalls within a public cloud data center infrastructure. It assumes the reader is familiar with the basic concepts of applications, networking, virtualization, security, and high availability, as well as a basic understanding of network and data center architectures.

To be successful, you must have a working knowledge of networking and policy in PAN-OS.

RELATED DOCUMENTATION

The following documents support this deployment guide:

- **Securing Applications in Azure: Design Guide**—Provides a detailed discussion of the available design considerations and options for the next-generation VM-Series firewall on Microsoft Azure.
- **Panorama on Azure: Deployment Guide**—Provides architectural guidance and deployment details for using a Palo Alto Networks Panorama virtual appliance, deployed on Microsoft Azure, to provide a single location from which you can create network configurations and security policies that enable visibility, control, and protection to your applications built in an Azure public cloud. The *Panorama on Azure: Deployment Guide* is a prerequisite for this guide.
- **SASE for Securing Private Applications: Design Guide**—Presents a detailed discussion of the available design considerations and options for Prisma® Access and Prisma SD-WAN when used for securing access to private applications.
- **SASE for Securing Private Applications: Deployment Guide**—Details deployment scenarios and step-by-step guidance for the Securing Private Applications design. This design includes securing private applications for mobile-users and for remote sites.
- **SASE for Securing Private Applications: Design Guide**—Presents a detailed discussion of the available design considerations and options for Prisma Access and Prisma SD-WAN when used for securing access to private applications.
- **SASE for Securing Private Applications: Deployment Guide**—Details deployment scenarios and step-by-step guidance for the Securing Private Applications design. This design includes securing private applications for mobile-users and for remote sites.

Deployment Overview

There are many ways to use the concepts discussed in [Securing Applications in Azure: Design Guide](#) to achieve an architecture that secures applications deployed on Azure. Each deployment option in that guide provides an example architecture for securing the communication between private virtual machines and workloads in different VNets, inbound access to an application in Azure, and the connection to your enterprise networks.

This guide is specific to the Transit VNet design model, covering both the dedicated inbound and common firewall deployment options.

CHOOSING A DESIGN MODEL OPTION

As discussed in the reference architecture guide, when choosing a deployment option, consider the following factors:

- **Scale**—What are the expected number of sessions and bandwidth required for the applications? Is this deployment for a proof-of-concept? Are the traffic requirements for inbound, outbound, east-west, and enterprise communication balanced? The Transit VNet model dedicated inbound deployment option allows you to scale inbound resources linearly and independently based on business need. The outbound, east-west, and enterprise network performance provided from the transit VNet scales linearly by adding additional firewalls to the load-balancer backend pools.
- **Complexity**—Is it more important to keep individual device configuration simple and permit easier troubleshooting, or is it acceptable to take on a somewhat higher administrative workload in order to reduce the total number of deployed devices? The Transit VNet model dedicated inbound option uses one set of firewalls for outbound, east-west, and enterprise traffic profiles and a separate set of firewalls for inbound traffic profiles. The proper function of each application VNet depends on the transit VNet, so careful consideration of any changes is necessary in order to evaluate overall impact.
- **Resiliency and high availability**—Are there differentiated availability requirements for different traffic profiles? The transit VNet model provides the same level of availability for the outbound, east-west, and enterprise traffic profiles. The Transit VNet dedicated inbound deployment option allows you to inherently provide a differentiated level of availability for inbound traffic flows, which have dedicated firewall resources.

Design Model

There are many ways to use the concepts discussed in the previous sections to secure application deployments in Azure. The options described in this section offer example architectures that secure inbound and outbound traffic flows, traffic between virtual networks, and the connection to your enterprise networks.

In this model, you allocate the security and application functions and resources across multiple VNets that are connected in a hub-and-spoke topology. The hub of the topology, or *transit VNet*, is the central point of connectivity for all inbound, outbound, east-west, and enterprise traffic. You deploy all VM-Series firewalls within the transit VNet. The spokes contain application workloads arranged in their own VNets. This model encompasses two different firewall deployment options. The options presented here differ in how they provide resiliency, scale, and services. Consider which option best fits your needs and use it as a starting point for your design:

- The *dedicated inbound option* separates traffic flows across two separate sets of VM-Series firewalls. One set of VM-Series firewalls is dedicated to inbound traffic flows, allowing for greater flexibility and scaling of inbound traffic loads. The second set of VM-Series firewalls services all outbound, east-west, and enterprise network traffic flows. This deployment choice offers increased scale and operational resiliency and reduces the chances of high bandwidth use from the inbound traffic flows affecting other traffic flows within the deployment. We recommend this model for your production deployments.
- The *common firewall option* leverages a single set of VM-Series firewalls. The sole set of firewalls operates as a shared resource and may present scale limitations with all traffic flowing through a single set of firewalls due to the performance degradation that occurs when traffic crosses virtual routers. This option is suitable for proof-of-concepts and smaller scale deployments because the number of firewalls is low. However, the technical integration complexity is high.

The transit VNet provides security for inbound, outbound, and enterprise traffic flows as a shared service and does not typically contain any application virtual compute resources. The transit VNet also controls and secures east-west traffic flows between application VNets. The model requires that you establish a VNet peering connection between each application VNet and the transit VNet. You cannot use overlapping IP address space within the collection of peered VNets.

If you are using the dedicated inbound option, one set of VM-Series firewalls secures all inbound traffic from the internet. Another set of firewalls provides visibility and control for outbound, east-west, and enterprise traffic flows. The separation of functions allows you to scale inbound traffic volume independently and simplifies the configuration of the inbound firewalls.

If you are using the common firewall option, a single set of VM-Series firewalls provides visibility and control for all inbound outbound, east-west, and enterprise traffic flows. Combining all traffic flows onto a single set of firewalls reduces the total number of firewalls that the design requires. However, this reduces the overall scalability and increases the complexity of the design because the deployment requires that you deploy multiple virtual routers on each firewall because both the public and private interfaces receive health probes.

Regardless of the deployment option you choose, employing availability sets to distribute the VM-Series virtual machines across the Azure physical infrastructure ensures that you avoid downtime caused by infrastructure maintenance or failure.

This design model uses a separate management VNet to centralize management so that a single Panorama deployment can manage VM-Series firewalls deployed across all of your organization's virtual networks. Panorama streamlines and consolidates core tasks and capabilities, enabling you to view all your firewall traffic, manage all aspects of device configuration, push global policies, and generate reports on traffic patterns or security incidents. You deploy Panorama in Management-Only mode and securely access it through the enterprise network connection or over the public internet.

Bootstrapping speeds up the process of configuring and licensing the VM-Series firewalls and making them operational on the network. This process allows you to deploy the firewall using custom data with only a basic configuration. After deployment, the VM-Series connects to Panorama and obtains the complete operational configuration. After they are operational and securing outbound, east-west, and inbound flows, the VM-Series firewalls encrypt and send all firewall logs to Strata Logging Service (SLS) over TLS/SSL connections.

Figure 1 Transit VNet model—dedicated inbound option

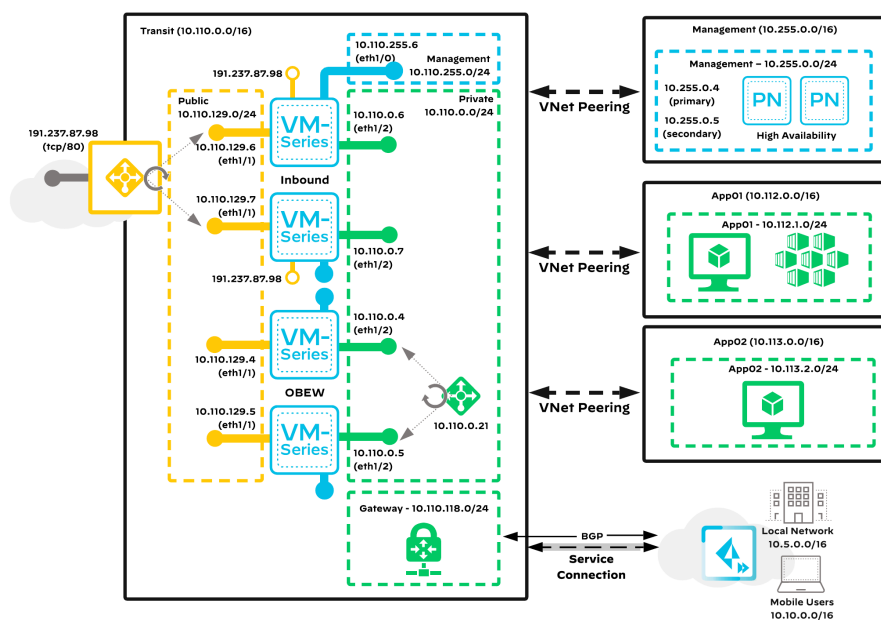
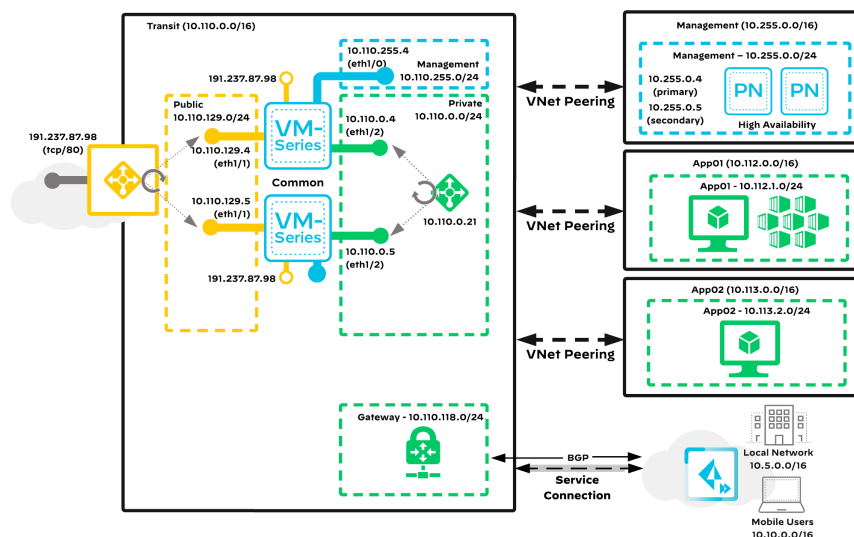


Figure 2 Transit VNet model—common firewall option

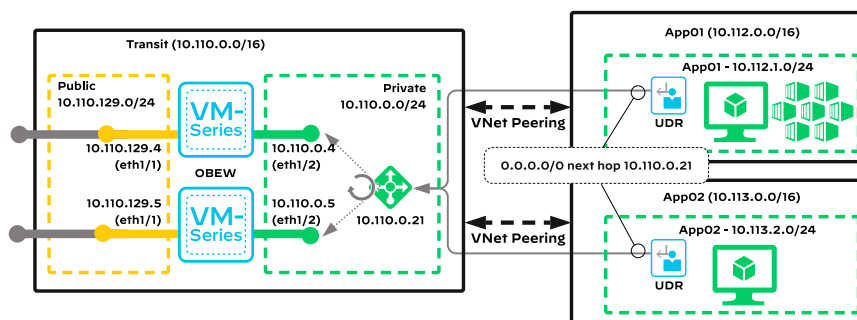


OUTBOUND TRAFFIC

Each VM-Series firewall has an interface in the transit VNet public, private, and management subnet. The VM-Series public interface contains a static default route to the internet. To reach any private or internal subnets, you configure static routes for all peered application and enterprise network subnets.

User-defined routes in the private subnets of the peered application VNets direct traffic to the internal load balancer's frontend IP address, which shares a subnet with the firewall private interfaces. The internal load balancer distributes traffic to a collection of VM-Series firewalls. Load-balancer rules forward all TCP and UDP ports to the firewalls. The internal load balancer's health probes monitor availability of each firewall through the HTTPS service, which you enable on the private interface by using a interface management profile. A security policy entry allows only traffic sourced from the Azure health probe IP address to connect to the HTTPS service.

Figure 3 Outbound and east-west traffic



To egress out of the private interface, you define static routes for the health probe IP address and for the application network ranges. Additionally, a static default route forwards traffic out of the public interface. If you choose the common firewall option, the same static routes are used but you configure them in different virtual routers.

The firewall applies source NAT to outbound internet traffic. The firewall translates the original source address to the IP address of its public interface. Azure then automatically translates the interface IP address to the public IP address associated to the firewall's public interface when the traffic leaves the transit VNet.

The firewall security policy allows appropriate application traffic from the resources in the application virtual networks to the internet. You should implement the security policy by using positive security policies (*allow listing*). Security profiles prevent known malware and vulnerabilities from entering the network in return traffic allowed by the security policy. URL filtering, data loss prevention, file blocking, and data filtering protect against data exfiltration.

EAST-WEST TRAFFIC

The same internal load balancer that distributes outbound traffic to the firewalls also distributes the spoke-to-spoke *east-west traffic*, which is the traffic between subnets in different application virtual networks. You apply user-defined routes for the private network subnets to the private subnets and direct traffic to the transit VNet's internal load balancer's frontend IP address. The existing load-balancer rules for outbound traffic also apply to east-west traffic and to all TCP/UDP ports.

For traffic between private application subnets, the firewall does not translate either the source or destination. A positive control security policy should allow only appropriate application traffic between private resources and requires that you create corresponding security policy rules to permit specific traffic. You must then override the default intrazone security policy rule and modify it to deny traffic. Security profiles should also be enabled to prevent known malware and vulnerabilities from moving laterally in the private network through traffic allowed by the security policy.

INBOUND TRAFFIC

There are two options for inbound traffic:

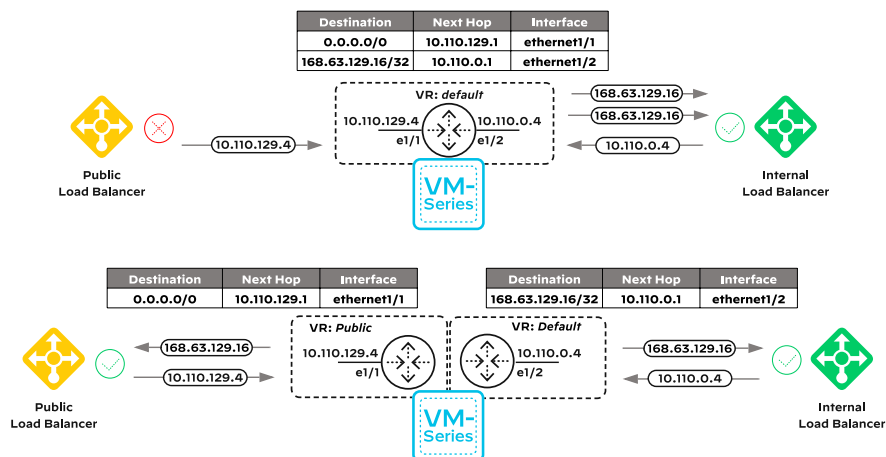
- **Public load balancer**—Choose this option if you require load balancing at Layer 4 only (TCP/UDP). Health probes in this design monitor the firewall resources and are not directly monitoring the health of the web server resources.
- **Application gateway**—Choose this option if you require load balancing at Layer 7 (application layer) for HTTP and HTTPS. Capabilities include URL path-based routing and SSL offload. Health probes in this design directly monitor the health of the web server resources.

Public Load Balancer

A public load balancer distributes incoming application traffic to the firewalls. To simplify firewall configuration, the frontend public IP address is associated with a DNS name, and floating IP is enabled on the load-balancer rules. Load-balancer rules forward the required web service ports to the firewalls. Common ports required for inbound traffic include TCP/80 (HTTP) and TCP/443 (HTTPS). The public load balancer's health probes monitor availability of each firewall through the HTTPS service which you enable on the public interface using a interface management profile. A security policy entry only allows traffic sourced from the Azure health probe IP address to connect to the HTTPS service.

If you are using the dedicated inbound option, a separate set of VM-Series firewalls secures all inbound traffic from the internet. If you are using the common firewall option, inbound traffic is secured on the same set of firewalls as outbound, east-west, and enterprise traffic and you must configure multiple virtual routers to support health probes from the Azure load balancers on multiple interfaces. Dedicated virtual routers allow the firewall interface that received the health probe to source responses. Additional static route entries define a default route out of the public interface as well as a route entries to all private networks through the virtual router dedicated to the private interface.

Figure 4 Health probes using multiple virtual routers



The firewall applies both a destination and source NAT to inbound traffic. Destination NAT translates the FQDN address object associated with the load-balancer public DNS name to the virtual machine or load balancer on the application network. The source NAT translates the source to be the IP address of the private interface of the firewall, ensuring return traffic flows symmetrically.

The firewall security policy allows appropriate application traffic to the resources in the private network while security profiles prevent known malware and vulnerabilities from entering the network in traffic allowed by the security policy.

Network security groups applied to the public subnet permit inbound internet traffic and deny any traffic destined for the other subnets in the transit VNet. This ensures that only inbound traffic forwarded through the public load balancer can communicate to private application resources through the firewall.

Application Gateway

For inbound traffic, an application gateway with a public frontend terminates incoming connections and initiates corresponding new connections to the configured HTTP/HTTPS backends. You assign unique TCP ports for all backends. The application gateway sources all new connections from the private IP addresses of the application gateway instances and distributes the connections to the public interfaces of the inbound firewalls, which you have configured as the backend pool targets for the application gateway. The application gateway's health probes monitor backend availability on all specified HTTP/HTTPS ports.

You must create an additional public subnet for the application gateway. Specify a minimum of two application gateway instances in order to ensure that you distribute the instances across Azure update and fault domains.

Figure 5 Transit VNet model—dedicated inbound option with application gateway

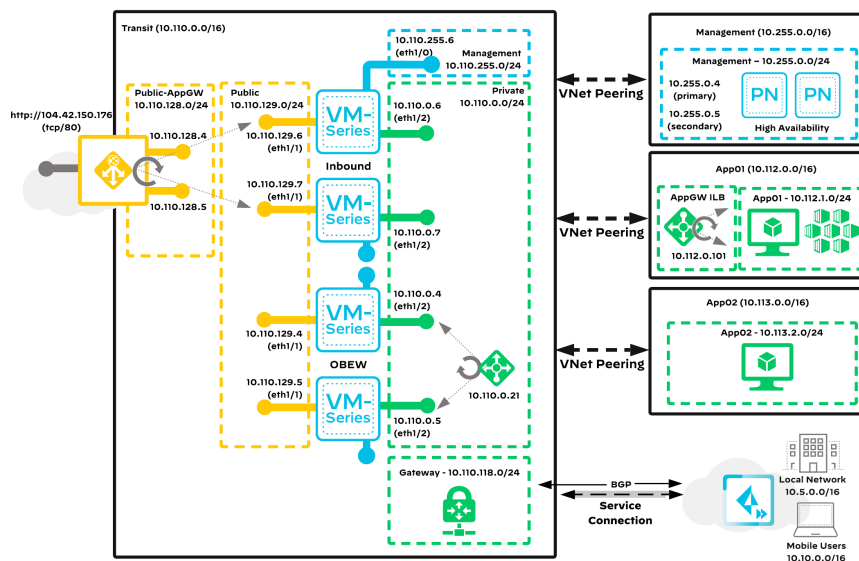
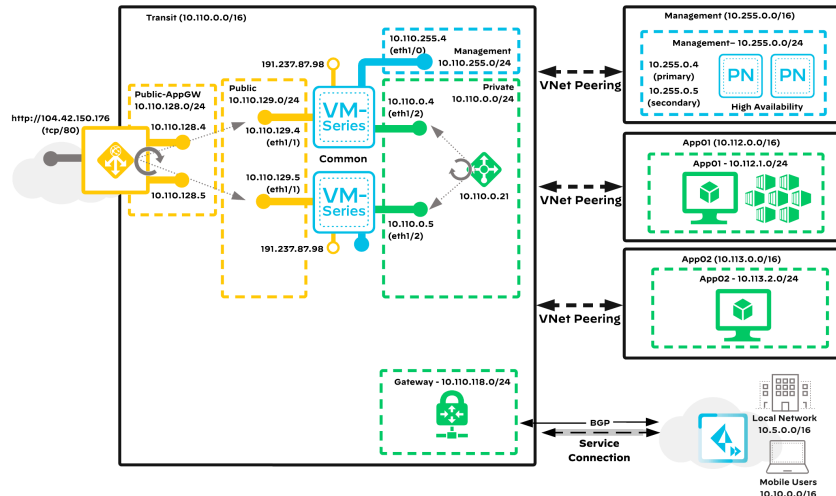


Figure 6 Transit VNet model—common firewall options with application gateway



You use application gateway destination NAT rules on the firewalls to map to backend resources directly or through one or more internal load balancers.

This model supports any combination of the following methods:

- **Firewall destination port NAT to backend resource**—No internal load balancer required, uses port-based NAT policy rules associated to the backend resource. The firewall NAT policy contains all resource-mapping parameters.
- **Internal load balancer with one or more frontend IP addresses**—Uses port-based NAT policy rules associated to the load balancer's frontend IP addresses. You also configure port mapping on the load balancer. This option uses the load-balancer for resiliency and scaling of the backend resources.
- **Multiple internal load balancers**—Uses port-based NAT policy rules associated to each load balancer's frontend IP addresses. This option supports more granular separation of both the load balancers and the backend resources.

The firewall also applies a source NAT to inbound traffic. The source NAT translates the source to be the IP address of the private interface of the firewall, ensuring return traffic flows symmetrically.

The firewall security policy allows HTTP/HTTPS application traffic from the application gateway instances to the resources in the private network, and security profiles prevent known malware and vulnerabilities from entering the network in traffic allowed by the security policy. To support the use of HTTP/HTTPS backends on ports other than 80/443, you should configure the service for security policy rules to include the specific service ports in use instead of *application-default*.

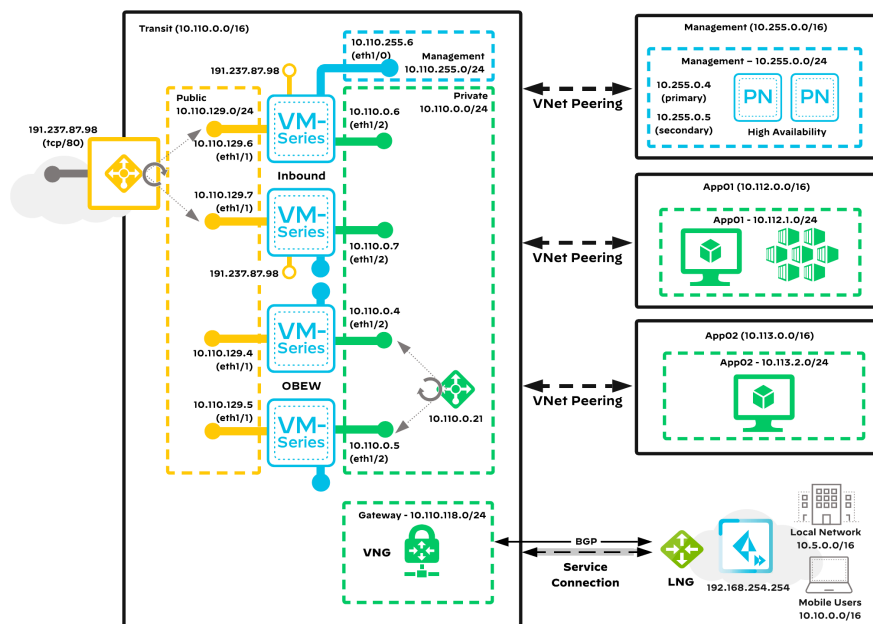
Network security groups applied to the application gateway subnet permit inbound internet traffic, allow that traffic to exit to the public subnet, and deny traffic destined for the other subnets in the transit VNet. This ensures that only inbound traffic forwarded through the application gateway can communicate to private application resources through the firewall.

ENTERPRISE NETWORK AND MANAGEMENT TRAFFIC

A virtual network gateway (VNG) deployed in the transit VNet connects the Azure virtual networks to your enterprise networks. You can enable BGP to facilitate dynamic routing between the application networks deployed in Azure and your existing enterprise networks.

To get traffic from your remote site and mobile users to application VNets in Azure, you use a VNG in a dedicated subnet along with a local network gateway (LNG). This VNG service establishes IPsec connectivity to Prisma Access via the LNG. In Prisma Access, you configure a service connection to allow access to your enterprise resources in Azure. You can use one or more IPsec tunnels between the Azure VNG and Prisma Access. BGP dynamic routing is enabled between the VNG and Prisma Access.

Figure 7 Enterprise network connections



The internal load balancer that distributes outbound and east-west traffic to the firewalls also distributes traffic to and from the enterprise networks. Traffic originating from the application virtual network subnets and destined to enterprise networks follow the same path as outbound and east-west traffic. User-defined routes applied to the gateway subnet direct traffic that has an application virtual network destination to the internal load balancer. The existing load-balancer rules for outbound and east-west traffic also apply to enterprise traffic for all TCP and UDP ports.

Traffic from the enterprise networks communicates to the transit management subnet and management VNet through the Outbound/East-West firewalls. Enterprise administrators will need to ensure proper access is allowed for Panorama to and from the desired networks. Network security groups are also applied to the management subnet to ensure that only management traffic from the peered management VNet and the enterprise network subnets can reach Panorama and VM-Series management interfaces. The management NSG also contains rules that limit internet management to a set of trusted public source addresses from within your organization.

Assumptions and Prerequisites



Automation

If you would like to deploy and manage software firewalls with familiar cloud-automation tools taking an infrastructure-as-code approach, visit the **software-firewall automation hub**. The automation hub allows you to leverage Terraform modules and examples built specifically to provide easy deployment of the software firewall reference architectures.

Microsoft Azure:

- Your organization has a valid active subscription associated with your Azure user account.
- Azure uses standard-SKU IP addresses and load-balancers, except where specifically noted in the guide.
- This deployment was tested predominantly in the US West region, although deploying this design should be possible in any Azure region.

Palo Alto Networks next-generation firewalls and Panorama:

- You have already deployed Panorama and completed all procedures in the **Panorama on Azure: Deployment Guide**.
- Using templates and device groups, you use Panorama to centrally manage device configuration.
- Firewall logging uses SLS.
- The tested PAN-OS version in this deployment guide is 10.2.10-h9 for all devices.
- The Prisma Access service for Enterprise network connectivity is configured and available.

Palo Alto Networks licensing:

- Your organization has a Panorama license for the current and expected number of managed VM-Series firewalls.
- Sufficient VM-Series licensing for the current and expected number of VM-Series firewalls. This guide assumes you are using the BYOL licensing option.
- The SLS instance used in this guide is hosted in the Americas region.

Deploying Azure Infrastructure

Although an account might have existing Azure VNets, this section describes configuring new resource groups and virtual networks for initial deployment or proof of concept. Larger organizations with an existing footprint in Azure might find value in deployment details regarding features or platforms that they have not yet deployed or experienced.

Procedures

Configuring the Application Resource Groups, VNet, Subnets, and Services

- 1.1 Create the Application Resource Groups
- 1.2 Create the Application VNet and Subnets
- 1.3 Create the NSGs for the Application Subnets

All resources in this guide were created and tested in the West US region. You should change to the Azure region most suitable for your deployment. In these procedures, you create the resource groups, virtual networking, and network security groups for the application compute resources.

1.1 Create the Application Resource Groups

First, you create two example application resource groups that contain Azure resources and the application virtual machines.

Table 1 Application resource groups

Resource group name	Function
app01	Example resource group for web servers
app02	Example resource group for DB servers

Step 1: Log in to the Azure Resource Manager at <https://portal.azure.com>.

Step 2: Navigate to **Home > Resource groups**, and then click **Create**.

Step 3: In the **Resource group** box, enter **app01**, and then in the **Region** list, choose **(US) West US** or the appropriate value.

Step 4: Click **Review + create**, and then click **Create**.

The screenshot shows the 'Review + create' tab in the Azure portal. It includes a description of a Resource group, followed by 'Project details' with dropdowns for 'Subscription' (Example) and 'Resource group' (app01), and 'Resource details' with a dropdown for 'Region' ((US) West US).

Step 5: Repeat this procedure for the **app02** resource group.

1.2 Create the Application VNet and Subnets

You create two application VNets with an IP address space and a subnet that must be within the IP address space. As necessary, to add additional IP address spaces and subnets, you can modify the VNets after creation.

Table 2 Application virtual network IP addressing and subnets

Resource group	VNet	IP address range	Subnet	Subnet address range
app01	app01	10.112.0.0/16	app01	10.112.1.0/24
app02	app02	10.113.0.0/16	app02	10.113.2.0/24

Step 1: Navigate to **Home > Virtual networks**, and then click **Create**.

Step 2: In the **Resource group** list, choose **app01**.

Step 3: In the **Name** box, enter **app01**.

Step 4: In the **Region** list, choose **(US) West US** or the appropriate value, and then click **Next: IP Addresses**.

Basics IP Addresses Security Tags Review + create

Azure Virtual Network (VNet) is the fundamental building block for your private network in Azure. VNet enables many types of Azure resources, such as Azure Virtual Machines (VM), to securely communicate with each other, the internet, and on-premises networks. VNet is similar to a traditional network that you'd operate in your own data center, but brings with it additional benefits of Azure's infrastructure such as scale, availability, and isolation. [Learn more about virtual network](#)

Project details

Subscription * ⓘ

Resource group * ⓘ [Create new](#)

Instance details

Name *

Region *

Step 5: Next to the prepopulated IPv4 address space, click the trash can.

Basics **IP Addresses** Security Tags Review + create

The virtual network's address space, specified as one or more address prefixes in CIDR notation (e.g. 192.168.1.0/24).

IPv4 address space

IPv4 address space	Remove
172.23.0.0/16 172.23.0.0 - 172.23.255.255 (65536 addresses)	

☐ Add IPv6 address space ⓘ

The subnet's address range in CIDR notation (e.g. 192.168.1.0/24). It must be contained by the address space of the virtual network.

[+ Add subnet](#) [Remove subnet](#)

Subnet name	Subnet address range
☐ default	172.23.0.0/24

Step 6: In the **IPv4 Address space** box, enter **10.112.0.0/16**.



Note

If the proposed address space overlaps with address space already assigned in another VNet within the same subscription, Azure Resource Manager provides a warning. If communication between these VNets is not required, you can ignore this warning. Otherwise, choose a different, non-overlapping address space.

Step 7: Click **Add Subnet**. The Add Subnet pane opens.

Step 8: In the **Subnet name** box, enter **app01**.

Step 9: In the **Subnet Address Range** box, enter **10.112.1.0/24**, and then click **Add**.

Step 10: Click **Review + create**, and then click **Create**.

Step 11: Repeat this procedure for the **app02** VNet and subnet.

1.3 Create the NSGs for the Application Subnets

In this procedure, you create network security groups (NSGs) for use with each application subnet. Each NSG includes default rules that allow for traffic within the VNet and from the peered VNets.



Note

Azure requires that you apply an NSG on a subnet or network interface card (NIC) of your virtual machine resource. If you associate the resource with a standard-SKU public IP address and you do not apply an NSG, traffic is not permitted to reach the resource.

First, you create the network security group.

Step 1: Navigate to **Home > Network Security groups**, and then click **Create**.

Step 2: In the **Resource group** list, choose **app01**.

Step 3: In the **Name** box, enter **app01**.

Step 4: In the **Region** list, choose **(US) West US** or the appropriate value.

Step 5: Click **Review + create**, and then click **Create**.

Next, you associate the network security group to the subnet.

Step 6: Navigate to **Home > Network security groups > app01**, and then in the Settings section, click **Subnets**.

Step 7: Click **Associate**, and then in the **Virtual Network** list, choose **app01**.

Step 8: In the **Subnet** list, choose **app01**, and then click **OK**.

Step 9: Repeat this procedure in order to create and apply the NSG for the **app02** subnet.

You can now deploy application compute resources in the app01 and app02 resource groups.

Deploying Outbound and East-West Security

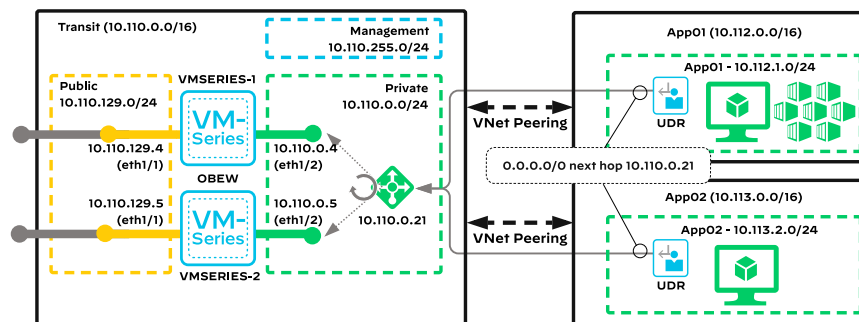
In these procedures, you create the transit resource group, virtual networking, and necessary services to secure outbound, east-west, and enterprise traffic. You configure VNet peering connections to support management and application flows. You then create device groups and templates in Panorama and finally deploy the VM-Series firewalls.



Note

If you are deploying the Transit VNet Model with common firewall option, follow the additional instructions in this group of procedures in order to prepare the outbound and east-west VM-Series firewalls for concurrent inbound security.

Figure 8 Outbound and east-west security deployment



Procedures

Configuring the Transit Resource Group, VNet, Subnets, and Services

- 2.1 Create the Transit Resource Group
- 2.2 Create the Transit VNet and Subnets
- 2.3 Create the NSG for the Private Subnet
- 2.4 Create the NSG for the Management Subnet
- 2.5 Create the NSG for the Public Subnet
- 2.6 Configure Transit and Management VNet Peering
- 2.7 Configure Transit and App VNet Peering

In these procedures, you create the transit resource group, virtual networking, and network security to support the VM-Series virtual machines.

2.1 Create the Transit Resource Group

First, you create a resource group that will contain Azure resources and the VM-Series firewalls.

Step 1: Log in to the Azure Resource Manager at <https://portal.azure.com>.

Step 2: Navigate to **Home > Resource groups**, and then click **Create**.

Step 3: In the **Resource group** box, enter **transit**, and then in the **Region** list, choose **(US) West US** or the appropriate value.

Step 4: Click **Review + create**, and then click **Create**.

2.2 Create the Transit VNet and Subnets

Next, you create the transit VNet with an IP address space and multiple subnets that must be within the IP address space. As necessary, to add additional IP address spaces and subnets, you can modify the transit VNet after creation.

Table 3 Transit virtual network subnets

Resource group	VNet	IP address range	Subnet	Subnet address range
transit	transit	10.110.0.0/16	management	10.110.255.0/24
transit	transit	10.110.0.0/16	public	10.110.129.0/24
transit	transit	10.110.0.0/16	private	10.110.0.0/24

Step 1: Navigate to **Home > Virtual networks**, and then click **Create**.

Step 2: In the **Resource group** list, choose **transit**.

Step 3: In the **Name** box, enter **transit**.

Step 4: In the **Region** list, choose **(US) West US** or the appropriate value, and then click **Next: IP Addresses**.

Step 5: Next to the prepopulated IPv4 address space, click the trash can.

Step 6: In the **IPv4 Address space** box, enter **10.110.0.0/16**.



Note

If the proposed address space overlaps with address space already assigned in another VNet within the same subscription, Azure Resource Manager provides a warning. If communication between these VNets is not required, you can ignore this warning. Otherwise, choose a different, non-overlapping address space.

Step 7: Click **Add Subnet**. The Add Subnet pane opens.

Step 8: In the **Subnet name** box, enter **management**.

Step 9: In the **Subnet Address Range** box, enter **10.110.255.0/24**, and then click **Add**.

Step 10: Repeat Step 7 through Step 9 for the rest of the subnets listed in Table 3.

Basics **IP Addresses** Security Tags Review + create

The virtual network's address space, specified as one or more address prefixes in CIDR notation (e.g. 192.168.1.0/24).

IPv4 address space

10.110.0.0/16 ✓

☐ Add IPv6 address space ⓘ

The subnet's address range in CIDR notation (e.g. 192.168.1.0/24). It must be contained by the address space of the virtual network.

+ Add subnet Remove subnet

☐ Subnet name	Subnet address range	NAT gateway
☐ management	10.110.255.0/24	-
☐ public	10.110.129.0/24	-
☐ private	10.110.0.0/24	-

i Use of a NAT gateway is recommended for outbound internet access from a subnet. You can deploy a NAT gateway and assign it to a subnet after you create the virtual network. [Learn more](#)

Step 11: Click **Review + create**, and then click **Create**.

2.3 Create the NSG for the Private Subnet

In this procedure, you create NSGs for use with the private subnet. The private NSG includes default rules that allow for traffic within the VNet and from the peered VNets.

First, you create the network security group for the private subnet.

Step 1: Navigate to **Home > Network Security groups**, and then click **Create**.

Step 2: In the **Resource group** list, choose **transit**.

Step 3: In the **Name** box, enter **private**.

Step 4: In the **Region** list, choose **(US) West US** or the appropriate value.

Step 5: Click **Review + create**, and then click **Create**.

Next, you associate the NSG with the private subnet.

Step 6: Navigate to **Home > Network security groups > private**, and then in the Settings section, click **Subnets**.

Step 7: Click **Associate**, and then in the **virtual network** list, choose **transit**.

Step 8: In the **Subnet** list, choose **private**, and then click **OK**.

2.4 Create the NSG for the Management Subnet

In this procedure, you create NSGs for use with the transit management subnet. The management NSG includes rules that allow your organization's firewall administrators to connect to the VM-Series firewall's management interface from a specific set of trusted enterprise IP addresses.



Note

Azure requires that you apply an NSG on a subnet or NIC of your virtual machine resource. If you associate the resource with a standard-SKU public IP address and do not apply an NSG, traffic is not permitted to reach the resource.

Table 4 Management inbound rules

Priority	Name	Destination port	Protocol	Source	Destination	Action
1000	AllowInternet	443	TCP	<TrustedIPs>	Any	Allow
1010	AllowManagement	22,443	TCP	10.255.0.0/24	Any	Allow
1020	AllowOnPremises	22,443	TCP	10.5.0.0/16	Any	Allow



Note

Azure uses an asterisk (*) to set a port/protocol to Any. Where you see Any listed, you enter an * in the field.

First, you create the network security group for the management subnet.

Step 1: Navigate to **Home > Network Security groups**, and then click **Create**.

Step 2: In the **Resource group** list, choose **transit**.

Step 3: In the **Name** box, enter **management**.

Step 4: In the **Region** list, choose **(US) West US** or the appropriate value.

Step 5: Click **Review + create**, and then click **Create**.

Next, you create rules that allow inbound management traffic for the Transit resource group.

Step 6: Navigate to **Home > Network security groups > management**, and then in the Settings section, click **Inbound security rules**.

Step 7: Click **Add**. The Add Inbound Security Rule pane appears.

Step 8: In the **Source** box, select **IP Addresses**.

Step 9: In the **Source IP addresses/CIDR ranges** box, enter [<TrustedIPs>](#). This represents a list of trusted IP addresses (Public IPs) that your organization uses to directly access the firewall management interfaces.

**Note**

Multiple public IP addresses may be entered by using a comma to separate the entries.

Step 10: In the **Destination** list, select **Any**.

Step 11: In the **Destination port ranges** box, enter [443](#).

Step 12: In the Protocol section, choose **TCP**.

Step 13: In the Action section, choose **Allow**.

Step 14: In the **Priority** box, enter [1000](#).

Step 15: In the **Name** box, enter [AllowInternet](#), and then click **Add**.

Step 16: Repeat Step 7 through Step 15 for the rules in Table 4.

Filter by name		Port == all	Protocol == all	Source == all	Destination == all	Action == all	
Priority ↑↓	Name ↑↓	Port ↑↓	Protocol ↑↓	Source ↑↓	Destination ↑↓	Action ↑↓	
Inbound Security Rules							
1000	AllowInternet	443	TCP	100.64.1.0/24	Any	Allow	
1010	AllowManagement	22,443	TCP	10.255.0.0/24	Any	Allow	
1020	AllowOnPremises	22,443	TCP	10.5.0.0/16	Any	Allow	
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow	
65001	AllowAzureLoadBalance...	Any	Any	AzureLoadBalancer	Any	Allow	
65500	DenyAllInBound	Any	Any	Any	Any	Deny	

Next, you associate the network security group with the management subnet in the Transit resource group.

Step 17: Navigate to **Home > Network security groups > [management](#)**, and then in the Settings section, click **Subnets**.

Step 18: Click **Associate**, and then in the **virtual network** list, choose [transit](#).

Step 19: In the **Subnet** list, choose [management](#), and then click **OK**.

2.5 Create the NSG for the Public Subnet

In this procedure, you create an NSG for use with the public subnet. You will configure a distinct set of inbound and outbound rules.

**Note**

Azure requires that you apply an NSG on a subnet or NIC of your virtual machine resource. If you associate the resource with a standard-SKU public IP address and you do not apply an NSG, traffic is not permitted to reach the resource.

Table 5 Public inbound rules

Priority	Name	Destination port	Protocol	Source	Destination	Action
1000	AllowInternet	80,443	TCP	Internet	Any	Allow
1040	AllowAzureLoadBalancer	Any	Any	AzureLoadBalancer	Any	Allow

Table 6 Public outbound rules

Priority	Name	Destination port	Protocol	Source	Destination	Action
1000	AllowInternetOutbound	Any	Any	Any	Internet	Allow
1010	AllowPublicToPublic	Any	Any	10.110.128.0/23	10.110.128.0/23	Allow

First, you create the network security group for the public subnet.

Step 1: Navigate to **Home > Network Security groups**, and then click **Create**.

Step 2: In the **Resource group** list, choose **transit**.

Step 3: In the **Name** box, enter **public**.

Step 4: In the **Region** list, choose **(US) West US** or the appropriate value.

Step 5: Click **Review + create**, and then click **Create**.

Next, you create rules that allow inbound traffic.

Step 6: Navigate to **Home > Network security groups > public**, and then in the Settings section, click **Inbound security rules**.

Step 7: Click **Add**. The Add Inbound Security Rule pane appears.

Step 8: In the **Source** box, select **Service Tag**.

Step 9: In the **Source service tag** list, select **Internet**.

Step 10: In the **Destination** list, select **Any**.

Step 11: In the **Destination port ranges** box, enter **80, 443**.

Step 12: In the Protocol section, choose **TCP**.

Step 13: In the Action section, choose **Allow**.

Step 14: In the **Priority** box, enter **1000**.

Step 15: In the **Name** box, enter **AllowInternet**, and then click **Add**.

Step 16: Repeat Step 7 through Step 15 with the remaining inbound values in Table 5.

Next, you create rules that allow outbound traffic.

Step 17: Repeating this procedure starting at Step 6, select **Outbound security rules**.

Step 18: Repeat Step 7 through Step 15, using the outbound values in Table 6.

Filter by name		Port == all	Protocol == all	Source == all	Destination == all	Action == all
Priority ↑↓	Name ↑↓	Port ↑↓	Protocol ↑↓	Source ↑↓	Destination ↑↓	Action ↑↓
Inbound Security Rules						
1000	AllowInternet	80,443	TCP	Internet	Any	✓ Allow
1040	AllowAzureLoadBalan...	Any	Any	AzureLoadBalancer	Any	✓ Allow
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	✓ Allow
65001	AllowAzureLoadBalan...	Any	Any	AzureLoadBalancer	Any	✓ Allow
65500	DenyAllInBound	Any	Any	Any	Any	✗ Deny
Outbound Security Rules						
1000	AllowInternetOutbound	Any	Any	Any	Internet	✓ Allow
1010	AllowPublicToPublic	Any	Any	10.110.128.0/23	10.110.128.0/23	✓ Allow
65000	AllowVnetOutBound	Any	Any	VirtualNetwork	VirtualNetwork	✓ Allow
65001	AllowInternetOutBound	Any	Any	Any	Internet	✓ Allow
65500	DenyAllOutBound	Any	Any	Any	Any	✗ Deny

Next, you associate the network security group with the public subnet.

Step 19: Navigate to **Home > Network security groups > public**, and then in the Settings section, click **Subnets**.

Step 20: Click **Associate**, and then in the **virtual network** list, choose **transit**.

Step 21: In the **Subnet** list, choose **public**, and then click **OK**.

2.6 Configure Transit and Management VNet Peering

This guide assumes that you have already deployed Panorama in high-availability mode in a dedicated management resource group and VNet. To manage devices deployed in the transit VNet, you must create a VNet peering connection between the transit VNet and the management VNet.

You need to configure only the peering relationship from the transit VNet. Azure then makes the necessary configuration for the management VNet.



Note

In the Configure Forwarded Traffic Settings section, do not enable traffic forwarding from the Panorama management VNet to the transit VNet. The Panorama management VNet does not include any devices configured for forwarding.

Enable traffic forwarding from the transit VNet to the Panorama management VNet. This model assumes that traffic from enterprise networks accesses the Panorama management VNet through the transit VNet.

Step 1: Navigate to **Home > Virtual networks > transit**, click **Peerings**, and then click **Add**.

Step 2: In the This Virtual Network section, in the **Peering link name** box, enter **transit-to-management**.

Step 3: In **Traffic to remote virtual network**, select **Allow (default)**.

Step 4: In **Traffic forwarded from remote virtual network**, select **Block traffic that originates from outside this virtual network**.

Step 5: In the Remote Virtual Network section, in the **Peering link name** box, enter **management-to-transit**.

Step 6: In the **Virtual network** list, choose **management**, and then click **Add**.

This virtual network

Peering link name *

transit-to-management ✓

Traffic to remote virtual network ⓘ

☒ Allow (default)

☐ Block all traffic to the remote virtual network

Traffic forwarded from remote virtual network ⓘ

☐ Allow (default)

☒ Block traffic that originates from outside this virtual network

Virtual network gateway or Route Server ⓘ

☐ Use this virtual network's gateway or Route Server

☐ Use the remote virtual network's gateway or Route Server

☒ None (default)

Remote virtual network

Peering link name *

management-to-transit ✓

Virtual network deployment model ⓘ

☒ Resource manager

☐ Classic

2.7 Configure Transit and App VNet Peering

To access application services and resources through the transit VNet, you must create a VNet peer connection between the transit VNet and each application VNet.

You need to configure only the peering relationship from the transit VNet. Azure then makes the necessary configuration for the application VNet.

Step 1: Navigate to **Home > Virtual networks > transit**, and then click **Peerings**.

Step 2: Click **Add**.

Step 3: In the This Virtual Network section, in the **Peering link name** box, enter **transit-to-app01**.

Step 4: In **Traffic to remote virtual network**, select **Allow (default)**.

Step 5: In **Traffic forwarded from remote virtual network**, select **Block traffic that originates from outside this virtual network**.

Step 6: In the Remote Virtual Network section, in the **Peering link name** box, enter **app01-to-transit**.

Step 7: In the **Virtual network** list, choose **app01**, and then click **Add**.

Step 8: Repeat Step 2 through Step 7 for the **app02** VNet.

Procedures

Configuring Panorama Templates, Template Stacks, and Device Groups

- 3.1 Create the Templates and Template Stack
- 3.2 Configure the Baseline Settings Template
- 3.3 Create Device Groups
- 3.4 Create Log Forwarding Objects
- 3.5 Create the Interface Management Profiles
- 3.6 Configure the Network Settings Template
- 3.7 Add Static Routes to the Virtual Routers
- 3.8 Configure the Azure Plugin
- 3.9 Create Dynamic Address Groups
- 3.10 Configure the Outbound NAT Policy
- 3.11 Configure the Health Probe Security Policy
- 3.12 Configure the Outbound Security Policy
- 3.13 Configure the East-West Security Policy
- 3.14 Generate VM Auth Key

These procedures assume that you have already deployed Panorama in a peered VNet, following the procedures in

Panorama on Azure: Deployment Guide.

Panorama provides the following tools for centralized administration:

- **Templates and template stacks**—Panorama manages common device and network configuration through templates. To manage configuration centrally and then push the changes to all managed firewalls, you can use templates. This approach avoids making the same individual firewall change repeatedly across many devices. To make things easier, you can stack templates and use them as building blocks for device and network configuration.
- **Hierarchical device groups**—Panorama manages common policies and objects through hierarchical device groups. To centrally manage the policies across all deployment locations with common requirements, you can use multi-level device groups.

3.1 Create the Templates and Template Stack

The templates include configurations for common functions and separate configurations for specific deployments across the VM-Series devices.

You create the following Panorama templates:

- **VMSeries-Baseline**—Includes basic networking services, including DNS and NTP, and also includes device functions that enable log forwarding to SLS.
- **Transit-2-Zone**—Includes firewall networking functions for outbound, east-west, and enterprise, including interfaces, zones, and the virtual router. You also use this template if you are deploying the inbound common firewall option.

You apply templates to devices by using Panorama template stacks, which logically merges the assigned templates and associates them with the relevant devices.

First, you create the templates.

Step 1: Log in to the primary Panorama virtual appliance's web interface.

Step 2: Navigate to **Panorama > Templates**, and then click **Add**.

Step 3: In the **Name** box, enter **VMSeries-Baseline**.

Step 4: In the **Description** box, enter a valid description, and then click **OK**.

Step 5: Click **Add**.

Step 6: In the **Name** box, enter **Transit-2-Zone**.

Step 7: In the **Description** box, enter a valid description, and then click **OK**.

Next, you create the template stack.

Step 8: Navigate to **Panorama > Templates**, and then click **Add Stack**.

Step 9: In the **Name** box, enter **Transit**.

Step 10: In the **Description** box, enter an appropriate description.

Step 11: In the Templates pane, click **Add**, select **VMSeries-Baseline** and **Transit-2-Zone**, and then click **OK**.

☐	NAME ^	DESCRIPTION	TYPE	STACK
☐	Transit		template-stack	VMSeries-Baseline Transit-2-Zone
☐	Transit-2-Zone		template	
☐	VMSeries-Baseline		template	

3.2 Configure the Baseline Settings Template

In the baseline template, you configure uniform DNS and NTP across all devices and ensure all devices forward logs to Strata Logging Service.

Step 1: Navigate to **Device**, and then in the **Template** list, choose **VMSeries-Baseline**.

Step 2: Navigate to **Setup > Services > Global > Services**, and then click the edit cog.

Step 3: In the **Primary DNS Server** box, enter **168.63.129.16**.

Step 4: In the **Secondary DNS Server** box, enter **8.8.8.8**.

Step 5: On the NTP tab, in the Primary NTP Server section, in the **NTP Server Address** box, enter **0.pool.ntp.org**.

Step 6: In the Secondary NTP Server section, in the **NTP Server Address** box, enter **1.pool.ntp.org**, and then click **OK**.

Step 7: Navigate to **Device > Setup > Management > Cortex Data Lake**, and then click the edit cog.

Step 8: Select **Enable Cortex Data Lake**.

Step 9: Select **Enable Enhanced Application Logging**.

Step 10: In **Region** list, choose **United States - Americas**, and then click **OK**.

Step 11: Navigate to **Device > Log Settings > System**, and then click **Add**. The Log Settings—System dialog box appears.

Step 12: In the **Name** box, enter **System Logs**.

Step 13: Select **Panorama/Cortex Data Lake**, and then click **OK**.

Step 14: In **Device > Log Settings > Configuration**, click **Add**. The Log Settings—Configuration dialog box appears.

Step 15: In the **Name** box, enter **Configuration Logs**.

Step 16: Select **Panorama/Cortex Data Lake**, and then click **OK**.

3.3 Create Device Groups

After the templates, you create the Panorama device-groups:

- **Azure-Baseline**—This parent device-group includes the log-forwarding object required for log forwarding to SLS. All child device-groups inherit configuration from this parent-device group.
- **Transit**—This child device-group includes objects and policies for outbound, east-west, and enterprise traffic. You also use this device group if you are deploying the inbound common firewall option.

First, you create the device groups and specify the parent device-group in order to properly inherit device group settings. Later, you create the objects and policies in the procedures that require them.

When a device group includes references to zones within a policy, you must include the template that you originally used to define the zone. This template serves as a reference template for the device group.

Table 7 Panorama outbound east/west device groups

Device group	Reference templates	Parent device group
Azure-Baseline	N/A	Shared
Transit	Transit-2-Zone	Azure-Baseline

Step 1: Navigate to **Panorama > Device Groups**, and then click **Add**.

Step 2: In the **Name** box, enter **Azure-Baseline**.

Step 3: In the **Description** box, enter a valid description.

Step 4: In the **Parent Device Group** box, verify the value is set to **Shared**, and then click **OK**.

Step 5: Navigate to **Panorama > Device Groups**, and then click **Add**.

Step 6: In the **Name** box, enter **Transit**.

Step 7: In the **Description** box, enter a valid description.

Step 8: In the **Parent Device Group** list, choose **Azure-Baseline**.

Step 9: In the Reference Templates pane, click **Add**, choose **Transit-2-Zone**, and then click **OK**.

☐	NAME ^	DESCRIPTION	REFERENCE TEMPLATES
☐	Shared		
☐	Azure-Baseline	Parent DG with Baseline configurations for all Azure devices	
☐	Transit	Child DG with objects and policies for OBEW VM-Series devices in the Transit VNet	Transit-2-Zone

3.4 Create Log Forwarding Objects

Initially, you create the log-forwarding profile to send security policy logs to SLS. This profile is associated with any security policy rules that you create that use SLS.

Step 1: Navigate to **Objects**, in the **Device Group** list, choose **Azure-Baseline**.

Step 2: Navigate to **Log Forwarding**, and then click **Add**.

Step 3: In the **Name** box, enter **Forward-to-Strata-Logging-Service**.

Step 4: Select **Enable enhanced application logging to Logging Service (including traffic and url logs)**, and then click **OK**.

Log Forwarding Profile

Name: Forward-to-Strata-Logging-Service

☐ Shared

☒ Enable enhanced application logging to Cortex Data Lake (including traffic and url logs)

☐ Disable override

Description:

NAME	LOG TYPE	FILTER	FORWARD METHOD	BUILT-IN ACTIONS
☐ traffic-enhanced-app-logging	traffic	All Logs	• Panorama/Cortex Data Lake	
☐ threat-enhanced-app-logging	threat	All Logs	• Panorama/Cortex Data Lake	
☐ wildfire-enhanced-app-logging	wildfire	All Logs	• Panorama/Cortex Data Lake	
☐ url-enhanced-app-logging	url	All Logs	• Panorama/Cortex Data Lake	
☐ data-enhanced-app-logging	data	All Logs	• Panorama/Cortex Data Lake	
☐ tunnel-enhanced-app-logging	tunnel	All Logs	• Panorama/Cortex Data Lake	
☐ auth-enhanced-app-logging	auth	All Logs	• Panorama/Cortex Data Lake	
☐ decryption-enhanced-app-logging	decryption	All Logs	• Panorama/Cortex Data Lake	

+ Add - Delete Clone

OK Cancel

3.5 Create the Interface Management Profiles

The load-balancer health-checks use HTTPS probes targeting the firewall's dataplane interfaces. By default, the firewall blocks responses to these probes. You use interface management profiles in order to allow the health checks and filter the sources.



Note

You can apply a single management profile to multiple interfaces. Palo Alto Networks recommends separate management profiles per interface, if required, to allow for different management policies.

First, you create a management profile for the private-facing VM-Series interfaces, allowing HTTPS from the Azure health-check source address.

Step 1: Navigate to **Network**, and then in the **Template** list, choose **Transit-2-Zone**.

Step 2: Navigate to **Network Profiles > Interface Mgmt**, and then click **Add**. The Interface Management Profile dialog box appears.

Step 3: In the **Name** box, enter **MP-private**.

Step 4: In the Administrative Management Services section, select **HTTPS**.

Step 5: In the Permitted IP Addresses pane, click **Add**.

Step 6: Enter **168.63.129.16/32**, and then click **OK**.

If you are deploying the inbound common firewall option, create a management profile for the public-facing VM-Series interfaces, allowing HTTPS from the Azure health check source address.

Step 7: Navigate to **Network**, and then in the **Template** list, choose **Transit-2-Zone**.

Step 8: Navigate to **Network Profiles > Interface Mgmt**, and then click **Add**. The Interface Management Profile dialog box appears.

Step 9: In the **Name** box, enter **MP-public**.

Step 10: In the Administrative Management Services section, select **HTTPS**.

Step 11: In the Permitted IP Addresses pane, click **Add**.

Step 12: Enter **168.63.129.16/32**, and then click **OK**.

3.6 Configure the Network Settings Template

In this procedure, you create the network settings template that configures interfaces, zones, and routing for the VM-Series firewalls. All interfaces obtain their IP addressing through DHCP.

If you are deploying the dedicated inbound option, create a single default virtual router by using the configuration values shown in Table 8.

If you are deploying the common firewall option, create two virtual routers by using the configuration values shown in Table 9.

Table 8 Template interface settings for Transit-2-Zone dedicated inbound option

Slot	Interface	Interface type	Virtual router	Security zone	IPv4	Management profile
Slot 1	ethernet1/1	Layer3	vr-default	public	DHCP Client	—
Slot 1	ethernet1/2	Layer3	vr-default	private	DHCP Client	mp-private

Table 9 Template interface settings for Transit-2-Zone common firewall option

Slot	Interface	Interface type	Virtual router	Security zone	IPv4	Management profile
Slot 1	ethernet1/1	Layer3	vr-public	public	DHCP Client	mp-public
Slot 1	ethernet1/2	Layer3	vr-private	private	DHCP Client	mp-private

Step 1: Navigate to **Network**, and then in the **Template** list, choose **Transit-2-Zone**.

Step 2: Navigate to **Interfaces > Ethernet**, and then click **Add Interface**. The Ethernet Interface dialog box appears.

Step 3: In the **Slot** list, choose **Slot 1**.

Step 4: In the **Interface Name** list, choose **ethernet1/1**.

Step 5: In the **Interface Type** list, choose **Layer3**.

Step 6: On the Config tab, in the **Virtual Router** list, choose **New Virtual Router**. The Virtual Router dialog box appears.

Step 7: In the **Name** box, enter the virtual router name from the table you are following, and then click **OK**.

Step 8: In the **Security Zone** list, choose **New Zone**. The Zone dialog box appears.

Step 9: In the **Name** box, enter **public**, and then click **OK**.

Ethernet Interface ⓘ

Slot: Slot 1

Interface Name: ethernet1/1

Comment:

Interface Type: Layer3

Netflow Profile: None

Config | IPv4 | IPv6 | SD-WAN | Advanced

Assign Interface To

Virtual Router: vr-default

Virtual System: vsys1

Security Zone: public

OK Cancel

Step 10: On the IPv4 tab, for **Type**, select **DHCP client**.

Step 11: Clear **Automatically create default route pointing to default gateway provided by server**.

Step 12: On the Advanced tab, in the **Management Profile** list, choose the management profile from the table you are following, and then click **OK**.

Step 13: On the warning, click **Yes**.

Step 14: Repeat this procedure for the second interface listed in the table you are following, making sure to create the correct virtual router(s) and select the assigned management profile.

3.7 Add Static Routes to the Virtual Routers

In this procedure, you configure the static routes in the firewall virtual router. Even though Azure does not use default gateways in the traditional way, you configure the static routes in the NGFW to point at a default gateway to forward the traffic into the correct VNet subnet.

- Default gateway for public zone: **10.110.129.1**
- Default gateway for private zone: **10.110.0.1**

If you are deploying the dedicated inbound option, use the static routes shown in Table 10.

If you are deploying the common firewall option, use the static routes shown in Table 11.

Table 10 Static routes for Transit-2-Zone template with the dedicated inbound option

Virtual router	Name	Destination prefix	Interface	Next-hop	Next-hop value
vr-default	default	0.0.0.0/0	ethernet1/1	IP Address	10.110.129.1
vr-default	public	10.110.128.0/23	ethernet1/1	IP Address	10.110.129.1
vr-default	azure-probe	168.63.129.16/32	ethernet1/2	IP Address	10.110.0.1
vr-default	private	10.110.0.0/17	ethernet1/2	IP Address	10.110.0.1
vr-default	app01	10.112.0.0/16	ethernet1/2	IP Address	10.110.0.1
vr-default	app02	10.113.0.0/16	ethernet1/2	IP Address	10.110.0.1
vr-default	management	10.255.0.0/16	ethernet1/2	IP Address	10.110.0.1
vr-default	on-premises	10.5.0.0/16	ethernet1/2	IP Address	10.110.0.1

Table 11 Static routes for Transit-2-Zone template with the common firewall option

Virtual router	Name	Destination prefix	Interface	Next-hop	Next-hop value
vr-public	default	0.0.0.0/0	ethernet1/1	IP Address	10.110.129.1
vr-public	public	10.110.128.0/23	ethernet1/1	IP Address	10.110.129.1
vr-public	azure-probe	168.63.129.16/32	ethernet1/1	IP Address	10.110.129.1
vr-public	private	10.110.0.0/17	none	Next VR	vr-private
vr-public	app01	10.112.0.0/16	none	Next VR	vr-private
vr-public	app02	10.113.0.0/16	none	Next VR	vr-private
vr-private	default	0.0.0.0/0	none	Next VR	vr-public
vr-private	private	10.110.0.0/17	ethernet1/2	IP Address	10.110.0.1
vr-private	azure-probe	168.63.129.16/32	ethernet1/2	IP Address	10.110.0.1
vr-private	app01	10.112.0.0/16	ethernet1/2	IP Address	10.110.0.1
vr-private	app02	10.113.0.0/16	ethernet1/2	IP Address	10.110.0.1
vr-private	management	10.255.0.0/16	ethernet1/2	IP Address	10.110.0.1
vr-private	on-premises	10.5.0.0/16	ethernet1/2	IP Address	10.110.0.1

Step 1: Navigate to **Network**, and then in the **Template** list, choose **Transit-2-Zone**.

Step 2: Navigate to **Virtual Routers**, and then click the virtual router for your deployment.

If you are deploying the dedicated inbound option, click **vr-default**.

If you are deploying the common firewall option, click **vr-public**.

Step 3: On the Static Routes tab, click **Add**.

Step 4: In the **Name** box, enter **default**.

Step 5: In the **Destination** box, enter **0.0.0.0/0**.

Step 6: In the **Interface** list, choose **ethernet1/1**.

Step 7: In the **Next Hop** list, choose **IP Address**, enter **10.110.129.1**, and then click **OK**.

Step 8: Repeat Step 3 through Step 7 for the remaining route values for the virtual router.

Step 9: After adding all routes for the selected virtual router, in the Virtual Router dialog box, click **OK**.

The screenshot shows the 'Virtual Router - vr-default' dialog box with the 'Static Routes' tab selected. The 'IPv4' tab is active, and a search bar shows '7 items'. The table below lists the static routes:

	NAME	DESTINATION	INTERFACE	Next Hop		ADMIN DISTANCE	METRIC	BFD	ROUTE TABLE
				TYPE	VALUE				
☐	azure-probe	168.63.129.16/32	ethernet1/2	ip-address	10.110.0.1	default	10	None	unicast
☐	app01	10.112.0.0/16	ethernet1/2	ip-address	10.110.0.1	default	10	None	unicast
☐	app02	10.113.0.0/16	ethernet1/2	ip-address	10.110.0.1	default	10	None	unicast
☐	on-premises	10.5.0.0/16	ethernet1/2	ip-address	10.110.0.1	default	10	None	unicast
☐	default	0.0.0.0/0	ethernet1/1	ip-address	10.110.129.1	default	10	None	unicast
☐	private	10.110.0.0/17	ethernet1/2	ip-address	10.110.0.1	default	10	None	unicast
☐	public	10.110.128.0/23	ethernet1/1	ip-address	10.110.129.1	default	10	None	unicast

At the bottom, there are buttons for 'Add', 'Delete', and 'Clone', and 'OK' and 'Cancel' buttons.

Step 10: If you are deploying the common firewall option, repeat this procedure for the vr-private virtual router and route values in Table 11.

3.8 Configure the Azure Plugin

In this procedure, you configure the Panorama Azure plugin. First, you enable monitoring, and then you add a monitoring definition that allows the Azure plugin to monitor Azure resources in your app01 and app02 VNets.

Step 1: On the primary Panorama virtual appliance, navigate to **Panorama > Azure > Setup**.

Step 2: On the General tab, select **Enable Monitoring**.

Step 3: On the Notify Groups tab, click **Add**.

Step 4: In the **Name** box, enter **Azure**.

Step 5: In the Notify Group box, select the **Transit** device group, and then click **OK**.

The screenshot shows the 'Notify Group' dialog box. The 'Name' field is set to 'Azure'. Below it, the 'Notify Group' list shows '3 items'. The 'DEVICE GROUP' section is expanded, showing 'Azure-Baseline' and 'Transit'. The 'Transit' group is selected with a checkmark. At the bottom, there are 'OK' and 'Cancel' buttons.

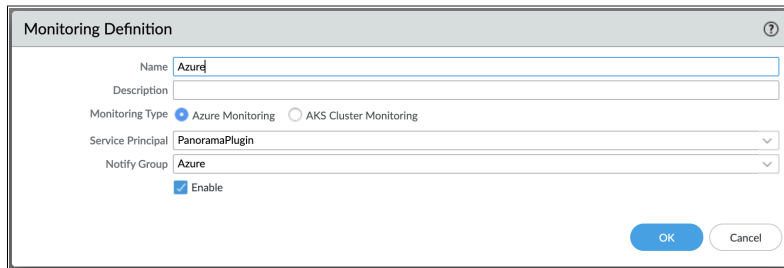
Next, you create a monitoring definition.

Step 6: Navigate to **Panorama > Azure > Monitoring Definition**, and then click **Add**.

Step 7: In the **Name** box, enter **Azure**.

Step 8: In the **Service Principal** list, select **PanoramaPlugin**.

Step 9: In the **Notify Group** list, select **Azure**, and then click **OK**.



The image shows a 'Monitoring Definition' dialog box. It has a title bar with a question mark icon. Inside, there are several fields: 'Name' with the value 'Azure', 'Description' (empty), 'Monitoring Type' with 'Azure Monitoring' selected (radio button), 'Service Principal' with a dropdown showing 'PanoramaPlugin', and 'Notify Group' with a dropdown showing 'Azure'. There is also an 'Enable' checkbox which is checked. At the bottom right, there are 'OK' and 'Cancel' buttons.

3.9 Create Dynamic Address Groups

In this procedure, you add Azure resource information into Panorama dynamic address groups (DAGs), using the object information provided through the Azure plugin.

First, you create a DAG for the app01 subnet.

Step 1: Navigate to **Objects**, and then in the **Device Group** list, choose **Transit**.

Step 2: Navigate to **Address Groups**, and then click **Add**.

Step 3: In the **Name** box, enter **app01**.

Step 4: In the **Type** list, choose **Dynamic**.

Step 5: Click **Add Match Criteria**.

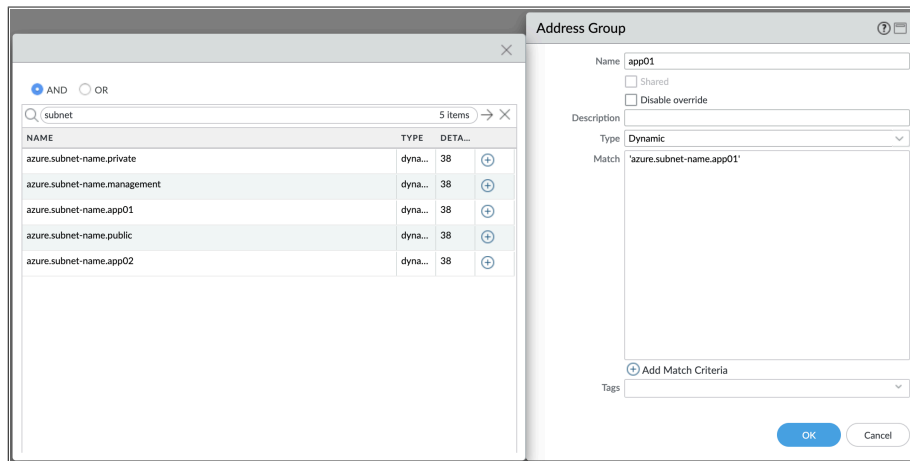


Note

Some browser extensions might block API calls between Panorama and Azure, which prevents Panorama from receiving match criteria. If Panorama displays no match criteria and you are using browser extensions, disable the extensions and synchronize dynamic objects. This populates the tags available to Panorama.

Step 6: In the filter box, type **subnet**, and then click the right arrow.

Step 7: Select **azure.subnet-name.app01**, click the + to add the entry into the **Match** box, and then click **OK**.



Step 8: Repeat this procedure for the **app02** subnet.

Step 9: On the **Commit** menu, select **Commit to Panorama**, and then click **Commit**.

3.10 Configure the Outbound NAT Policy

For virtual machines behind the firewall to communicate with devices on the internet, the firewall must translate the source IP address of the outbound traffic to an IP address on the public subnet. The simplest method is to use dynamic IP and port translation to the firewall's public interface IP address.

In this procedure, you use NAT Pre Rules, which are evaluated prior to local rules and cannot be overridden.

Step 1: Navigate to **Policies**, and then in the **Device Group** list, choose **Transit**.

Step 2: Navigate to **NAT > Pre Rules**, and then click **Add**.

Step 3: In the **Name** box, enter **Outbound-Internet**.

Step 4: On the **Original Packet** tab, in the **Source Zone** pane, click **Add**, and then select **private**.

Step 5: In the **Destination Zone** list, choose **public**.

Step 6: In the **Destination Interface** list, choose **ethernet1/1**.

Step 7: In the **Source Address** pane, click **Add**, and then select **app01**.

Step 8: In the Source Address pane, click **Add**, and then select **app02**.

NAT Policy Rule

General | **Original Packet** | Translated Packet | Active/Active HA Binding | Target

Source Zone: **public** (Destination Zone: **Any**)

Destination Interface: **ethernet1/1** (Service: **any**)

Source Address: **app01**, **app02** (Destination Address: **Any**)

+ Add - Delete (Source Address) + Add - Delete (Destination Address)

OK Cancel

Step 9: On the Translated Packet tab, in the Source Address Translation section, in the **Translation Type** list, choose **Dynamic IP And Port**.

Step 10: In the Source Address Translation section, in the **Address Type** list, choose **Interface Address**.

Step 11: In the Source Address Translation section, in the **Interface** list, choose **ethernet1/1**.

NAT Policy Rule

General | Original Packet | **Translated Packet** | Active/Active HA Binding | Target

Source Address Translation:

- Translation Type: **Dynamic IP And Port**
- Address Type: **Interface Address**
- Interface: **ethernet1/1**
- IP Type: **None**

Destination Address Translation:

- Translation Type: **None**

OK Cancel

Step 12: On the Target tab, verify that **Any (target to all devices)** is selected, and then click **OK**.

Caution

Make sure to target all devices in the device group. Otherwise, the policy rule is not automatically applied to new group members.

3.11 Configure the Health Probe Security Policy

You must permit health probes from the Azure load-balancer on the firewall interfaces. The Azure load-balancer sends probes at 3-second intervals, which would generate a significant amount of firewall log activity. In this procedure, you create a security policy rule that explicitly permits the probes and suppresses logging from matches to the rule.

**Note**

To permit the probes instead of relying on the default intrazone rule, Palo Alto Networks recommends the use of an explicit rule. The rule permitting the probes is configured in the parent device group with a security Pre Rule and is inherited by the child device groups.

First, you create an address object for the Azure health check source address.

Step 1: Navigate to **Objects**, and then in the **Device Group** list, choose **Azure-Baseline**.

Step 2: Navigate to **Addresses**, and then click **Add**.

Step 3: In the **Name** box, enter **azure-probe**.

Step 4: In the **Type** list, choose **IP Netmask**.

Step 5: In the **Type value** box, enter **168.63.129.16/32**, and then click **OK**.

Address

Name:

☐ Shared

☐ Disable override

Description:

Type: IP Netmask 168.63.129.16/32 [Resolve](#)

Enter an IP address or a network using the slash notation (Ex. 192.168.80.150 or 192.168.80.0/24). You can also enter an IPv6 address or an IPv6 address with its prefix (Ex. 2001:db8:123:1::1 or 2001:db8:123:1::/64)

Tags:

OK **Cancel**

Next, you create the security policy that allows the health check probes.

Step 6: Navigate to **Policies**, and then in the **Device Group** list, choose **Azure-Baseline**.

Step 7: Navigate to **Security > Pre Rules**, and then click **Add**.

Step 8: In the **Name** box, enter **Permit-Azure-Probes**.

Step 9: In the **Rule Type** list, choose **intrazone**.

Step 10: On the **Source** tab, in the **Source Zone** pane, select **Any**.

Step 11: In the **Source Address** pane, click **Add**, and then select **azure-probe**.

Step 12: On the **Service/URL Category** tab, in the **Service** pane, select **any**.

Step 13: On the **Actions** tab, in the **Action Setting** section, in the **Action** list, choose **Allow**.

Step 14: In the **Log Setting** section, clear **Log at Session End**.

Step 15: On the Target tab, verify that **Any (target to all devices)** is selected, and then click **OK**.

Caution

Make sure to target all devices (any) in the device group. Otherwise, the policy rule is not automatically applied to new group members.

	NAME	LOCATION	TAGS	TYPE	Source				Destination			APPLICATION	SERVICE	ACTION
					ZONE	ADDRESS	USER	DEVICE	ZONE	ADDRESS	DEVICE			
1	Permit-Azure-Probes	Azure-Baseline	none	intrazone	any	azure-probe	any	any	(intrazone)	any	any	any	application-default	Allow

3.12 Configure the Outbound Security Policy

This example uses a common outbound policy for all private subnets. If you wish to use a differentiated policy, create separate rules for each subnet. In this procedure, you use security Pre Rules, which are logically evaluated prior to local rules and cannot be overridden on the device.

The security policy example for outbound access permits yum and NTP. As required, you can add additional applications to your policy.

Step 1: Navigate to **Policies**, and then in the **Device Group** list, choose **Transit**.

Step 2: Navigate to **Security > Pre Rules**, and then click **Add**.

Step 3: In the **Name** box, enter **Outbound-Internet**.

Step 4: On the Source tab, in the Source Zone pane, click **Add**, and then select **private**.

Step 5: In the Source Address pane, click **Add**, and then select **app01**.

Step 6: In the Source Address pane, click **Add**, and then select **app02**.

Step 7: On the Destination tab, in the Destination Zone pane, click **Add**, and then select **public**.

Step 8: On the Application tab, in the Applications pane, click **Add**.

Step 9: In the search box, enter **yum**, and then in the results list, choose **yum**.

Step 10: In the Applications pane, click **Add**.

Step 11: In the search box, enter **ntp**, and then in the results list, choose **ntp**.

Step 12: On the Service/URL Category tab, in the Service pane, select **application-default**.

Step 13: On the Actions tab, in the Action Setting section, in the **Action** list, choose **Allow**.

Step 14: In the Log Setting section, in the **Log Forwarding** list, choose **Forward-to-Strata-Logging-Service**.

Step 15: On the Target tab, verify that **Any (target to all devices)** is selected, and then click **OK**.

	NAME	LOCATION	TAGS	TYPE	Source				Destination			APPLICATION	SERVICE	ACTION	PROFILE	OPTIONS
					ZONE	ADDRESS	USER	DEVICE	ZONE	ADDRESS	DEVICE					
1	Permit-Azure-Probes	Azure-Baseline	none	intrazone	any	azure-probe	any	any	(intrazone)	any	any	any	application-default	Allow	none	none
2	Outbound-Internet	Transit	none	universal	private	app01 app02	any	any	public	any	any	ntp yum	application-default	Allow	none	

Step 16: On the **Commit** menu, select **Commit to Panorama**, and then click **Commit**.

3.13 Configure the East-West Security Policy

In this procedure, you configure an east-west security policy rule that allows MySQL traffic from app01 to app02. As required, you can add additional east-west applications to your policy. You use security Pre Rules, which are logically evaluated prior to local rules and cannot be overridden on the device.



Note

Azure networking does not require the use of source NAT on the firewall in order to enforce symmetry if both directions of the flow pass through the same internal load-balancer frontend IP and backend pool.

Step 1: Navigate to **Policies**, and then in the **Device Group** list, choose **Transit**.

Step 2: Navigate to **Security > Pre Rules**, and then click **Add**.

Step 3: In the **Name** box, enter **App01-to-App02**.

Step 4: In the **Rule Type** list, choose **intrazone**.

Step 5: On the Source tab, in the Source Zone pane, click **Add**, and then select **private**.

Step 6: In the Source Address pane, click **Add**, and then select **app01**.

Step 7: On the Destination tab, in the Destination Address pane, click **Add**, and then select **app02**.

Step 8: On the Application tab, in the Applications pane, click **Add**.

Step 9: In the search box, enter **mysql**, and then in the results list, choose **mysql**.

Step 10: On the Service/URL Category tab, in the Service pane, select **application-default**.

Step 11: On the Actions tab, in the Action Setting section, in the **Action** list, choose **Allow**.

Step 12: In the Log Setting section, in the **Log Forwarding** list, choose **Forward-to-Strata-Logging-Service**.

Step 13: On the Target tab, verify that **Any (target to all devices)** is selected, and then click **OK**.

	NAME	LOCATION	TAGS	TYPE	Source				Destination			APPLICATION	SERVICE	ACTION	PROFILE	OPTIONS
					ZONE	ADDRESS	USER	DEVICE	ZONE	ADDRESS	DEVICE					
1	Permit-Azure-Probes	Azure-Baseline	none	intrazone	any	azure-probe	any	any	(intrazone)	any	any	any	application-default	Allow	none	none
2	Outbound-Internet	Transit	none	universal	private	app01	any	any	public	any	any	ntp yum	application-default	Allow	none	
3	App01-to-App02	Transit	none	intrazone	private	app01	any	any	(intrazone)	app02	any	mysql	application-default	Allow	none	

Next, you override the default intrazone rule and configure it to deny traffic.

Step 14: Navigate to **Policies > Security > Default Rules**.

Step 15: Select the row **intrazone-default**, and then at the bottom of the screen, click **Override**.

Step 16: On the Actions tab in the Action Setting section, in the **Action** list, choose **Deny**.

Step 17: In the Log Setting section, check **Log at Session End**.

Step 18: In the Log Setting section, in the **Log Forwarding** list, choose **Forward-to-Strata-Logging-Service**, and then click **OK**.

Step 19: On the **Commit** menu, select **Commit to Panorama**, and then click **Commit**.

3.14 Generate VM Auth Key

In this procedure, you generate a VM auth key on Panorama, to be used during VM-Series bootstrapping.

Step 1: Using SSH from a trusted source, log in to the Panorama command line.

Step 2: Request the VM auth key.

```
request bootstrap vm-auth-key generate lifetime 8760
```

```
VM auth key 123456789012345 generated. Expires at: 2023/06/09 14:15:56
```



Note

The lifetime of the key can vary between 1 hour and 8760 hours (1 year). After the specified time, the key expires. Panorama does not register VM-Series firewalls without a valid auth key in the connection request.

Step 3: Save the auth key for use later in this guide.

Procedures

Deploying the VM-Series Virtual Machines

- 4.1 Create a VM-Series Deployment Profile
- 4.2 Create the Storage Account
- 4.3 Create the Availability Set
- 4.4 Create the Management Public IPs
- 4.5 Deploy Using a Custom ARM Template

4.1 Create a VM-Series Deployment Profile

To create a deployment profile, you must have a Customer Support Portal account and you must have activated your credit pool.

Step 1: Log in to the Customer Support Portal at <https://support.paloaltonetworks.com/Support/Index>.

Step 2: In **Assets > Software NGFW Credits**, in the Prisma NGFW Credits pane, click **Create Deployment Profile**. The Create Deployment Profile dialog box opens.

Step 3: For **Select firewall type**, choose **VM-Series**.

Step 4: Select **PAN-OS 10.0.4 and above**, and then click **Next**.

Step 5: In the **Profile Name** box, enter **Azure Transit Outbound VM-Series**.

Step 6: In the **Number of Firewalls** box, enter **2**.

Step 7: In the **Planned vCPU/Firewall** box, enter **4**.

Step 8: In the **Security Use Case** list, choose **Data Center/Cloud Outbound Security**.

Step 9: Next to Use Credits to Enable VM Panorama, clear the **For Management** and **As Dedicated Log Collector** check boxes, and then click **Create Deployment Profile**.

Create Deployment Profile

VM-Series

Profile Name: Azure Transit Outbound VM-Series

* Number of Firewalls: 2

* Planned vCPU per Firewall: 4

* Security Use Case: Data Center/Cloud Outbound Security

Customize Subscriptions:

- ☒ Threat Prevention
- ☐ Advanced URL Filtering
- ☒ DNS
- ☐ Global Protect
- ☐ DLP
- ☐ Wildfire
- ☐ SD-WAN
- ☐ Intelligent Traffic Offload
- ☒ URL Filtering
- ☐ Advanced Threat Prevention
- ☐ Web Proxy (Promotional Offer)
- ☐ Advanced Wildfire

Use Credits to Enable VM Panorama:

- ☐ For Management
- ☐ As Dedicated Log Collector

The deployment profile takes a couple of minutes to deploy.

Step 10: Scroll to the far right of the Current Deployment Profiles list and capture the auth code for the deployment profile.

4.2 Create the Storage Account

The VM-Series firewalls require general-purpose storage for their virtual file systems and bootstrapping.

Step 1: Log in to the Azure Resource Manager at <https://portal.azure.com>.

Step 2: Navigate to **Home > Storage accounts**, and then click **Create**.

Step 3: In the **Resource group** list, choose **transit**.

Step 4: In the **Storage account name** box, enter **transitv2vmseries**.

Step 5: In the **Region** list, choose **(US) West US** or the appropriate value.

Step 6: In the **Performance** list, choose **Standard**.

Step 7: In the **Redundancy** list, choose **Locally-redundant storage (LRS)**.

Instance details

If you need to create a legacy storage account type, please click [here](#).

Storage account name: transitv2vmseries

Region: (US) West US

Performance: ☒ Standard: Recommended for most scenarios (general-purpose v2 account)
☐ Premium: Recommended for scenarios that require low latency.

Redundancy: Locally-redundant storage (LRS)

Step 8: Click **Review**, and then click **Create**.

4.3 Create the Availability Set

To provide resilience in the firewall deployment, you use an availability set with two fault domains. This distributes the VM-Series virtual machines across different fault domains.



Note

You can configure an availability set on a virtual machine only during its initial deployment. After the virtual machine is deployed, you can't modify a virtual machine's availability set configuration.

Step 1: Navigate to **Home > Availability sets**, and then click **Create**.

Step 2: In the **Resource group** list, choose **transit**.

Step 3: In the **Name** box, enter **transit-vmseries**.

Step 4: In the **Region** list, choose **(US) West US** or the appropriate value.

Step 5: For **Use managed disks**, select **No (classic)**. This is required for the Azure Resource Manager (ARM) template.

The screenshot shows the 'Instance details' configuration page. The 'Name' field is set to 'transit-vmseries'. The 'Region' is set to '(US) West US'. The 'Fault domains' slider is set to 2. The 'Update domains' slider is set to 5. The 'Use managed disks' section has two radio buttons: 'No (Classic)' which is selected, and 'Yes (Aligned)'.

Step 6: Click **Review + create**, and then click **Create**.

4.4 Create the Management Public IPs

You manage the VM-Series firewalls deployed on Azure by using public IP addresses or an enterprise connection. The process to configure enterprise network connectivity is included later in this guide.

In this procedure, you create a public IP address that is associated with the management interface of the VM-Series device at deployment time. Take note of the FQDN that is defined by adding the location-specific suffix to your DNS name label.

Step 1: Navigate to **Home > Public IP addresses**, and then click **Create**.

Step 2: In the SKU section, select **Standard**.

Step 3: In the **Name** box, enter **vmseries-1-mgmt**.

Step 4: In the **DNS name label** box, enter **vmseries-1-mgmt**.

Step 5: In the **Resource group** list, choose **transit**.

Step 6: In the **Location** list, choose **(US) West US** or the appropriate value.

Step 7: Click **Review + create**, and then click **Create**.

IP Version * ⓘ
☒ IPv4 ☐ IPv6 ☐ Both

SKU * ⓘ
☒ Standard ☐ Basic

Tier
☒ Regional ☐ Global

IPv4 IP Address Configuration

Name *
 vmseries-1-mgmt ✓

IP address assignment
☐ Dynamic ☒ Static

Routing preference ⓘ
☒ Microsoft network ☐ Internet

Idle timeout (minutes) * ⓘ
 4

DNS name label ⓘ
 vmseries-1-mgmt ✓

.westus.cloudapp.azure.com

Step 8: Repeat this procedure to create a management public IP address for **vmseries-2**.

4.5 Deploy Using a Custom ARM Template

Azure has removed the option to select an existing resource group for marketplace solutions that enable multiple NICs. To deploy a VM-Series firewall into an existing resource group, use the ARM templates in the [GitHub Repository](#) or your own custom ARM template.

The custom Azure Resource Manager template used in this procedure has been developed and validated specifically to support the [Securing Applications in Azure: Design Guide](#).

For details and features, see the [deployment template readme](#).

Table 12 VM-Series deployment parameters for outbound, east-west, and common firewall option

Parameter	Value	Description
Resource group	transit	—
Location	—	Tested in West US
VM name	vmseries-1 vmseries-2	Outbound, east-west, and common firewall option deployment
Storage account name	transit2vmseries	—
Storage account existing RG	transit	—

(Table 12 continues on the following page)

Table 12 (continued)

Parameter	Value	Description
Fw Av Set	transit-vmseries	Use for all outbound, east-west, and common firewall option firewalls
VM size	Standard_D3_v2	For more information, see Minimum System Requirements for the VM-Series on Azure
Public IP type	standard	Standard-SKU IP addressing required for use with Azure Standard load-balancer
Image version	10.2.6	—
Image SKU	byol	—
Virtual network name	transit	—
Virtual network address prefix	10.110.0.0/16	Transit IP address space
Virtual network existing RG name	transit	—
Subnet0Name	management	—
Subnet1Name	public	—
Subnet2Name	private	—
Subnet0Prefix	10.110.255.0/24	—
Subnet1Prefix	10.110.129.0/24	—
Subnet2Prefix	10.110.0.0/24	—
Subnet0Start Address	10.110.255.4 10.110.255.5	vmseries-1 vmseries-2
Subnet1Start Address	10.110.129.4 10.110.129.5	vmseries-1 vmseries-2
Subnet2Start Address	10.110.0.4 10.110.0.5	vmseries-1 vmseries-2
Admin username	refarchadmin	—
Admin password	<password>	—
Public IP address name	vmseries-1-mgmt vmseries-2-mgmt	vmseries-1 vmseries-2

(Table 12 continues on the following page)

Table 12 (continued)

Parameter	Value	Description
Nsg name	None	NSG is applied at the subnet level
Custom data	<pre> type=dhcp-client; panorama- server=10.255.0.4; panorama- server-2=10.255.0.5; tplname=Transit; dgname=Transit; dns-primary=168.63.129.16; authcodes=<authcode>; vm-auth-key=<authkey>; dhcp-accept-server- hostname=yes; dhcp-accept-server- domain=yes; </pre>	Custom data is added for bootstrapping. Values are separated by semicolons. Replace <authcode> with the value you created in Procedure 4.1 Replace <authkey> with the value you created in Procedure 3.14

Step 1: Log in to the Azure Resource Manager at <https://portal.azure.com>. If you are already signed in to the Azure Resource Manager, the deployment from GitHub uses the same session authorization.

Step 2: In separate tab or web browser, open the [deployment template readme](#).

Step 3: On the template page, click **Deploy to Azure**. This opens the deployment template in Azure.

Step 4: In the **Resource group** list, choose [transit](#).

Step 5: In the **VM Name** box, enter [vmseries-1](#).

Step 6: In the **Storage Account Name** box, enter [transitv2vmseries](#).

Step 7: In the **Storage Account Existing RG** box, enter [transit](#).

Step 8: In the **Fw Av Set** box, enter [transit-vmseries](#).

Step 9: In the **Vm Size** list, choose **Standard_D3_v2**.

Step 10: In the **Public IP Type** list, choose **standard**.

Step 11: In the **Image Version** list, choose **10.2.6**.

Step 12: In the **Image Sku** list, choose **byol**.

Step 13: In the **Bootstrap Firewall** list, choose **no**.

Step 14: In the **Virtual Network Name** box, enter [transit](#).

Step 15: In the **Virtual Network Address Prefix** box, enter [10.110.0.0/16](#).

Step 16: In the **Virtual Network Existing RG Name** box, enter **transit**.

Step 17: In the **Subnet0Name** box, enter **management**.

Step 18: In the **Subnet1Name** box, enter **public**.

Step 19: In the **Subnet2Name** box, enter **private**.

Step 20: In the **Subnet0Prefix** box, enter **10.110.255.0/24**.

Step 21: In the **Subnet1Prefix** box, enter **10.110.129.0/24**.

Step 22: In the **Subnet2Prefix** box, enter **10.110.0.0/24**.

Step 23: In the **Subnet0Start Address** box, enter **10.110.255.4**.

Step 24: In the **Subnet1Start Address** box, enter **10.110.129.4**.

Step 25: In the **Subnet2Start Address** box, enter **10.110.0.4**.

Step 26: In the **Admin Username** box, enter **refarchadmin**.

Step 27: In the **Admin Password** box, enter the password.

Step 28: In the **Public IP Address Name** box, enter **vmseries-1-mgmt**.

Step 29: In the **Nsg Name** box, enter **None**.

Step 30: In the **Custom Data** box, enter the following text:

```
type=dhcp-client;
panorama-server=10.255.0.4;
panorama-server-2=10.255.0.5;
tplname=Transit;
dgname=Transit;
dns-primary=168.63.129.16;
authcodes=<authcode>;
vm-auth-key=<authkey>;
dhcp-accept-server-hostname=yes;
dhcp-accept-server-domain=yes;
```

Step 31: Click **Review + create**, and then click **Create**. The deployment can take several minutes to complete.

Step 32: Repeat this procedure for **vmseries-2**, using the second set of parameters in Table 12.

Procedures

Configuring the VM-Series After Deployment

- 5.1 Set the Azure Device Hostname from Panorama
- 5.2 Retrieve Device Certificate
- 5.3 Deploy Updates from Panorama
- 5.4 Validate Deployment and Refresh Licenses

After the deployment is complete, you must perform the final device configuration tasks.

5.1 Set the Azure Device Hostname from Panorama

In this procedure, you use Panorama to remotely access the Azure device context so you can make local device configuration changes. You set the hostname and domain name, assign a time zone, and change the default admin password for the device.

Step 1: Log in to the primary Panorama virtual appliance's web interface.

Step 2: From any Panorama screen, in the top-left corner, in the Panorama list, click the down arrow.

Step 3: In the device list, choose the Azure device (example: **PA-VM**). To confirm that you are starting with the primary NGFW, you can hover over the device name and view the serial number value.

After you select the device, Panorama loads the device context. This process might take a few minutes.

Step 4: In **Device > Setup > Management > General Settings**, click the edit cog.

Step 5: In the **Hostname** box, enter **VMSeries-1**.

Step 6: In the **Domain** box, enter **example.com**.

Step 7: In the **Time Zone** list, choose the appropriate time zone (example: **US/Pacific**), and then click **OK**.

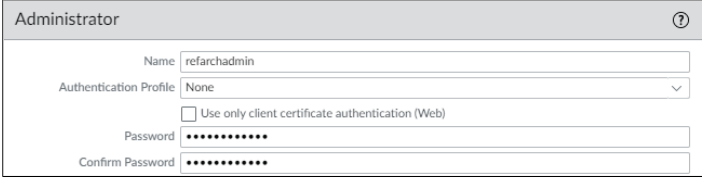
General Settings	
Hostname	VMSeries-1
Domain	example.com
	☐ Accept DHCP server provided Hostname ☐ Accept DHCP server provided Domain
Login Banner	
	☐ Force Admins to Acknowledge Login Banner
SSL/TLS Service Profile	None
Time Zone	US/Pacific

Step 8: In **Device > Administrators**, click **Add**. The Administrator dialog box appears.

Step 9: In the **Name** box, enter **refarchadmin**.

Step 10: In the **Password** box, enter your password.

Step 11: In the **Confirm Password** box, enter your password, and then click **OK**.



The image shows a dialog box titled "Administrator" with a help icon in the top right corner. It contains the following fields and options:

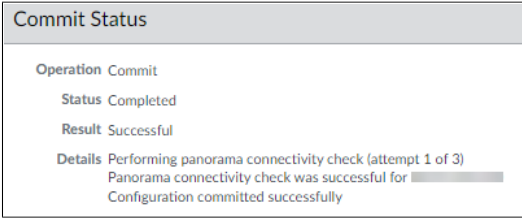
- Name:** A text box containing "refarchadmin".
- Authentication Profile:** A dropdown menu showing "None".
- ☐ Use only client certificate authentication (Web)
- Password:** A text box with masked characters (dots).
- Confirm Password:** A text box with masked characters (dots).

Next, you commit all configuration changes.

Step 12: Click **Commit**. The Commit dialog box appears.

Step 13: Select **Commit All Changes**, and then click **Commit**. The Commit Status dialog box appears.

Step 14: Verify that the Panorama connectivity check succeeded.



The image shows a dialog box titled "Commit Status" with the following information:

- Operation:** Commit
- Status:** Completed
- Result:** Successful
- Details:**
 - Performing panorama connectivity check (attempt 1 of 3)
 - Panorama connectivity check was successful for [Progress Bar]
 - Configuration committed successfully

Step 15: After the commit status is Completed, click **Close**.

5.2 Retrieve Device Certificate

For all firewall features to operate properly, they need to have a valid device certificate.

In this procedure, you generate a one-time password on the customer support portal and then enter the one-time password on your device in order to retrieve the device certificate.

Step 1: Continuing in the device, navigate to **Dashboard > General Information**, record the value for **Serial #**.

Step 2: Log in to the customer support portal at <https://support.paloaltonetworks.com>.

Step 3: In **Assets > Device Certificates**, in the NGFW pane, click **Generate OTP**.

Step 4: On the One Time Password screen, select **Generate OTP for Next-Gen Firewall**, and then click **Generate OTP**.

Step 5: In the **PAN-OS Device** list, choose the serial number for your device, and then click **Generate OTP**.

The screenshot shows a web interface titled "One Time Password". At the top, there is a progress bar with three steps: "Device Type", "Device Number", and "One Time Password", with the third step being active. Below the progress bar, the heading is "Generate OTP for Next-Gen Firewalls". A message states: "Your one time password has been created and is available below. The password will be valid for 60 minutes." There is a dropdown menu for "PAN OS Device:" and a text box for "Password:". Below the password box, it says "Expires On: 10:03:11 AM". At the bottom, there are four buttons: "Generate OTP", "Download OTP", "Copy to Clipboard", and "Done".

Step 6: To copy the one-time password, click **Copy to Clipboard**.

Step 7: On your device, in **Device > Setup > Management**, in the Device Certificate pane, click **Get certificate**. The Device Certificate dialog box appears.

Step 8: In the **One-time Password** box, enter the one-time password you retrieved from the customer support portal, and then click **OK**. The Getting Device Certificate dialog box replaces the Device Certificate dialog box.

Step 9: In the Getting Device Certificate dialog box, click **Close**.

When the task completes, the **Device > Setup > Management** screen updates and displays the valid device certificate information.

The screenshot shows a "Device Certificate" dialog box. It contains the following information: "Current Device Certificate Status: Valid", "Not Valid Before: 2023/04/04 11:49:26 PDT", "Not Valid After: 2023/07/03 11:49:26 PDT", "Last Fetched Message: Successfully fetched Device Certificate", "Last Fetched Status: success", and "Last Fetched Timestamp: 2023/04/04 11:59:26 PDT". At the bottom right, there is a blue link that says "Get certificate".

Next, you exit the device context and return to Panorama.

Step 10: From any device screen, in the top-left corner, in the **PA-VM** list, click the down arrow.

Step 11: In the device list, choose **Panorama**.

Step 12: After you select Panorama, Panorama returns to its standard context. This process might take a few minutes.

Step 13: Repeat Procedure 5.1 and Procedure 5.2 for **VMSeries-2**.

5.3 Deploy Updates from Panorama

In this procedure, you apply the latest application and threat updates and then upgrade the PAN-OS software.

Step 1: In **Panorama > Device Deployment > Dynamic Updates**, click **Check Now**.

Step 2: In the Application and Threat section, in the row for the latest panupv2-all-contents release (example: panupv2-all-contents-8688-7937), in the Action column, click **Download**.

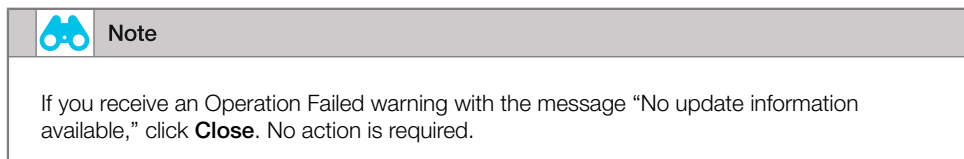
Step 3: After the download is completed, click **Close**.

Step 4: In the Application and Threat section, in the row for the latest panupv2-all-contents release (example: panupv2-all-contents-8688-7937), in the Action column, click **Install**. The Deploy App/Threat Content screen appears.

Step 5: In the Devices pane, select your device (example: **VMSeries-1**), and then click **OK**. The Install Content dialog box appears. The operation might take several minutes.

Step 6: After the content file is uploaded, click **Close**.

Step 7: Navigate to **Panorama > Device Deployment > Software**.



Step 8: In **Panorama > Device Deployment > Software**, click **Check Now**.

Step 9: To limit the choices to those valid for your device, in the search box, enter **VM**, and then click **Apply Filter**.

Step 10: If you have not already downloaded PAN-OS **10.2.0** for this platform, for version **10.2.0**, in the Actions column, click **Download**, and then when the download is complete, click **Close**.

Step 11: If you have not already downloaded PAN-OS **10.2.10-h9** for this platform, for version **10.2.10-h9**, in the Actions column, click **Download**, and then when the download is complete, click **Close**.

Step 12: In the row for **10.2.0**, after the status in the Available column has changed to Downloaded, in the **Action** column, click **Install**.

Step 13: In the Devices pane, select your device (example: **VMSeries-1**).

Step 14: Select **Upload only to device (do not install)**, and then click **OK**. This operation might take several minutes to complete.

Step 15: In the row for **10.2.10-h9**, after the status in the Available column has changed to Downloaded, in the **Action** column, click **Install**.

Step 16: In the Devices pane, select your device (example: **VMSeries-1**).

Step 17: Select **Reboot device after install**, and then click **OK**. This operation might take several minutes to complete.

Step 18: In the Install Software dialog box, click **Close**.

Step 19: Repeat this procedure for **VMSeries-2**.

5.4 Validate Deployment and Refresh Licenses

After the VM-Series content release and software is updated on each device, the firewalls reboot and register with Panorama. This process can take several minutes to complete.

Step 1: On the primary Panorama virtual appliance, navigate to **Panorama > Managed Devices > Summary**.

	DEVICE NAME	MODEL	SERIAL NUMBER	IP Address	TEMPLATE	Status				
				IPV4		DEVICE STATE	DEVICE CERTIFICATE	DEVICE CERTIFICATE EXPIRY DATE	SHARED POLICY	TEMPLATE
✓ ☐ Transit (2/2 Devices Connected): Shared > Azure-Baseline > Transit										
☐	VMSeries-1	PA-VM		10.110.255.4 (DHCP)	Transit	Connected	Valid	2023/05/09 14:39:20 PDT	In Sync Panorama pushed version: 39	In sync Panorama pushed version: 43
☐	VMSeries-2	PA-VM		10.110.255.5 (DHCP)	Transit	Connected	Valid	2023/05/09 14:45:05 PDT	In Sync Panorama pushed version: 43	In sync Panorama pushed version: 43

Next, you refresh the VM-Series firewall's license in order to enable Strata Logging Service.

Step 2: Navigate to **Panorama > Device Deployment > Licenses**, and then click **Refresh**. The Refresh License Deployment dialog box appears.

Step 3: In the **Device Name** column, select the VM-Series firewalls, and then click **Refresh**.

Procedures

Configuring Azure After Deployment

- 6.1 Create the Public IPs
- 6.2 Create the Internal Load-Balancer
- 6.3 Configure the Internal Load-Balancer
- 6.4 Configure User-Defined Routes for the Transit VNet
- 6.5 Configure User-Defined Routes for the App VNets

After the NGFWs are deployed and updated, you complete the next set of procedures in Azure.

6.1 Create the Public IPs

For the public interface of each firewall used for outbound access, you create a public IP address. Azure network uses this IP address for traffic leaving the VNet. As the outbound traffic leaves the transit VNet, Azure translates the firewall public-subnet source IP address.

First, you create the public IP address.

Step 1: Log in to the Azure Resource Manager at <https://portal.azure.com>.

Step 2: Navigate to **Home > Public IP addresses**, and then click **Create**.

Step 3: In the SKU section, select **Standard**.

Step 4: In the **Name** box, enter **vmseries-1-public**.

Step 5: In the **DNS name label** box, enter **vmseries-1-public**.

Step 6: In the **Resource group** list, choose **transit**.

Step 7: In the **Region** list, choose **(US) West US** or the appropriate value.

Step 8: Click **Review + create**, and then click **Create**.

Next, you associate the public IP address with the public subnet interface of the VM-Series firewall.

Step 9: Navigate to **Home > Public IP address > vmseries-1-public**, and then click **Associate**.

Step 10: In the Associate Public IP address pane, in the **Resource type** list, choose **Network interface**.

In the Choose Network Interface pane, select the public interface of vmseries-1 (example: **vmseries-1-eth1**), and then click **OK**.

Step 11: Repeat this procedure for the **vmseries-2** public IP address.

6.2 Create the Internal Load-Balancer

To support outbound and east-west traffic flows, an internal load-balancer is required. When you create the Azure internal load-balancer, you assign it a single private frontend IP address. You use the frontend IP address as the routing next-hop for user-defined routes (UDRs) in the private and application VNets.



Note

The internal load balancer's front end IP address is **10.110.0.21**, which must be in the same subnet as the private IP addresses for the NGFWs behind it.

Step 1: Navigate to **Home > Load Balancers**, and then click **Create**.

Step 2: In the **Resource group** list, choose **transit**.

Step 3: In the **Name** box, enter **transit-internal**.

Step 4: In the **Region** list, choose **(US) West US** or the appropriate value.

Step 5: In the SKU section, select **Standard**.

Step 6: In the Type section, select **Internal**, and then click **Next : Frontend IP configuration**.

Instance details

Name *

Region *

SKU * ☒ Standard
☐ Gateway
☐ Basic

Type * ☐ Public
☒ Internal

Tier * ☒ Regional
☐ Global

Microsoft recommends Standard SKU load balancer for production workloads. [Learn more about pricing differences between Standard and Basic SKU](#)

Step 7: Click **Add a frontend IP configuration**. The Add Frontend IP Configuration pane opens.

Step 8: In the **Name** box, enter **transit-internal-frontend**.

Step 9: In the **Virtual network** list, choose **transit**.

Step 10: In the **Subnet** list, choose **private**.

Step 11: In the IP Address Assignment section, select **Static**.

Step 12: In the **IP address** box, enter **10.110.0.21**, and then click **Add**.

Name *

Virtual network *

Subnet *

Assignment
☐ Dynamic ☒ Static

IP address *

As necessary, after you create the load balancer, you can add additional frontend IP addresses.

Step 13: Click **Review + create**, and then click **Create**.

6.3 Configure the Internal Load-Balancer

In this procedure, you configure the internal load-balancer health probes, the backend firewall pool, and a load-balancing rule to direct traffic to the VM-Series firewalls.

First, you add an HTTPS probe.

Step 1: Navigate to **Home > Load Balancers > transit-internal**, click **Health probes**, and then click **Add**.

Step 2: In the **Name** box, enter **HTTPS-probe**.

Step 3: In the **Protocol** list, select **HTTPS**.

Step 4: In the **Path** box, enter **/php/login.php**, and then click **Add**.

Next, you create the backend pool that contains the VM-Series firewalls.

Step 5: Navigate to **Home > Load Balancers > transit-internal**, click **Backend pools**, and then click **Add**.

Step 6: In the **Name** box, enter **vmseries**.

Step 7: In the IP configurations section, click **Add**. The Add IP configurations to backend pool pane opens.

Step 8: Select the rows for the virtual machine and IP configuration that are associated with the two VM-Series firewalls and **private** subnet (example: **ipconfig-trust**), and then click **Add**.

IP configurations
IP configurations associated to virtual machines and virtual machine scale sets must be in same location as the load balancer and be in the same virtual network.

+ Add | X Remove

☐	Resource Name	Resource group	Type	IP configuration	IP Address	Availability set
☐	vmseries-1	TRANSIT	Virtual machine	ipconfig-trust	10.110.0.4	TRANSIT-VMSEI
☐	vmseries-2	TRANSIT	Virtual machine	ipconfig-trust	10.110.0.5	TRANSIT-VMSEI

Step 9: Click **Save**.

Next, you add a load-balancing rule to map the components together and complete the internal load-balancer configuration.

Step 10: Navigate to **Home > Load Balancers > transit-internal**, click **Load balancing rules**, and then click **Add**.

Step 11: In the **Name** box, enter **private-all-ports**.

Step 12: In the **Frontend IP address** list, choose **transit-internal-frontend**.

Step 13: In the **Backend pool** list, choose **vmseries**.

Step 14: Select **HA ports**.

Step 15: In the **Health probe** list, choose **HTTPS-probe**, and then click **Add**.

Name *	private-all-ports
IP Version *	☒ IPv4 ☐ IPv6
Frontend IP address * ⓘ	transit-internal-frontend (10.110.0.21) ▼
Backend pool * ⓘ	vmseries ▼
HA Ports ⓘ	☒
Health probe * ⓘ	HTTPS-probe (HTTPS:443/php/login.php) ▼
Session persistence ⓘ	None ▼
Idle timeout (minutes) * ⓘ	4
Enable TCP Reset	☐
Enable Floating IP ⓘ	☐

6.4 Configure User-Defined Routes for the Transit VNet

In this procedure, you create and apply a route table for the private subnet in the transit VNet. Then you create a default UDR that redirects traffic to the frontend IP of the internal load-balancer.

Azure networking automatically creates system routes for the address space defined in the VNet. To insert virtual devices such as load balancers and firewalls into the traffic forwarding path, you override the Azure system routes with UDRs. After adding UDRs, Azure evaluates the effective route table using traditional routing-rules based on longest prefix match of the destination address.

If you add a UDR with the same prefix and prefix-length as a system route, the UDR becomes the active route, and the state of the original system route changes to an Invalid state.

If you add a UDR with a more specific prefix that falls within the address space of a system route, the UDR becomes an active route, and the original system route also remains in an Active state.



Caution

The use of UDR summary routes can have unexpected consequences. If you apply a UDR summary to a subnet that falls within the summary but does not have a more specific UDR, the UDR controls traffic within the subnet (host-to-host traffic).

Azure networking does not have a concept of equal cost paths. You cannot add multiple UDRs with same prefix and prefix-length with different next-hops to perform traffic load balancing. The only method by which you can perform load balancing is by using UDRs to forward traffic to an Azure load-balancing resource.

First, you create a route table for the transit VNet private subnet.

Step 1: Navigate to **Home > Route tables**, and then click **Create**.

Step 2: In the **Resource group** list, choose **transit**.

Step 3: In the **Region** list, choose **(US) West US** or the appropriate value.

Step 4: In the **Name** box, enter **private**.

Step 5: Click **Review + create**, and then click **Create**.

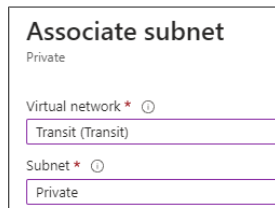
Next, you associate the route table with the private subnet.

Step 6: Navigate to **Home > Route tables**, and then click **private**.

Step 7: Select Subnets, and then click **Associate**. The Associate Subnet page appears.

Step 8: In the **Virtual network** box, select **Transit (Transit)**.

Step 9: In the **Subnet** box, choose **private**, and then click **OK**.



The 'Associate subnet' dialog box shows the 'Virtual network' dropdown set to 'Transit (Transit)' and the 'Subnet' dropdown set to 'Private'.

Next, you add a default route with a next-hop of the transit internal load-balancer. This UDR overrides the automatically created system default route and directs all traffic to the internal load-balancer.

Step 10: Navigate to **Home > Route tables > private**, click **Routes**, and then click **Add**. The Add Route page appears.

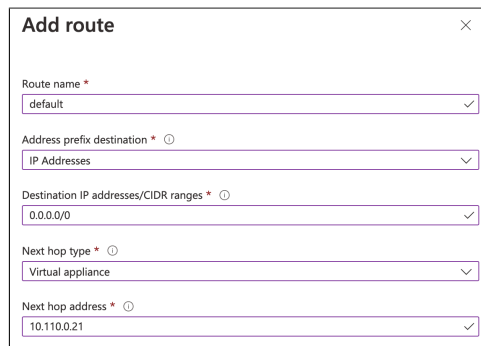
Step 11: In the **Route name** box, enter **default**.

Step 12: In the **Address prefix destination** box, select **IP Addresses**.

Step 13: In the **Destination IP addresses/CIDR ranges** box, enter **0.0.0.0/0**.

Step 14: In the **Next hop type** list, choose **Virtual Appliance**.

Step 15: In the **Next hop address** box, enter **10.110.0.21**, and then click **Add**. This is the IP address of the internal load-balancer in the transit VNet.



The 'Add route' dialog box shows the following configuration: Route name: default; Address prefix destination: IP Addresses; Destination IP addresses/CIDR ranges: 0.0.0.0/0; Next hop type: Virtual appliance; Next hop address: 10.110.0.21.

Finally, you verify that the effective routes for your transit VNet private subnet.

Step 16: Navigate to **Home > Route tables > private**, and then click **Effective Routes**.

Step 17: In the **Network interface** list, choose any interface from a running virtual machine with an interface associated with the **private** subnet (example: **vmseries-1-eth2**).

Step 18: In the **Effective routes** table for your transit VNet private subnet, verify that the address space from peered VNets and the default UDR are **Active**.

Effective routes											
Source	↑↓	State	↑↓	Address Prefixes	↑↓	Next Hop Type	↑↓	Next Hop IP Address	↑↓	User Defined Route Name	↑↓
Default		Active		10.110.0.0/16		Virtual network		-		-	
Default		Active		10.255.0.0/16		VNet peering		-		-	
Default		Active		10.112.0.0/16		VNet peering		-		-	
Default		Active		10.113.0.0/16		VNet peering		-		-	
Default		Invalid		0.0.0.0/0		Internet		-		-	
User		Active		0.0.0.0/0		Virtual appliance		10.110.0.21		default	

6.5 Configure User-Defined Routes for the App VNets

In this procedure, you create and apply a route table for the application subnets. Then you create a default UDR that redirects traffic to the frontend IP of the internal load-balancer in the transit VNet.

First, you create a route table for the app01 subnet.

Step 1: Navigate to **Home > Route tables**, and then click **Create**.

Step 2: In the **Resource group** list, choose **app01**.

Step 3: In the **Region** list, choose **(US) West US** or the appropriate value.

Step 4: In the **Name** box, enter **app01**.

Step 5: Click **Review + create**, and then click **Create**.

Next, you associate the route table with the app01 subnet.

Step 6: Navigate to **Home > Route tables**, and then click **app01**.

Step 7: Select **Subnets**, and then click **Associate**. The Associate Subnet page appears.

Step 8: In the **Virtual network** box, select **app01 (app01)**.

Step 9: In the **Subnet** box, choose **app01**, and then click **OK**.

Associate subnet

app01

Virtual network * ⓘ

app01 (app01)

Subnet * ⓘ

app01

Finally, you add a default route with a next-hop of the transit internal load-balancer. This UDR overrides the automatically created system default route and directs all traffic to the internal load-balancer.

Step 10: Navigate to **Home > Route tables > app01**, click **Routes**, and then click **Add**. The Add Route page appears.

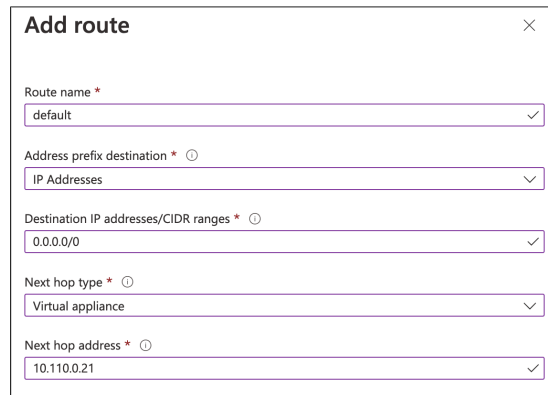
Step 11: In the **Route name** box, enter **default**.

Step 12: In the **Address prefix destination** box, select **IP Addresses**.

Step 13: In the **Destination IP addresses/CIDR ranges** box, enter **0.0.0.0/0**.

Step 14: In the **Next hop type** list, choose **Virtual Appliance**.

Step 15: In the **Next hop address** box, enter **10.110.0.21**, and then click **Add**.



The screenshot shows a dialog box titled "Add route" with a close button (X) in the top right corner. It contains five input fields, each with a checkmark icon on the right:

- Route name ***: A text input field containing "default".
- Address prefix destination * ⓘ**: A dropdown menu with "IP Addresses" selected.
- Destination IP addresses/CIDR ranges * ⓘ**: A text input field containing "0.0.0.0/0".
- Next hop type * ⓘ**: A dropdown menu with "Virtual appliance" selected.
- Next hop address * ⓘ**: A text input field containing "10.110.0.21".

Step 16: Repeat this procedure for the **app02** subnet.

Deploying Inbound Security

There are two options for securing inbound traffic in the Transit VNet model: the dedicated inbound or common firewall options. The dedicated inbound option uses a devoted set of firewalls for inbound traffic. The common firewall option uses a single set of firewalls for all outbound, east-west, inbound, and enterprise traffic. Choose the inbound security option that is appropriate for your organization.



Note

If you are deploying the dedicated inbound option, follow the procedures to configure and deploy the dedicated firewalls.

If you are deploying the common firewall option, you use the instructions in this section to prepare the outbound and east-west VM-Series firewalls for concurrent inbound security.

Figure 9 Dedicated inbound security deployment option

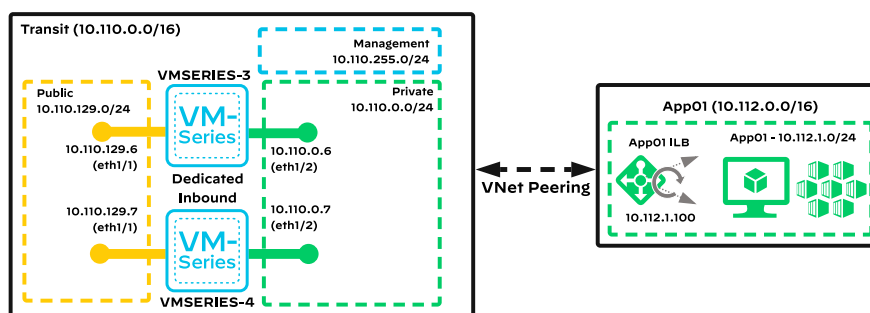
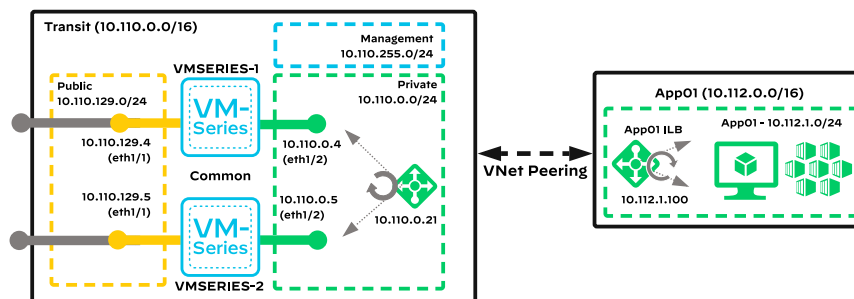


Figure 10 Common firewall security deployment option



There are two additional deployment options for inbound application traffic: a public load-balancer or an application gateway. Choose the inbound application traffic option that is appropriate for your organization. Palo Alto Networks tested this guide with both public load-balancer and application gateway options configured concurrently.

Procedures

Configuring Panorama Templates, Template Stacks, and Device Groups

- 7.1 Create the Dedicated Inbound Template and Template Stack
- 7.2 Create the Dedicated Inbound Device Group
- 7.3 Create the Interface Management Profile
- 7.4 Configure the Network Settings Template
- 7.5 Add Static Routes to the Virtual Router

If you are deploying the dedicated inbound firewall option, complete this section.

If you are deploying the common firewall option, skip to the "Deploying the Public Load-Balancer Option" section.

Panorama provides the following tools for centralized administration:

- **Templates and template stacks**—Panorama manages common device and network configuration through templates. To manage configuration centrally and then push the changes to all managed firewalls, you can use templates. This approach avoids making the same individual firewall change repeatedly across many devices. To make things easier, you can stack templates and use them as building blocks for device and network configuration.
- **Hierarchical device groups**—Panorama manages common policies and objects through hierarchical device groups. To centrally manage the policies across all deployment locations with common requirements, you can use multi-level device groups.

7.1 Create the Dedicated Inbound Template and Template Stack

The templates include configurations for common functions and separate configurations for specific deployments across the VM-Series devices.

In this procedure, you create the **Transit-2-Zone-Inbound** Panorama templates. These templates include firewall networking functions for the dedicated inbound firewalls, including interfaces, zones, and the virtual router.

You apply templates to devices by using Panorama template stacks. This logically merges the assigned templates and associates them with the relevant devices.

Within the relevant procedures, you create the specific configurations for the templates. You create the template stack later in this guide, when associating the first device to the templates.

Step 1: Navigate to **Panorama > Templates**, and then click **Add**.

Step 2: In the **Name** box, enter **Transit-2-Zone-Inbound**.

Step 3: In the **Description** box, enter a valid description, and then click **OK**.

Next, you create the template stack.

Step 4: Navigate to **Panorama > Templates**, and then click **Add Stack**.

Step 5: In the **Name** box, enter **Transit-Inbound**.

Step 6: In the **Description** box, enter an appropriate description.

Step 7: In the Templates pane, click **Add**, select **VMSeries-Baseline** and **Transit-2-Zone-Inbound**, and then click **OK**.

☐	NAME ^	DESCRIPTION	TYPE	STACK
☐	Transit		template-stack	VMSeries-Baseline Transit-2-Zone
☐	Transit-2-Zone		template	
☐	Transit-2-Zone-Inbound		template	
☐	Transit-Inbound		template-stack	VMSeries-Baseline Transit-2-Zone-Inbound
☐	VMSeries-Baseline		template	

7.2 Create the Dedicated Inbound Device Group

First, you create the device group and specify the parent device group to properly inherit device group settings. Later, you create the objects and policies in the procedures that require them.

When a device group includes references to zones within a policy, you must include the template that you originally used to define the zone. This template serves as a reference template for the device group.

Table 13 Panorama inbound device group

Device group	Reference templates	Parent device group
Transit-Inbound	Transit-2-Zone-Inbound	Azure-Baseline

In this procedure, you create the **Transit-Inbound** Panorama device group. This is the device group for objects and policies for the dedicated inbound firewall option.

Step 1: Log in to the primary Panorama virtual appliance's web interface.

Step 2: Navigate to **Panorama > Device Groups**, and then click **Add**.

Step 3: In the **Name** box, enter **Transit-Inbound**.

Step 4: In the **Description** box, enter a valid description.

Step 5: In the **Parent Device Group** list, choose **Azure-Baseline**.

Step 6: In the Reference Templates pane, click **Add**, choose **Transit-2-Zone-Inbound**, and then click **OK**.

☐	NAME ^	DESCRIPTION	REFERENCE TEMPLATES
☐	Shared		
☐	Azure-Baseline	Parent DG with Baseline configurations for all Azure devices	
☐	Transit	Child DG with objects and policies for OBEW VM-Series devices in the Transit VNet	Transit-2-Zone
☐	Transit-Inbound	Child DG with objects and policies for Inbound VM-Series devices in the Transit VNet	Transit-2-Zone-Inbound

7.3 Create the Interface Management Profile

The load-balancer health-checks use HTTPS probes towards the firewall's dataplane interfaces. By default, the firewall blocks responses to these probes. To override the default block operation, you use interface management profiles. In this procedure, you create a management profile for the public-facing VM-Series interfaces, allowing HTTPS from the Azure health check source address.

Step 1: Navigate to **Network**, and then in the **Template** list, choose **Transit-2-Zone-Inbound**.

Step 2: Navigate to **Network Profiles > Interface Mgmt**, and then click **Add**. The Interface Management Profile dialog box appears.

Step 3: In the **Name** box, enter **MP-public**.

Step 4: In the Administrative Management Services section, select **HTTPS**.

Step 5: In the Permitted IP Addresses pane, click **Add**.

Step 6: Enter **168.63.129.16/32**, and then click **OK**.

7.4 Configure the Network Settings Template



Note

Although the VM-Series device is not a modular hardware platform, assign interfaces to Slot 1 when using Panorama templates for the VM-Series devices.

Table 14 Template interface settings for Transit-2-Zone-Inbound

Slot	Interface	Interface type	Virtual router	Security zone	IPv4	Management profile
Slot 1	ethernet1/1	Layer3	vr-default	public	DHCP Client	MP-public
Slot 1	ethernet1/2	Layer3	vr-default	private	DHCP Client	—

Step 1: Navigate to **Network**, and then in the **Template** list, choose **Transit-2-Zone-Inbound**.

Step 2: Navigate to **Interfaces > Ethernet**, and then click **Add Interface**. The Ethernet Interface dialog box appears.

Step 3: In the **Slot** list, choose **Slot 1**.

Step 4: In the **Interface Name** list, choose **ethernet1/1**.

Step 5: In the **Interface Type** list, choose **Layer3**.

Step 6: On the Config tab, in the **Virtual Router** list, choose **New Virtual Router**. The Virtual Router dialog box appears.

Step 7: In the **Name** box, enter **vr-default**, and then click **OK**.

Step 8: In the **Security Zone** list, choose **New Zone**. The Zone dialog box appears.

Step 9: In the **Name** box, enter **public**, and then click **OK**.

Step 10: On the IPv4 tab, for **Type**, select **DHCP client**.

Step 11: Clear **Automatically create default route pointing to default gateway provided by server**.

Step 12: On the Advanced tab, in the **Management Profile** list, choose **MP-public**, and then click **OK**.

Step 13: On the warning, click **Yes**.

Step 14: Repeat this procedure for the second interface listed in Table 14.

7.5 Add Static Routes to the Virtual Router

In this procedure, you configure the static routes in the firewall virtual router. Even though Azure does not use default gateways in the traditional way, you configure the static routes in the NGFW to point at a default gateway in order to forward the traffic into the correct VNet subnet.

- Default gateway for public zone: 10.110.129.1
- Default gateway for private zone: 10.110.0.1

Table 15 Static routes for Transit-2-Zone-Inbound firewalls

Name	Destination prefix	Interface	Next-hop value
default	0.0.0.0/0	ethernet1/1	10.110.129.1
azure-probe	168.63.129.16/32	ethernet1/1	10.110.129.1

(Table 15 continues on the following page)

Table 15 (continued)

Name	Destination prefix	Interface	Next-hop value
public	10.110.128.0/23	ethernet1/1	10.110.129.1
private	10.110.0.0/17	ethernet1/2	10.110.0.1
app01	10.112.0.0/16	ethernet1/2	10.110.0.1
app02	10.113.0.0/16	ethernet1/2	10.110.0.1
management	10.255.0.0/16	ethernet1/2	10.110.0.1
on-premises	10.5.0.0/16	ethernet1/2	10.110.0.1

Step 1: Navigate to **Network**, and then in the **Template** list, choose **Transit-2-Zone-Inbound**.

Step 2: Navigate to **Virtual Routers**, and then click **vr-default**. The Virtual Router dialog box appears.

Step 3: On the Static Routes tab, click **Add**. The Virtual Router—Static Route—IPv4 dialog box appears.

Step 4: In the **Name** box, enter **default**.

Step 5: In the **Destination** box, enter **0.0.0.0/0**.

Step 6: In the **Interface** list, choose **ethernet1/1**.

Step 7: In the **Next Hop** list, choose **IP Address**, enter **10.110.129.1**, and then click **OK**.

Step 8: Repeat Step 3 through Step 7 for the remaining routes in Table 15.

	NAME	DESTINATION	INTERFACE	Next Hop		ADMIN DISTANCE	METRIC	BFD	ROUTE TABLE
				TYPE	VALUE				
☐	default	0.0.0.0/0	ethernet1/1	ip-address	10.110.129.1	default	10	None	unicast
☐	azure-probe	168.63.129.16/32	ethernet1/1	ip-address	10.110.129.1	default	10	None	unicast
☐	public	10.110.128.0/23	ethernet1/1	ip-address	10.110.129.1	default	10	None	unicast
☐	private	10.110.0.0/17	ethernet1/2	ip-address	10.110.0.1	default	10	None	unicast
☐	app01	10.112.0.0/16	ethernet1/2	ip-address	10.110.0.1	default	10	None	unicast
☐	app02	10.113.0.0/16	ethernet1/2	ip-address	10.110.0.1	default	10	None	unicast
☐	management	10.255.0.0/16	ethernet1/2	ip-address	10.110.0.1	default	10	None	unicast
☐	on-premises	10.5.0.0/16	ethernet1/2	ip-address	10.110.0.1	default	10	None	unicast

Step 9: After adding all routes for the virtual router, on the Virtual Router dialog box, click **OK**.

Step 10: On the **Commit** menu, click **Commit to Panorama**.

Procedures

Deploying the VM-Series Virtual Machines

- 8.1 Create a VM-Series Deployment Profile
- 8.2 Create the Availability Sets
- 8.3 Create the Management Public IPs
- 8.4 Deploy Using a Custom ARM Template

If you are deploying the dedicated inbound firewall option, complete this section.

If you are deploying the common firewall option, skip to the "Deploying the Public Load-Balancer Option" section.

8.1 Create a VM-Series Deployment Profile

To create a deployment profile, you must have a Customer Support Portal account and you must have activated your credit pool.

Step 1: Log in to the Customer Support Portal at <https://support.paloaltonetworks.com/Support/Index>.

Step 2: In **Assets > Software NGFW Credits**, in the Prisma NGFW Credits pane, click **Create Deployment Profile**. The Create Deployment Profile dialog box opens.

Step 3: For **Select firewall type**, choose **VM-Series**.

Step 4: Select **PAN-OS 10.0.4 and above**, and then click **Next**.

Step 5: In the **Profile Name** box, enter **Azure Transit Inbound VM-Series**.

Step 6: In the **Number of Firewalls** box, enter **2**.

Step 7: In the **Planned vCPU/Firewall** box, enter **4**.

Step 8: In the **Security Use Case** list, choose **Data Center/ Cloud Inbound Security**.

Step 9: Next to Use Credits to Enable VM Panorama, clear **For Management**, and then click **Create Deployment Profile**.

The deployment profile takes a couple of minutes to deploy.

Step 10: Scroll to the far right of the Current Deployment Profiles list and capture the auth code for the deployment profile.

8.2 Create the Availability Sets

The VM-Series resiliency model for Azure benefits from the use of an availability set with two fault domains. This distributes the VM-Series firewalls across different fault domains. The Transit VNet design model uses two firewall sets, one for inbound and another for east-west, outbound, and enterprise traffic. To increase the resiliency of the design, each firewall set uses a dedicated Azure availability set.



Note

You can configure an availability set on a virtual machine during its initial deployment only. You can't modify a virtual machine's availability set configuration after the virtual machine is deployed.

Step 1: Log in to the Azure Resource Manager at <https://portal.azure.com>.

Step 2: Navigate to **Home > Availability sets**, and then click **Create**.

Step 3: In the **Resource group** list, choose **transit**.

Step 4: In the **Name** box, enter **transit-vmseries-inbound**.

Step 5: In the **Region** list, choose **(US) West US** or the appropriate value.

Step 6: For **Use managed disks**, select **No (classic)**. This is required for the ARM template.

Step 7: Click **Review + create**, and then click **Create**.

The screenshot shows the 'Instance details' section of an Azure resource creation form. It contains the following fields and values:

- Name:** transit-vmseries-inbound
- Region:** (US) West US
- Fault domains:** 2
- Update domains:** 5
- Use managed disks:** Yes (Aligned)

8.3 Create the Management Public IPs

You manage the dedicated inbound VM-Series firewalls deployed on Azure by using public IP addresses or an enterprise network connection. The process to configure enterprise network connectivity is included later in this guide.

In this procedure, you create a public IP address that is associated with the management interface of the VM-Series device at deployment time. Take note of the FQDN that is defined by adding the location-specific suffix to your DNS name label. Palo Alto Networks recommends managing your devices by using the DNS name rather than the public IP address, which might change.

Step 1: Navigate to **Home > Public IP addresses**, and then click **Create**.

Step 2: In the SKU section, select **Standard**.

Step 3: In the **Name** box, enter **vmseries-3-mgmt**.

Step 4: In the **DNS name label** box, enter **vmseries-3-mgmt**.

Step 5: In the **Resource group** list, choose **transit**.

Step 6: In the **Region** list, choose **(US) West US** or the appropriate value.

Step 7: Click **Review + create**, and then click **Create**.

Step 8: Repeat this procedure for **vmseries-4**.

8.4 Deploy Using a Custom ARM Template

Azure has removed the option to select an existing resource group for marketplace solutions that enable multiple NICs. To deploy a VM-Series firewall into an existing resource group, use the ARM templates in the [GitHub Repository](#) or your own custom ARM template.

The custom Azure Resource Manager template used in this procedure has been developed and validated specifically to support the [Securing Applications in Azure: Design Guide](#).

For details and features, see the [deployment template readme](#).

Table 16 VM-Series deployment parameters for dedicated inbound firewalls

Parameter	Value	Description
Resource group	transit	—
Location	—	Tested in West US
VM name	vmseries-3 vmseries-4	Dedicated inbound firewalls
Storage account name	transitv2vmseries	—
Storage account existing RG	transit	—
Fw Av Set	transit-vmseries-inbound	Use for all inbound firewalls
VM size	Standard_D3_v2	For more information, see Minimum System Requirements for the VM-Series on Azure.
Public IP type	standard	Standard-SKU IP addressing required for use with Azure Standard load-balancer
Image version	10.2.6	—
Image SKU	byol	—
Virtual network name	transit	—
Virtual network address prefix	10.110.0.0/16	Match the IP address space from Transit-VNet
Virtual network existing RG name	transit	—
Subnet0Name	management	—
Subnet1Name	public	—
Subnet2Name	private	—
Subnet0Prefix	10.110.255.0/24	—
Subnet1Prefix	10.110.129.0/24	—
Subnet2Prefix	10.110.0.0/24	—
Subnet0Start Address	10.110.255.6 10.110.255.7	vmseries-3 vmseries-4
Subnet1Start Address	10.110.129.6 10.110.129.7	vmseries-3 vmseries-4

(Table 16 continues on the following page)

Table 16 (continued)

Parameter	Value	Description
Subnet2Start Address	10.110.0.6 10.110.0.7	vmseries-3 vmseries-4
Admin username	refarchadmin	—
Admin password	<password>	—
Public IP address name	vmseries-3-mgmt vmseries-4-mgmt	vmseries-3 vmseries-4
Nsg name	None	NSG is applied at the subnet level
Custom data	type=dhcp-client; panorama-server=10.255.0.4; panorama-server-2=10.255.0.5; tplname=Transit-Inbound; dgrname=Transit-Inbound; dns-primary=168.63.129.16; authcodes=<authcode>; vm-auth-key=<authkey>; dhcp-accept-server-hostname=yes; dhcp-accept-server-domain=yes;	Custom data is added for bootstrapping. Values are separated by semicolons. Replace <authcode> with the value you created in Procedure 4.1 Replace <authkey> with the value you created in Procedure 3.14

Step 1: Log in to the Azure Resource Manager at <https://portal.azure.com>. If you are already signed in to the Azure Resource Manager the deployment from GitHub uses the same session authorization.

Step 2: In a separate tab or web browser, open the [deployment template](#).

Step 3: On the template page, click **Deploy to Azure**. This opens the deployment template.

Step 4: In the **Resource group** list, choose [transit](#).

Step 5: In the **Vm Name** box, enter [vmseries-3](#).

Step 6: In the **Storage Account Name** box, enter [transitv2vmseries](#).

Step 7: In the **Storage Account Existing RG** box, enter [transit](#).

Step 8: In the **Fw Av Set** box, enter [transit-vmseries-inbound](#).

Step 9: In the **Vm Size** list, choose **Standard_D3_v2**.

Step 10: In the **Public IP Type** list, choose **standard**.

Step 11: In the **Image Version** list, choose **10.2.6**.

- Step 12:** In the **Image Sku** list, choose **byol**.
- Step 13:** In the **Bootstrap Firewall** list, choose **no**.
- Step 14:** In the **Virtual Network Name** box, enter **transit**.
- Step 15:** In the **Virtual Network Address Prefix** box, enter **10.110.0.0/16**.
- Step 16:** In the **Virtual Network Existing RG Name** box, enter **transit**.
- Step 17:** In the **Subnet0Name** box, enter **management**.
- Step 18:** In the **Subnet1Name** box, enter **public**.
- Step 19:** In the **Subnet2Name** box, enter **private**.
- Step 20:** In the **Subnet0Prefix** box, enter **10.110.255.0/24**.
- Step 21:** In the **Subnet1Prefix** box, enter **10.110.129.0/24**.
- Step 22:** In the **Subnet2Prefix** box, enter **10.110.0.0/24**.
- Step 23:** In the **Subnet0Start Address** box, enter **10.110.255.6**.
- Step 24:** In the **Subnet1Start Address** box, enter **10.110.129.6**.
- Step 25:** In the **Subnet2Start Address** box, enter **10.110.0.6**.
- Step 26:** In the **Admin Username** box, enter **refarchadmin**.
- Step 27:** In the **Admin Password** box, enter the password.
- Step 28:** In the **Public IP Address Name** box, enter **vmseries-3-mgmt**.
- Step 29:** In the **Nsg Name** box, enter **None**.
- Step 30:** In the **Custom Data** box, enter the following text:

```
type=dhcp-client;
panorama-server=10.255.0.4;
panorama-server-2=10.255.0.5;
tplname=Transit-Inbound;
dgname=Transit-Inbound;
dns-primary=168.63.129.16;
authcodes=<authcode>;
vm-auth-key=<authkey>;
dhcp-accept-server-hostname=yes;
dhcp-accept-server-domain=yes;
```

- Step 31:** Click **Review + create**, and then click **Create**. The deployment can take several minutes to complete.
- Step 32:** Repeat this procedure for **vmseries-4**, using the second set of parameters in Table 16.

Procedures

Configuring the VM-Series After Deployment

- 9.1 Set the Azure Device Hostname from Panorama
- 9.2 Retrieve Device Certificate
- 9.3 Deploy Updates from Panorama
- 9.4 Validate Deployment and Refresh Licenses

After the deployment is complete, you must perform the final device-configuration tasks.

9.1 Set the Azure Device Hostname from Panorama

In this procedure, you use Panorama to remotely access the Azure device context so that you can make local device configuration changes. You set the hostname and domain name, assign a time zone, and change the default admin password for the device.

Step 1: Log in to the primary Panorama virtual appliance's web interface.

Step 2: From any Panorama screen, in the top-left corner, in the Panorama list, click the down arrow.

Step 3: In the device list, choose the Azure device (example: **PA-VM**). To confirm that you are starting with the primary NGFW, you can hover over the device name to view the serial number value.

After you select the device, Panorama loads the device context. This process might take a few minutes.

Step 4: In **Device > Setup > Management > General Settings**, click the edit cog.

Step 5: In the **Hostname** box, enter **VMSeries-3**.

Step 6: In the **Domain** box, enter **example.com**.

Step 7: In the **Time Zone** list, choose the appropriate time zone (example: **US/Pacific**), and then click **OK**.

General Settings	
Hostname	VMSeries-3
Domain	example.com
	☐ Accept DHCP server provided Hostname
	☐ Accept DHCP server provided Domain
Login Banner	
	☐ Force Admins to Acknowledge Login Banner
SSL/TLS Service Profile	None
Time Zone	US/Pacific

Step 8: In **Device > Administrators**, click **Add**. The Administrator dialog box appears.

Step 9: In the **Name** box, enter **refarchadmin**.

Step 10: In the **Password** box, enter your password.

Step 11: In the **Confirm Password** box, enter your password, and then click **OK**.



The image shows a dialog box titled "Administrator" with a help icon in the top right corner. It contains the following fields and options:

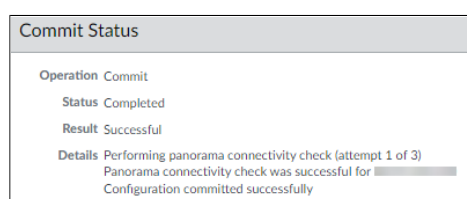
- Name:** A text box containing "refarchadmin".
- Authentication Profile:** A dropdown menu showing "None".
- ☐ Use only client certificate authentication (Web)
- Password:** A text box with masked characters (dots).
- Confirm Password:** A text box with masked characters (dots).

Next, you commit all configuration changes.

Step 12: Click **Commit**. The Commit dialog box appears.

Step 13: Select **Commit All Changes**, and then click **Commit**. The Commit Status dialog box appears.

Step 14: Verify that the Panorama connectivity check succeeded.



The image shows a dialog box titled "Commit Status" with the following information:

- Operation:** Commit
- Status:** Completed
- Result:** Successful
- Details:**
 - Performing panorama connectivity check (attempt 1 of 3)
 - Panorama connectivity check was successful for 
 - Configuration committed successfully

Step 15: After the commit status is Completed, click **Close**.

9.2 Retrieve Device Certificate

For all firewall features to operate properly, they need to have a valid device certificate.

In this procedure, you generate a one-time password on the customer support portal and then enter the one-time password on your device to retrieve the device certificate.

Step 1: Continuing in the device, navigate to **Dashboard > General Information**, record the value for **Serial #**.

Step 2: Log in to the customer support portal at <https://support.paloaltonetworks.com>.

Step 3: In **Assets > Device Certificates**, in the NGFW pane, click **Generate OTP**.

Step 4: On the One Time Password screen, select **Generate OTP for Next-Gen Firewall**, and then click **Generate OTP**.

Step 5: In the **PAN-OS Device** list, choose the serial number for your device, and then click **Generate OTP**.

The screenshot shows a web interface titled "One Time Password". At the top, there is a progress bar with three steps: "Device Type", "Device Number", and "One Time Password", with the third step being active. Below the progress bar, the heading is "Generate OTP for Next-Gen Firewalls". A message states: "Your one time password has been created and is available below. The password will be valid for 60 minutes." There is a dropdown menu for "PAN OS Device:" and a text box for "Password:". Below the password box, it says "Expires On: 10:03:11 AM". At the bottom, there are four buttons: "Generate OTP", "Download OTP", "Copy to Clipboard", and "Done".

Step 6: To copy the one-time password, click **Copy to Clipboard**.

Step 7: On your device, in **Device > Setup > Management**, in the Device Certificate pane, click **Get certificate**. The Device Certificate dialog box appears.

Step 8: In the **One-time Password** box, enter the one-time password you retrieved from the customer support portal, and then click **OK**. The Getting Device Certificate dialog box replaces the Device Certificate dialog box.

Step 9: In the Getting Device Certificate dialog box, click **Close**.

When the task completes, the **Device > Setup > Management** screen updates and displays the valid device certificate information.

The screenshot shows a "Device Certificate" dialog box. It contains the following information:

Current Device Certificate Status	Valid
Not Valid Before	2023/04/04 11:49:26 PDT
Not Valid After	2023/07/03 11:49:26 PDT
Last Fetched Message	Successfully fetched Device Certificate
Last Fetched Status	success
Last Fetched Timestamp	2023/04/04 11:59:26 PDT

At the bottom right, there is a blue link that says "Get certificate".

Next, you exit the device context and return to Panorama.

Step 10: From any device screen, in the top-left corner, in the **PA-VM** list, click the down arrow.

Step 11: In the device list, choose **Panorama**.

Step 12: After you select Panorama, Panorama returns to its standard context. This process might take a few minutes.

Step 13: Repeat Procedure 9.1 and Procedure 9.2 for **VMSeries-4**.

9.3 Deploy Updates from Panorama

In this procedure, you apply the latest application and threat updates and then upgrade the PAN-OS software.

Step 1: In **Panorama > Device Deployment > Dynamic Updates**, click **Check Now**.

Step 2: In the Application and Threat section, in the row for the latest panupv2-all-contents release (example: panupv2-all-contents-8688-7937), in the Action column, click **Download**.

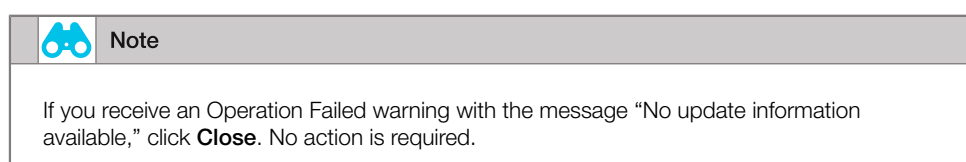
Step 3: After the download is completed, click **Close**.

Step 4: In the Application and Threat section, in the row for the latest panupv2-all-contents release (example: panupv2-all-contents-8688-7937), in the Action column, click **Install**. The Deploy App/Threat Content screen appears.

Step 5: In the Devices pane, select your device (example: **VMSeries-3**), and then click **OK**. The Install Content dialog box appears. The operation might take several minutes.

Step 6: After the content file is uploaded, click **Close**.

Step 7: Navigate to **Panorama > Device Deployment > Software**.



Step 8: In **Panorama > Device Deployment > Software**, click **Check Now**.

Step 9: To limit the choices to those valid for your device, in the search box, enter **VM**, and then click **Apply Filter**.

Step 10: If you have not already downloaded PAN-OS **10.2.0** for this platform, for version **10.2.0**, in the Actions column, click **Download**, and then when the download is complete, click **Close**.

Step 11: If you have not already downloaded PAN-OS **10.2.10-h9** for this platform, for version **10.2.10-h9**, in the Actions column, click **Download**, and then when the download is complete, click **Close**.

Step 12: In the row for **10.2.0**, after the status in the Available column has changed to Downloaded, in the **Action** column, click **Install**.

Step 13: In the Devices pane, select your device (example: **VMSeries-3**).

Step 14: Select **Upload only to device (do not install)**, and then click **OK**. This operation might take several minutes to complete.

Step 15: In the row for **10.2.10-h9**, after the status in the Available column has changed to Downloaded, in the **Action** column, click **Install**.

Step 16: In the Devices pane, select your device (example: **VMSeries-3**).

Step 17: Select **Reboot device after install**, and then click **OK**. This operation might take several minutes to complete.

Step 18: In the Install Software dialog box, click **Close**.

Step 19: Repeat this procedure **VMSeries-4**.

9.4 Validate Deployment and Refresh Licenses

After the VM-Series content release and software is updated on each device, the firewalls reboot and register with Panorama. This process can take several minutes to complete.

Step 1: On the primary Panorama virtual appliance, navigate to **Panorama > Managed Devices > Summary**.

Transit-Inbound (2/2 Devices Connected): Shared > Azure-Baseline > Transit-Inbound										
☐	VMSeries-3	PA-VM		10.110.255.6 (DHCP)	Transit-Inbound	Connected	Valid	2023/05/11 13:11:25 PDT	In Sync Panorama pushed version: 46	In sync Panorama pushed version: 45
☐	VMSeries-4	PA-VM		10.110.255.7 (DHCP)	Transit-Inbound	Connected	Valid	2023/05/11 13:40:02 PDT	In Sync Panorama pushed version: 46	In sync Panorama pushed version: 45

Next, you refresh the VM-Series firewall's license in order to enable Strata Logging Service.

Step 2: Navigate to **Panorama > Device Deployment > Licenses**, and then click **Refresh**. The Refresh License Deployment dialog box appears.

Step 3: In the **Device Name** column, select the VM-Series firewalls, and then click **Refresh**.

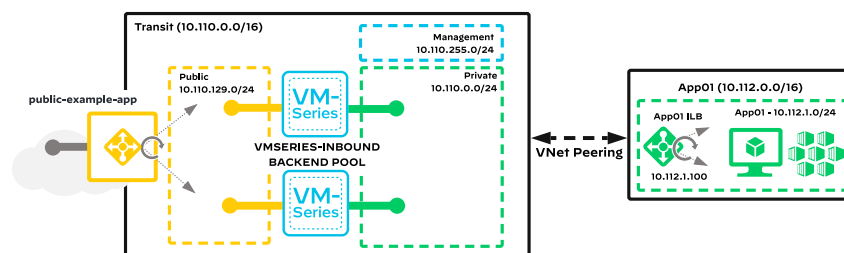
Procedures

Deploying the Public Load-Balancer Option

- 10.1 Create the Public Load-Balancer Front-End IP
- 10.2 Create the Public Load-Balancer
- 10.3 Configure the Public Load-Balancer
- 10.4 Create the Internal Load-Balancer
- 10.5 Configure the Internal Load-Balancer
- 10.6 Create Address Objects
- 10.7 Configure the Inbound NAT Policy
- 10.8 Configure the Inbound Security Policy

In this group of procedures, you configure the Azure public load-balancer option for inbound traffic.

Figure 11 Public load-balancer deployment option

**Note**

If you are deploying the common firewall option, follow the instructions in this group of procedures in order to prepare the outbound and east-west VM-Series firewalls for concurrent inbound security.

10.1 Create the Public Load-Balancer Front-End IP

In this procedure, you create a public IP address for an example web application. Note that you define a FQDN by adding the location-specific suffix to your DNS name label. You use this value in subsequent procedures as a load balancer frontend IP address and when you create Panorama IP address objects for the inbound NAT and security policies.

Step 1: Log in to the Azure Resource Manager at <https://portal.azure.com>.

Step 2: Navigate to **Home > Public IP addresses**, and then click **Create**.

Step 3: In the SKU section, select **Standard**.

Step 4: In the **Name** box, enter **public-example-app**.

Step 5: In the **DNS name label** box, enter **public-example-app**.

Step 6: In the **Resource group** list, choose **transit**.

Step 7: In the **Region** list, choose **(US) West US** or the appropriate value.

Step 8: Click **Review + create**, and then click **Create**.

Step 9: Record the value for the FQDN (example: **public-example-app.westus.cloudapp.azure.com**).

10.2 Create the Public Load-Balancer

Next, you create a public load-balancer and assign a public IP address as the frontend IP address for inbound traffic to the web server resources.

Step 1: Navigate to **Home > Load Balancers**, and then click **Create**.

Step 2: In the **Resource group** list, choose **transit**.

Step 3: In the **Name** box, enter **transit-public**.

Step 4: In the **Region** list, choose **(US) West US** or the appropriate value.

Step 5: In the **Type** section, select **Public**.

Step 6: In the **SKU** section, select **Standard**.

Step 7: Click **Next : Frontend IP configuration**.

Instance details

Name *

Region *

SKU * ☒ Standard ☐ Gateway ☐ Basic

Microsoft recommends Standard SKU load balancer for production workloads. [Learn more about pricing differences between Standard and Basic SKU](#)

Type * ☒ Public ☐ Internal

Tier * ☒ Regional ☐ Global

Step 8: Click **Add a frontend IP configuration**. The Add Frontend IP Configuration pane opens.

Step 9: In the **Name** box, enter **transit-public-frontend**.

Step 10: In the **Public IP address** list, choose **public-example-app**, and then click **Add**.

As necessary, you can add additional frontend IP addresses to the load balancer after you create it.

Name *

IP version ☒ IPv4 ☐ IPv6

IP type ☒ IP address ☐ IP prefix

Public IP address * [Create new](#)

Gateway Load balancer

Step 11: Click **Review + create**, and then click **Create**.

10.3 Configure the Public Load-Balancer

In this procedure, you configure the public load-balancer health probes, backend firewall pool, and a load-balancing rule that uses floating IP in order to direct traffic to the VM-Series firewalls.

First, you add an HTTPS probe.

Step 1: Navigate to **Home > Load Balancers > transit-public**, and then click **Health probes**.

Step 2: Click **Add**.

Step 3: In the **Name** box, enter **HTTPS-probe**.

Step 4: In the **Protocol** list, choose **HTTPS**.

Step 5: In the **Port** box, enter **443**.

Step 6: In the **Path** box, enter **/php/login.php**, and then click **Add**.

Next, you create the backend pool that contains the VM-Series firewalls.

Step 7: Navigate to **Home > Load Balancers > transit-public**, and then click **Backend pools**.

Step 8: Click **Add**.

Step 9: In the **Name** box, enter **vmseries-inbound**.

Step 10: In the **Virtual network** list, choose **transit**.

Step 11: In the IP Configurations section, click **Add**. The Add IP Configurations to Backend Pool pane opens.

Step 12: Select the rows for the Virtual Machine and IP configuration that are associated with the two firewalls **public** subnet (example: **ipconfig-untrust**), and then click **Add**.

If you are deploying the dedicated inbound option, select rows for **vmseries-3** and **vmseries-4**.

If you are deploying the common firewall option, select rows for **vmseries-1** and **vmseries-2**.

IP configurations						
IP configurations associated to virtual machines and virtual machine scale sets must be in same location as the load balancer and be in the same virtual network.						
+ Add X Remove						
☐	Resource Name	Resource group	Type	IP configuration	IP Address	Availability set
☐	vmseries-3	Transit	Virtual machine	ipconfig-untrust	10.110.129.6	TRANSIT-VM SERIES-INBOUND
☐	vmseries-4	Transit	Virtual machine	ipconfig-untrust	10.110.129.7	TRANSIT-VM SERIES-INBOUND

Step 13: Click **Save**.

Next, you add a load-balancing rule for each required TCP port (example: **TCP/80**, **TCP/443**).

Step 14: Navigate to **Home > Load Balancers > transit-public**, click **Load balancing rules**, and then click **Add**.

Step 15: In the **Name** box, enter **transit-public-frontend-80**.

Step 16: In the **Frontend IP address** list, choose **transit-public-frontend**.

Step 17: In the **Backend pool** list, choose **vmseries-inbound**.

Step 18: In the **Port** box, enter **80**.

Step 19: In the **Backend port** box, enter **80**.

Step 20: In the **Health probe** list, choose **HTTPS-probe**.

Step 21: Select **Enable Floating IP**, and then click **Save**.

The screenshot shows the configuration page for a service named 'transit-public-frontend-80'. The settings are as follows:

- Name:** transit-public-frontend-80
- IP Version:** IPv4 (selected)
- Frontend IP address:** transit-public-frontend (20.228.68.157)
- Backend pool:** transit-public
- Protocol:** TCP (selected)
- Port:** 80
- Backend port:** 80
- Health probe:** SSH-probe (TCP:22)
- Session persistence:** None
- Idle timeout (minutes):** 4
- Enable TCP Reset:** ☐
- Enable Floating IP:** ☒
- Outbound source network address translation (SNAT):** (Recommended) Use outbound rules to provide backend pool members access to the internet. [Learn more.](#)

Step 22: As necessary, repeat Step 14 through Step 21 for additional ports.

10.4 Create the Internal Load-Balancer

In this procedure, you create an internal load-balancer for the app01 web application with a single private frontend IP address and associate it with the app01 web server resources. You use the frontend IP address as the destination address in the inbound firewall NAT policy. This procedure is required for adding web server resiliency or to provide horizontal web-server scaling. Otherwise, each HTTP/HTTPS backend corresponds only to a single web server.

Step 1: Navigate to **Home > Load Balancers**, and then click **Create**.

Step 2: In the **Resource group** list, choose **app01**.

Step 3: In the **Name** box, enter **app01-internal**.

Step 4: In the **Region** list, choose **(US) West US** or the appropriate value.

Step 5: In the **Type** section, select **Internal**.

Step 6: In the **SKU** section, select **Standard**.

Step 7: Click **Next : Frontend IP configuration**.

Step 8: Click **Add a frontend IP configuration**. The Add Frontend IP Configuration pane opens.

Step 9: In the **Name** box, enter **app01-internal-frontend**.

Step 10: In the **Virtual network** list, choose **app01**.

Step 11: In the **Subnet** list, choose **app01**.

Step 12: In the IP Address Assignment section, select **Static**.

Step 13: In the **Private IP address** box, enter **10.112.1.100**.

As necessary, you can add additional frontend IP addresses to the load balancer after you create it.

Step 14: Click **Review + create**, and then click **Create**.

10.5 Configure the Internal Load-Balancer

In this procedure, you configure the app01 internal load-balancer health probes, backend web pool, and a load-balancing rule to direct traffic to the web servers.

Table 17 Example internal load-balancer rules

Rule	Frontend IP	Frontend port	Backend pool	Pool members	Backend port
public-example-app	10.112.1.100	TCP/80	web-pool-1	10.112.1.4 10.112.1.5	TCP/80

First, you add an HTTP probe.

Step 1: Navigate to **Home > Load Balancers > app01-internal**, click **Health probes**, and then click **Add**.

Step 2: In the **Name** box, enter **HTTP-probe**.

Step 3: In the **Port** box, enter **80**, click **OK**, and then click **Add**.

Next, you create the backend pool that contains the app01 web server resources.

Step 4: Navigate to **Home > Load Balancers > app01-internal**, and then click **Backend pools**.

Step 5: Click **Add**.

Step 6: In the **Name** box, enter **web-pool-1**.

Step 7: Select the rows for the Virtual Machine and IP configuration that are associated with the web server (example: **web1**) and **app01** subnet (example: **ipconfig**), and then click **Add**.

Next, you add a load-balancing rule to map the components together and complete the internal load-balancer configuration.

Step 8: Navigate to **Home > Load Balancers > app01-internal**, click **Load balancing rules**, and then click **Add**.

Step 9: In the **Name** box, enter **public-example-app**.

Step 10: In the **Frontend IP address** list, choose **app01-internal-frontend**.

Step 11: In the **Backend pool** list, choose **web-pool-1**.

Step 12: In the **Port** box, enter **80**.

Step 13: In the **Backend port** box, enter **80**.

Step 14: In the **Health probe** list, choose **HTTP-probe**, and then click **Add**.

10.6 Create Address Objects

This procedure assumes that you have already deployed a set of web server resources in the app01 subnet. In a resilient web server model, the web servers are in a backend pool of an Azure internal load-balancer. Security and NAT policy rules reference the load-balancer frontend IP, which should be defined as an address object (example: **10.112.1.100**). This guide does not include the procedure to create the web server resources.

Table 18 Inbound traffic address objects

Object name	Description	Type	Type value
public-example-app-fqdn	FQDN of public web server	FQDN	public-example-app.westus.cloudapp.azure.com
app01-internal-frontend	Frontend IP address of internal load-balancer in app01	IP netmask	10.112.1.100/32

Step 1: Log in to the primary Panorama virtual appliance's web interface.

Step 2: Navigate to **Objects**, and then choose a device group

If you are using the dedicated inbound firewall option, choose **Transit-Inbound**.

If you are using the common firewall option, choose **Transit**.

Step 3: Navigate to **Addresses**, and then click **Add**.

Step 4: In the **Name** box, enter **public-example-app-fqdn**.

Step 5: In the **Type** list, choose **FQDN**.

Step 6: In the **Type value** box, enter **public-example-app.westus.cloudapp.azure.com**, and then click **OK**.

Step 7: Repeat Step 3 through Step 6 for all rows in Table 18.

10.7 Configure the Inbound NAT Policy

In this procedure, you use security Pre Rules. These rules are logically evaluated prior to local rules and cannot be locally overridden on the local device.

Step 1: Navigate to **Policies**, and then choose a device group.

If you are using the dedicated inbound firewall option, choose **Transit-Inbound**.

If you are using the common firewall option, choose **Transit**.

Step 2: Navigate to **NAT > Pre Rules**, and then click **Add**.

Step 3: In the **Name** box, enter **public-example-app**.

Step 4: On the Original Packet tab, in the Source Zone pane, click **Add**, and then select **public**.

Step 5: In the **Destination Zone** list, choose **public**.

Step 6: In the Destination Address pane, click **Add**, and then select **public-example-app-fqdn**.

The screenshot shows the 'NAT Policy Rule' configuration window with the 'Original Packet' tab selected. The 'General' tab is also visible. The 'Source Zone' list contains 'public'. The 'Destination Zone' list contains 'public'. The 'Destination Interface' is set to 'any'. The 'Service' is set to 'any'. The 'Destination Address' list contains 'public-example-app-fqdn'. The 'Add' and 'Delete' buttons are visible at the bottom of each list. The 'OK' and 'Cancel' buttons are at the bottom right.

Step 7: On the Translated Packet tab, in the Source Address Translation section, in the **Translation Type** list, choose **Dynamic IP And Port**.

Step 8: In the Source Address Translation section, in the **Address Type** list, choose **Interface Address**.

Step 9: In the Source Address Translation section, in the **Interface** list, choose **ethernet1/2**.

Step 10: In the Destination Address Translation section, in the **Translation Type** list, choose **Static IP**.

Step 11: In the Destination Address Translation section, in the **Translated Address** list, choose **app01-internal-frontend**.

The screenshot shows the 'NAT Policy Rule' configuration window with the 'Translated Packet' tab selected. The 'Source Address Translation' section shows 'Translation Type' set to 'Dynamic IP And Port', 'Address Type' set to 'Interface Address', 'Interface' set to 'ethernet1/2', 'IP Type' set to 'IP', and 'None' selected. The 'Destination Address Translation' section shows 'Translation Type' set to 'Static IP', 'Translated Address' set to 'app01-internal-frontend', 'Translated Port' set to '[1 - 65535]', 'Enable DNS Rewrite' checked, and 'Direction' set to 'reverse'. The 'OK' and 'Cancel' buttons are at the bottom right.

Step 12: On the Target tab, verify that **Any (target to all devices)** is selected, and then click **OK**.

10.8 Configure the Inbound Security Policy

In this procedure, you use security Pre Rules. These rules are logically evaluated prior to local rules and cannot be locally overridden on the local device.

The security policy example for the inbound access traffic profile permits these applications:

- Web browsing (web-browsing)
- SSL (ssl)

As required, you can add additional applications to your policy.

Step 1: Navigate to **Policies**, and then choose a device group.

If you are using the dedicated inbound firewall option, choose **Transit-Inbound**.

If you are using the common firewall option, choose **Transit**.

Step 2: Navigate to **Security > Pre Rules**, and then click **Add**.

Step 3: In the **Name** box, enter **public-example-app-inbound**.

Step 4: On the Source tab, in the Source Zone pane, click **Add**, and then select **public**.

Step 5: On the Destination tab, in the Destination Zone pane, click **Add**, and then select **private**.

Step 6: In the Destination Address pane, click **Add**, and then select **public-example-app-fqdn**.

Step 7: On the Application tab, in the Applications pane, click **Add**.

Step 8: In the search box, enter **web-browsing**, and then in the results list, choose **web-browsing**.

Step 9: In the Applications pane, click **Add**.

Step 10: In the search box, enter **ssl**, and then in the results list, choose **ssl**.

Step 11: On the Service/URL Category tab, in the Service pane, select **application-default**.

Step 12: On the Actions tab, in the Action Setting section, in the **Action** list, choose **Allow**.

Step 13: In the Log Setting section, in the **Log Forwarding** list, choose **Forward-to-Strata-Logging-Service**.

Step 14: On the Target tab, verify that **Any (target to all devices)** is selected, and then click **OK**.

Step 15: On the **Commit** menu, click **Commit and Push**.

Procedures

Deploying the Application Gateway Option

- 11.1 Create the Application Gateway Subnet
- 11.2 Create the Application Gateway Frontend IP
- 11.3 Create the Application Gateway
- 11.4 Configure Additional Application Gateway Options
- 11.5 Create the Internal Load-Balancer
- 11.6 Configure the Internal Load-Balancer
- 11.7 Enable XFF
- 11.8 Create Address Objects
- 11.9 Configure the Inbound NAT Policy
- 11.10 Configure the Inbound Security Policy

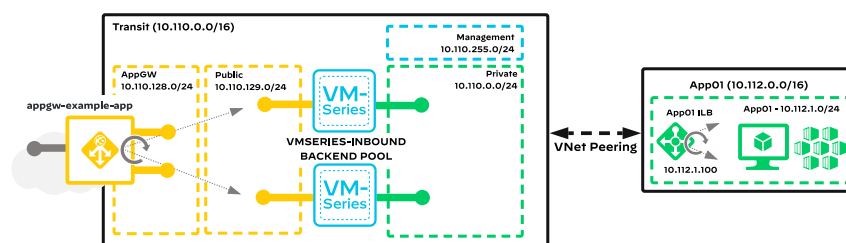
In this group of procedures, you configure the Azure application gateway option for inbound traffic.

Two configuration options are available for the application gateway:

- **With internal load-balancer**—You use a load-balancer frontend IP address for the application gateway backends. Port-based NAT policy rules on the firewall translate any traffic from the application gateways to the firewall public interface IP address to the frontend IP of the internal load-balancer. You also configure port mapping on the load balancer in order to direct traffic to the actual web server resources.
- **Without internal load-balancer**—You use web server resources for the application gateway backends. Port-based NAT policy rules on the firewall translate any traffic from the application gateways to the firewall public interface IP address to the IP addresses of the web server resources. No internal load-balancer is required.

In this procedure, you configure the application gateway with an internal load-balancer. If you would like to use a direct-to-web-server configuration, follow the example for direct-to-server configuration in Procedure 11.4 and skip the internal load-balancer procedures, Procedure 11.5 and Procedure 11.6.

Figure 12 Application gateway with internal load-balancer deployment option

**Note**

If you are deploying the common firewall option, follow the additional instructions in this group of procedures to prepare the outbound and east-west VM-Series firewalls for concurrent inbound security.

11.1 Create the Application Gateway Subnet

Azure does not permit you to configure route tables on the application gateway subnet, so you modify the transit VNet in order to add an additional subnet for the application gateway.

Step 1: Log in to the Azure Resource Manager at <https://portal.azure.com>.

Step 2: Navigate to **Home > Virtual networks > transit**, and then click **Subnets**.

Step 3: Click + **Subnet**.

Step 4: In the **Name** box, enter **public-appgw**.

Step 5: In the **Address Range (CIDR block)** box, enter **10.110.128.0/24**, and then click **Save**.

11.2 Create the Application Gateway Frontend IP

In this procedure, you create and assign a public IP address as the frontend IP address for the Azure application gateway for inbound traffic to the web server resources.

Note that you define a FQDN by adding the location-specific suffix to your DNS name label. You use this value in a subsequent procedure when you create Panorama IP address objects for the inbound access traffic profile.

Step 1: Navigate to **Home > Public IP addresses**, and then click **Create**.

Step 2: In the SKU section, select **Standard**.

Step 3: In the **Name** box, enter **appgw-example-app**.

Step 4: In the **DNS name label** box, enter **appgw-example-app**.

Step 5: In the **Resource group** list, choose **transit**.

Step 6: In the **Region** list, choose **(US) West US** or the appropriate value.

Step 7: Click **Review + create**, and then click **Create**.

Step 8: Record the value for the FQDN (example: appgw-example-app.westus.cloudapp.azure.com).

11.3 Create the Application Gateway

In this procedure, you create an application gateway with a public frontend IP address and VM-Series firewalls as backend targets. You then create a frontend listener with a default and path-based routing rule.

You use application gateway routing rules in order to associate the frontend listeners with the HTTP/HTTPS backends. There are two types of routing rules, basic and path based. Basic rules allow only a single HTTP/HTTPS backend for each listener. You use path-based rules to associate multiple HTTP/HTTPS backends to a single frontend listener. A path-based rule includes both default settings and additional path-based exceptions within a single rule. In this guide, Palo Alto Networks recommends that you use path-based rules in your application gateway configuration.

After creating the application gateway, you can create additional listeners and backends.

Table 19 Application gateway listener

Listener name	Listener protocol/ port	Use
appgw-listener-HTTP-80	HTTP/80	HTTP listener

Table 20 HTTP/HTTPS backends

Frontend listener (protocol/port)	Path	Use	HTTP/HTTPS backend	HTTP/HTTPS backend (protocol/port)
HTTP/80	All	Default	appgw-backend-HTTP-80	HTTP/80
HTTP/80	/images/*	URL path-based routing	appgw-backend-HTTP-8081	HTTP/8081

Step 1: Navigate to **Home > Application gateways**, and then click **Create**.

Step 2: In the **Resource group** list, choose [transit](#).

Step 3: In the **Application gateway name** box, enter [transit-appgw](#).

Step 4: In the **Region** list, choose **(US) West US** or the appropriate value.

Step 5: In the **Tier** list, choose **Standard V2**.

Step 6: In the Enable Autoscaling section, select **No**.



Note

This example does not use autoscaling and is explicitly configured for two instances only.

Step 7: In the Configure Virtual Network section, in the **Virtual network** list, choose **transit**.

Step 8: In the Configure Virtual Network section, in the **Subnet** list, choose **public-appgw**, and then click **Next: Frontends**.

The screenshot shows the 'Instance details' and 'Configure virtual network' sections of the Application Gateway configuration page. In the 'Instance details' section, 'Application gateway name' is 'transit-appgw', 'Region' is 'West US', 'Tier' is 'Standard V2', 'Enable autoscaling' is set to 'No', 'Instance count' is '2', 'Availability zone' is 'None', and 'HTTP2' is 'Disabled'. In the 'Configure virtual network' section, 'Virtual network' is 'transit' and 'Subnet' is 'appgw (10.110.128.0/24)'.

Step 9: In the **Public IP address** list, choose **appgw-example-app**, and then click **Next: Backends**.

Step 10: Click **Add a backend pool**.

Step 11: In the **Name** box, enter **vmseries-inbound**.

Step 12: In the Backend Targets section, in the **Target type** list, choose **Virtual machine**.

Step 13: In the Backend Targets section, in the **Target** list, choose the VM-Series interface associated with the public subnet (example: **vmseries-3-eth1**).

If you are deploying the dedicated inbound option, choose **vmseries-3-eth1** and **vmseries-4-eth1**.

If you are deploying the common firewall option, choose **vmseries-1-eth1** and **vmseries-2-eth1**.

Step 14: Repeat Step 12 and Step 13 for all VM-Series firewalls that you are assigning to this backend pool, and then click **Add**.

Step 15: Click **Next: Configuration**.

Step 16: In the Routing Rules section, click **Add a routing rule**.

Step 17: In the **Rule name** box, enter **HTTP-rule-80**.

Step 18: In the **Priority** box, enter **1000**.

Step 19: In the **Listener name** box, enter **appgw-listener-HTTP-80**.

Step 20: In the **Frontend IP** list, choose **Public**.

Add a routing rule ✕

Configure a routing rule to send traffic from a given frontend IP address to one or more backend targets. A routing rule must contain a listener and at least one backend target.

Rule name *

Priority *

* Listener * Backend targets

A listener "listens" on a specified port and IP address for traffic that uses a specified protocol. If the listener criteria are met, the application gateway will apply this routing rule.

Listener name *

Frontend IP *

Protocol ☒ HTTP ☐ HTTPS

Port *

Additional settings

Listener type ☒ Basic ☐ Multi site

Error page url ☐ Yes ☒ No

Step 21: On the Backend Targets tab, in the **Backend target** list, choose **vmseries-inbound**.

Step 22: In the Backend Setting section, click **Add new**. The Add Backend Setting pane appears.

Step 23: In the **Backend setting name** box, enter **appgw-backend-HTTP-80**.

Step 24: In **Backend protocol**, select **HTTP**, and then click **Add**.

Step 25: In the Backend Setting section, click **Add new**. The Add Backend Setting pane appears.

Step 26: In the **Backend setting name** box, enter **appgw-backend-HTTP-8081**.

Step 27: In **Backend protocol**, select **HTTP**.

Step 28: Click **Add**.

Step 29: Click **Add multiple targets to create a path-based rule**.

Step 30: In the **Path** box, enter **/images/***.

Step 31: In the **Target name** box, enter **Images**.

Step 32: In the **Backend target** list, select **vmseries-inbound**.

Step 33: In the **Backend settings** list, select **appgw-backend-HTTP-8081**, click **Add**, and then click **Add** again. This closes the Add Backend setting pane.

Step 34: On the Add a Routing Rule pane, click **Add**.

Step 35: Click **Next: Tags**.

Step 36: On the next screen, click **Next: Review + create**, and then click **Create**.

11.4 Configure Additional Application Gateway Options

(Optional)

In this procedure, you can create two additional example frontend listeners and backends. The examples include an HTTPS listener for TCP/443 and a direct-to-server HTTP listener for TCP/8000.

If you plan to use HTTPS, you can use these steps' example application gateway configuration that leverages SSL re-encryption and SSL offload. You add an additional HTTPS listener for TCP/443. SSL requires this listener. All HTTPS listeners require a X.509 digital certificate with a bundled private key in PKCS #12 format. The certificate must be issued by a trusted certificate authority, and the certificate file type must be .PFX.



Note

Certificate creation and management is beyond the scope of this document.

If your application gateway is providing access to `https://appgw-example-app.westus.cloudapp.azure.com`, then you need a certificate with a subject alternate name for the FQDN `transit-public-appgw.westus.cloudapp.azure.com` or you need a wildcard certificate for `*.westus.cloudapp.azure.com`.

Alternatively, if you have a DNS CNAME entry for your domain that maps `web.yourdomain.com` to `appgw-example-app.westus.cloudapp.azure.com`, then you could use a certificate with a subject alternate name for the FQDN `web.yourdomain.com` or you could use a wildcard certificate for `*.yourdomain.com`.

Table 21 HTTPS gateway listeners

Listener name	Listener port	Protocol	Use
appgw-listener-HTTPS-443	443	HTTPS	SSL re-encryption and SSL offload

Table 22 HTTPS backends

Frontend listener (protocol/port)	Path	Use	HTTP/HTTPS backend	HTTP/HTTPS backend (protocol/port)
HTTPS/443	All	Re-encrypt (requires server certificate)	appgw-backend-HTTPS-443	HTTPS/443
HTTPS/443	/images/*	SSL offload	appgw-backend-HTTP-8443	HTTP/8443

First, you add the HTTPS listener.

Step 1: Navigate to **Home > Application Gateways > transit-appgw**, click **Listeners**, and then click **Add listener**.

Step 2: In the **Name** box, enter **appgw-listener-HTTPS-443**.

Step 3: In the **Frontend IP** list, choose **Public**.

Step 4: In the Protocol section, select **HTTPS**. Verify that the value in the **Port** box updates to **443**.

Step 5: In the Choose a Certificate section, select **Upload a certificate**.

Step 6: Next to the **PFX certificate file** box, click the browse icon, and then select the public web server certificate.

Step 7: In the **Certificate name** box, enter **transit-appgw-certificate**.

Step 8: In the certificate **Password** box, enter the certificate password, and then click **Add**.

Add listener (transit-appgw)

Listener name *

Frontend IP *

Protocol ☐ HTTP ☒ HTTPS

Port *

Https Settings

Choose a certificate

☒ Upload a certificate ☐ Choose a certificate from Key Vault

Cert name *

PFX certificate file *

Password *

☐ Enable SSL Profile

Additional settings

Listener type ☒ Basic ☐ Multi site

Error page url ☐ Yes ☒ No

Because the application gateway backend pool includes only the public interfaces of the firewalls, to associate with the different HTTP/HTTPS backend resources, you must use multiple TCP ports on the firewall.



Note

If you are using a public load-balancer for inbound traffic with health probes on TCP/443, then you cannot also use TCP/443 for an application gateway HTTPS backend.

If you configure the application gateway to re-encrypt SSL traffic to an HTTPS backend, you must provide a DER or Base-64 encoded X.509 digital certificate for the actual server resource and you must save the certificate in .CER format.

Next, you create the HTTPS backend.

Step 9: Navigate to **Home > Application gateways > transit-appGW**, click **Backend settings**, and then click **Add**.

Step 10: In the **Name** box, enter **appgw-backend-HTTPS-443**.

Step 11: In the Protocol section, select **HTTPS**.

Step 12: In the **Port** box, enter **443**.

Step 13: In the Trusted Root Certificates section, select **Create new**.

Step 14: In the **Name** box, enter **trustedroot**.

Step 15: Next to the **Upload CER certificate** box, click the browse icon, and then select the trusted root certificate (example: **trustedroot.cer**).

Add Backend setting

Backend settings name *

Backend protocol
☐ HTTP ☒ HTTPS

Backend port *

Trusted root certificate
 For end-to-end SSL encryption, the backends must be in the allowlist of the application gateway. Upload the public certificate of the backend servers to this Backend setting.

Use well known CA certificate
☐ Yes ☒ No

Choose a certificate
☒ Create new ☐ Select existing

CER certificate *

Cert name *

[+ Add certificate](#)

Step 16: In the Host Name section, under **Override with new host name**, choose **Yes**.

Step 17: Select **Pick host name from backend address**.

Host name

By default, Application Gateway does not change the incoming HTTP host header from the client and sends the header unaltered to the backend. Multi-tenant services like App service or API management rely on a specific host header or SNI extension to resolve to the correct endpoint. Change these settings to overwrite the incoming HTTP host header.

Override with new host name
☒ Yes ☐ No

Host name override
☒ Pick host name from backend target
☐ Override with specific domain name

Step 18: Click **Save**.

Next, you create the HTTP/8443 backend.

Step 19: Navigate to **Home > Application gateways > transit-appGW**, click **Backend settings**, and then click **Add**.

Step 20: In the **Name** box, enter **appgw-backend-HTTPS-8443**.

Step 21: In the Protocol section, select **HTTP**.

Step 22: In the **Port** box, enter **8443**, and then click **Save**.

Next, you create the HTTPS routing rule.

Step 23: Navigate to **Home > Application gateways > transit-appgw**, and then click **Rules**.

Step 24: Click **+ Routing rule**.

Step 25: In the **Name** box, enter **HTTPS-rule-443**.

Step 26: In the **Priority** box, enter **1010**.

Step 27: In the **Listener** list, choose **appgw-listener-HTTPS-443**.

Step 28: On the Backend Targets tab, in the **Backend target** list, choose **vmseries-inbound**.

Step 29: In the **Backend setting list**, enter **appgw-backend-HTTPS-443**.

Step 30: Click **Add multiple targets to create a path-based rule**.

Step 31: In the **Path** box, enter **/images/***.

Step 32: In the **Target name** box, enter **SSL-offload**.

Step 33: In the **Backend settings** list, select **appgw-backend-HTTP-8443**.

Step 34: In the **Backend target** list, select **vmseries-inbound**, click **Add**, and then click **Add** again.

If you would like to use a direct-to-web-server configuration, these steps provide an example of an application gateway basic rule for the direct-to-server option. An additional HTTP listener for TCP/8000 is created to show the details for destination NAT configuration on the firewall that does not require the use of an internal load-balancer.

Table 23 Direct-to-server application gateway listeners

Listener name	Listener port	Protocol	Use
appgw-listener-HTTP-8000	8000	HTTP	Direct to server, uses destination NAT on firewall. No internal load-balancer required.

Table 24 Direct-to-server backends

Frontend listener (protocol/port)	Path	Use	HTTP backend	HTTP backend (protocol/port)
HTTP/8000	All	Direct to web server (no internal load-balancer used)	appgw-backend-HTTP-8000	HTTP/8000

Step 35: Navigate to **Home > Application Gateways > transit-appgw**, click **Listeners**, and then click **Add listener**.

Step 36: In the **Name** box, enter **appgw-listener-HTTP-8000**.

Step 37: In the **Frontend IP** list, choose **Public**.

Step 38: In the **Port** box, enter **8000**, and then click **Add**.



Note

If you are using the direct-to-server configuration, skip the internal load-balancer deployment and configuration and follow the instructions for the direct-to-server deployment in the remaining procedures.

Add listener transit-appgw ×

Listener name * ⓘ
 ✓

Frontend IP * ⓘ
 ✓

Protocol ⓘ
☒ HTTP ☐ HTTPS

Port * ⓘ
 ✓

Additional settings

Listener type ⓘ
☒ Basic ☐ Multi site

Error page url
☐ Yes ☒ No

Next, you create the HTTP/800 backend.

Step 39: Navigate to **Home > Application gateways > transit-appgw**, click **Backend settings**, and then click **Add**.

Step 40: In the **Name** box, enter **appgw-backend-HTTP-8000**.

Step 41: In the Protocol section, select **HTTP**.

Step 42: In the **Port** box, enter **8000**, and then click **Save**.

Next, you create the HTTP/8000 routing rule.

Step 43: Navigate to **Home > Application gateways > transit-appgw**, and then click **Rules**.

Step 44: Click **+ Routing rule**.

Step 45: In the **Name** box, enter **HTTP-rule-8000**.

Step 46: In the **Priority** box, enter **1020**.

Step 47: In the **Listener** list, choose **appgw-listener-HTTP-8000**.

Step 48: On the Backend Targets tab, in the **Backend target** list, choose **vmseries-inbound**.

Step 49: In the **Backend setting list**, enter **appgw-backend-HTTP-8000**, and then click **Add**.

11.5 Create the Internal Load-Balancer

You create an internal load-balancer for the app01 web applications with a single private frontend IP address and associate it with the app01 web server resources. In the inbound firewall NAT policy, you use the frontend IP address as the destination address. This procedure is required for adding web server resiliency or providing horizontal web-server scaling. Otherwise, each HTTP/HTTPS backend corresponds only to a single web server.

Step 1: Navigate to **Home > Load Balancers**, and then click **Create**.

Step 2: In the **Resource group** list, choose **app01**.

Step 3: In the **Name** box, enter **app01-internal**.

Step 4: In the **Region** list, choose **(US) West US** or the appropriate value.

Step 5: In the **Type** section, select **Internal**.

Step 6: In the **SKU** section, select **Standard**.

Step 7: Click **Next : Frontend IP configuration**.

Step 8: Click **Add a frontend IP configuration**. The Add Frontend IP Configuration pane opens.

Step 9: In the **Name** box, enter **app01-internal-frontend**.

Step 10: In the **Virtual network** list, choose **app01**.

Step 11: In the **Subnet** list, choose **app01**.

Step 12: In the **IP Address Assignment** section, select **Static**.

Step 13: In the **Private IP address** box, enter **10.112.1.100**.

Step 14: Click **Review + create**, and then click **Create**.

As necessary, you can add additional frontend IP addresses to the load balancer after you create it.

11.6 Configure the Internal Load-Balancer

This example uses separate web server backend pools for each use: HTTP default, images, and SSL. You use the backend pool for the image servers for cleartext HTTP access and for SSL offload. In this case, because there are two rules associated with the same backend pool, the pool members must listen on multiple ports (TCP/80 and TCP/8443). You configure load-balancer health probes for all backend web server ports.

Table 25 Example internal load-balancer rules

Rule	Frontend IP	Frontend port	Backend pool	Pool members	Backend port
appgw-80	10.112.1.100	TCP/80	web-pool-1	10.112.1.4 10.112.1.5	TCP/80
appgw-8081	10.112.1.100	TCP/8081	image-pool-2	10.112.1.10 10.112.1.11	TCP/80
appgw-8443	10.112.1.100	TCP/8443	image-pool-2	10.112.1.10 10.112.1.11	TCP/8443
appgw-443	10.112.1.100	TCP/443	SSL-pool-3	10.112.1.20 10.112.1.21	TCP/443

First, you add the necessary probes.

Step 1: Navigate to **Home > Load Balancers > app01-internal**, click **Health probes**, and then click **Add**.

Step 2: In the **Name** box, enter **HTTP-probe**.

Step 3: In the **Port** box, enter **80**, click **OK**, and then click **Add**.

Step 4: In the **Name** box, enter **HTTP-probe-8081**.

Step 5: In the **Port** box, enter **8081**, click **OK**, and then click **Add**.

Step 6: In the **Name** box, enter **HTTP-probe-8443**.

Step 7: In the **Port** box, enter **8443**, click **OK**, and then click **Add**.

Step 8: In the **Name** box, enter **HTTPS-probe-443**.

Step 9: In the **Port** box, enter **443**, and then click **OK**.

Next, you create the backend pools that contain the app01 web server resources.

Step 10: Navigate to **Home > Load Balancers > app01-internal**, click **Backend pools**, and then click **Add**.

Step 11: In the **Name** box, enter **web-pool-1**.

Step 12: For each pool member listed for this backend pool in Table 25, select the rows for the **Virtual Machine** and **IP configuration** that are associated with the web server (example: **web1**) and **app01** subnet (example: **ipconfig**), and then click **Add**.

Step 13: Repeat Step 10 through Step 12 for each backend pool listed in Table 25.

Next, you add the load-balancing rules to map the components together and complete the internal load-balancer configuration.

Step 14: Navigate to **Home > Load Balancers > app01-internal**, and then click **Load balancing rules**.

Step 15: For each rule listed in Table 25, perform the following sub-steps:

- Click **Add**.
- In the **Name** box, enter **appgw-80**.
- In the **Frontend IP address** list, choose **app01-internal-frontend**.
- For **Protocol**, select **TCP**.
- In the **Port** box, enter **80**.
- In the **Backend port** box, enter **80**.
- In the **Backend pool** list, choose **web-pool-1**.
- In the **Health probe** list, choose **HTTP-probe**, and then click **Add**.

11.7 Enable XFF

The application gateway operates as a proxy and masks the original source IP address of incoming connections so that the firewall sees only the source IP addresses of the application gateway instances. The application gateway adds the original source address information to the HTTP packet header by using the X-Forwarded-For (XFF) HTTP header field. The firewall is configured to extract XFF information from the session and add the original source IP address information to the logs.



Note

Palo Alto Networks recommends that you review and install the most recent application and content updates. If your Panorama virtual appliance is running with a more recent content release than your VM-Series devices, the URL filtering profile created in this procedure might contain new URL filtering categories. When you commit and push the configuration, these categories might not be recognized on the VM-Series devices running older content releases and the commit fails.

Step 1: Log in to the primary Panorama virtual appliance's web interface.

Step 2: Navigate to **Device**, and then choose a template.

If you are using the dedicated inbound firewall option, choose **Transit-2-Zone-Inbound**.

If you are using the common firewall option, choose **Transit-2-Zone**.

Step 3: Navigate to **Setup > Content-ID**.

Step 4: In the X-Forwarded-For Headers section, click the edit cog. The X-Forwarded-For Headers dialog box opens.

Step 5: In the **Use X-Forwarded-For Header** list, select **Enabled for User-ID**, and then click **OK**.

Step 6: Navigate to **Objects**, and then choose a device group.

If you are using the dedicated inbound firewall option, choose **Transit-Inbound**.

If you are using the common firewall option, choose **Transit**.

Step 7: Navigate to **Security Profiles > URL Filtering**, and then click **Add**.

Step 8: In the URL Filtering Profile pane, in the **Name** box, enter **Enable-XFF-Logging**.

Step 9: In the **Site Access** column, click the down arrow, and then select **Set All Actions > alert**. This sets all categories to alert.



Note

If the Site Access category action is Allow, URL filtering logs are not generated. Palo Alto Networks suggests that you initially configure all categories to Alert.

The application gateway health probes generate a significant amount of log messages with the category of private-ip-address. To suppress these messages, you can set the Site Access category action for this category to Allow.

URL Filtering Profile ⓘ

Name:

Description:

☐ Shared

☐ Disable override

Categories | URL Filtering Settings | User Credential Detection | HTTP Header Insertion | Inline ML

73 items → ×

☐ CATEGORY	SITE ACCESS	USER CREDENTIAL SUBMISSION
Pre-defined Categories		
☐ abortion	alert	allow
☐ abused-drugs	alert	allow
☐ adult	alert	allow
☐ alcohol-and-tobacco	alert	allow
☐ auctions	alert	allow
☐ business-and-economy	alert	allow

* Indicates a custom URL category. + Indicates external dynamic list

[Check URL Category](#)

OK Cancel

Step 10: On the URL Filtering Settings tab, in the HTTP Header Logging section, select **X-Forwarded-For**, and then click **OK**.

11.8 Create Address Objects

To simplify rule creation, you create address objects for the NAT policy rules and Security policy rules.

To match the application gateway instances, Palo Alto Networks recommends that you use a subnet range. We recommend this because Azure dynamically assigns an IP address for each application gateway instance but does not provide details about which addresses are assigned. If you use a subnet range, you can easily add additional application gateway instances (manually or autoscaled) without having to modify your firewall policies.

This procedure assumes that you have already deployed a set of web server resources in the app01 subnet. In a resilient web server model, the web servers are in a backend pool of an Azure internal load-balancer. The load-balancer frontend IP is referenced by security and NAT policy rules and should be defined as an address object (example: **10.112.1.100**).

Table 26 Inbound traffic address objects

Object name	Description	Type	Type value
vmseries-1-public	Public subnet interface of common option firewall (vmseries-1 ethernet1/1)	IP netmask	10.110.129.4/32
vmseries-2-public	Public subnet interface of common option firewall (vmseries-2 ethernet1/1)	IP netmask	10.110.129.5/32
vmseries-3-public	Public subnet interface of dedicated inbound firewall (vmseries-3 ethernet1/1)	IP netmask	10.110.129.6/32
vmseries-4-public	Public subnet interface of dedicated inbound firewall (vmseries-4 ethernet1/1)	IP netmask	10.110.129.7/32
appgw-subnet	Application gateway subnet (matches all instances)	IP netmask	10.110.128.0/24

(Table 26 continues on the following page)

Table 26 (continued)

Object name	Description	Type	Type value
app01-internal-frontend	IP address of app01 internal load-balancer	IP netmask	10.112.1.100/32
web-server	IP address of web server resource (direct-to-server deployment does not require the app01 internal load-balancer)	IP netmask	10.112.1.4/32

Step 1: Navigate to **Objects**, and then choose a device group:

- If you are using the dedicated inbound firewall option, choose **Transit-Inbound**.
- If you are using the common firewall option, choose **Transit**.

Next, you add objects. If you are deploying the dedicated inbound option, you add objects for **vmseries-3-public** and **vmseries-4-public** and the remaining objects in the table. If you are deploying the common firewall option, you add objects for **vmseries-1-public** and **vmseries-2-public** and the remaining objects in the table.

Step 2: Navigate to **Addresses**, and then click **Add**.

Step 3: In the **Name** box, enter **vmseries-3-public**.

Step 4: In the **Type** list, choose **IP Netmask**.

Step 5: In the **Type value** box, enter **10.110.129.6/32**, and then click **OK**.

Step 6: Repeat Step 2 through Step 5 for the remaining rows in Table 26.

11.9 Configure the Inbound NAT Policy

In this procedure, you use security Pre Rules. These rules are logically evaluated prior to local rules and cannot be locally overridden on the local device.

You configure the application gateway to send web traffic on multiple non-standard ports as well as the standard ports (80/443). To simplify the configuration of the NAT and security policies, you create a custom service for each non-standard port that is not predefined.



Note

Do not use TCP/8080 as a non-standard port. This port is already predefined as service-http. To create an explicit NAT rule that uses this port, you must manually remove this port from the predefined service and create a new service.

Table 27 Custom services for non-standard ports

Name	Type	Destination port	Description
service-http-8000	Custom	TCP/8000	Used for TCP/8000 direct-to-server deployment
service-http-8081	Custom	TCP/8081	Used for TCP/8081
service-http-8443	Custom	TCP/8443	Used for TCP/8443

First, you create the custom services.

Step 1: Navigate to **Objects**, and then choose a device group.

If you are using the dedicated inbound firewall option, choose **Transit-Inbound**.

If you are using the common firewall option, choose **Transit**.

Step 2: Navigate to **Services**, and then click **Add**.

Step 3: In the **Name** box, enter **service-http-8000**.

Step 4: In the **Destination Port** box, enter **8000**, and then click **OK**.

Step 5: Repeat Step 2 through Step 4 for all rows in Table 27.

Each application gateway HTTP/HTTPS backend as specified in Table 20, Table 22, and Table 24 must have a corresponding NAT translation rule on each firewall. The firewall that corresponds to a backend resource in the private networks maps the original packet service (or destination TCP port) to a translated address and translated port. These resources can be either a load-balancer frontend or actual web servers.

Each firewall needs a unique set of NAT policy rules. In each set, the destination address of the original packet is always set to the IP address of the public interface of each firewall. Table 28 lists only two firewalls as targets. If you deploy additional firewalls, then you must add a new set of rules for each additional firewall.

Table 28 NAT translation rules for the dedicated inbound firewall option

Name	Service	Destination address	Translated address/ translated port	Target firewall
appgw-example-app-HTTP80-3	service-http	vmseries-3-public	app01-internal-frontend/80	vmseries-3
appgw-example-app-HTTP80-4	service-http	vmseries-4-public	app01-internal-frontend/80	vmseries-4
appgw-example-app-HTTP8000-3	service-http-8000	vmseries-3-public	web-server/80	vmseries-3
appgw-example-app-HTTP8000-4	service-http-8000	vmseries-4-public	web-server/80	vmseries-4

(Table 28 continues on the following page)

Table 28 (continued)

Name	Service	Destination address	Translated address/ translated port	Target firewall
appgw-example-app-HTTP8081-3	service-http-8081	vmseries-3-public	app01-internal-frontent/8081	vmseries-3
appgw-example-app-HTTP8081-4	service-http-8081	vmseries-4-public	app01-internal-frontent/8081	vmseries-4
appgw-example-app-HTTPS443-3	service-https	vmseries-3-public	app01-internal-frontent/443	vmseries-3
appgw-example-app-HTTPS443-4	service-https	vmseries-4-public	app01-internal-frontent/443	vmseries-4
appgw-example-app-HTTP8443-3	service-http-8443	vmseries-3-public	app01-internal-frontent/8443	vmseries-3
appgw-example-app-HTTP8443-4	service-http-8443	vmseries-4-public	app01-internal-frontent/8443	vmseries-4

Table 29 NAT translation rules for the common firewall deployment option

Name	Service	Destination address	Translated address/ translated port	Target firewall
appgw-example-app-HTTP80-1	service-http	vmseries-1-public	app01-internal-frontent/80	vmseries-1
appgw-example-app-HTTP80-2	service-http	vmseries-2-public	app01-internal-frontent/80	vmseries-2
appgw-example-app-HTTP8000-1	service-http-8000	vmseries-1-public	web-server/80	vmseries-1
appgw-example-app-HTTP8000-2	service-http-8000	vmseries-2-public	web-server/80	vmseries-2
appgw-example-app-HTTP8081-1	service-http-8081	vmseries-1-public	app01-internal-frontent/8081	vmseries-1
appgw-example-app-HTTP8081-2	service-http-8081	vmseries-2-public	app01-internal-frontent/8081	vmseries-2
appgw-example-app-HTTPS443-1	service-https	vmseries-1-public	app01-internal-frontent/443	vmseries-1
appgw-example-app-HTTPS443-2	service-https	vmseries-2-public	app01-internal-frontent/443	vmseries-2

(Table 29 continues on the following page)

Table 29 (continued)

Name	Service	Destination address	Translated address/ translated port	Target firewall
appgw-example-app-HTTP8443-1	service-http-8443	vmseries-1-public	app01-internal-frontend/8443	vmseries-1
appgw-example-app-HTTP8443-2	service-http-8443	vmseries-2-public	app01-internal-frontend/8443	vmseries-2

Next, you create the NAT translation rules for the inbound traffic flows.

Step 6: Navigate to **Policies**, and then choose a device group:

- If you are using the dedicated inbound firewall option, choose **Transit-Inbound**.
- If you are using the common firewall option, choose **Transit**.

Step 7: If you are deploying the dedicated inbound option, configure the values shown in Table 28.

If you are deploying the common firewall option, configure the values shown in Table 29.

Step 8: For each rule shown in Table 28 or Table 29, perform the following sub-steps:

- Navigate to **NAT > Pre Rules**, and then click **Add**.
- In the **Name** box, enter the value from the table (example: **appgw-example-app-HTTP80-3**).
- On the Original Packet tab, in the Source Zone pane, click **Add**, and then select **public**.
- In the **Destination Zone** list, choose **public**.
- In the **Service** list, choose the value from the table (example: **service-http**).
- In the Source Address pane, click **Add**, and then select **appgw-subnet**.

- In the Destination Address pane, click **Add**, and then enter the value from the table (example: **vmseries-3-public**).

The screenshot shows the 'NAT Policy Rule' configuration window with the 'Original Packet' tab selected. The 'General' tab is also visible. The 'Original Packet' section includes a 'Destination Zone' dropdown set to 'public', a 'Destination Interface' dropdown set to 'any', and a 'Service' dropdown set to 'service-http'. The 'Destination Address' section has a list with one entry: 'vmseries-3-public'. The 'Source Address' section has a list with one entry: 'appgw-subnet'. The 'Add' and 'Delete' buttons are visible at the bottom of each list.

- On the Translated Packet tab, in the Source Address Translation section, in the **Translation Type** list, choose **Dynamic IP And Port**.
- In the Source Address Translation section, in the **Address Type** list, choose **Interface Address**.
- In the Source Address Translation section, in the **Interface list**, choose **ethernet1/2**.
- In the Destination Address Translation section, in the **Translation Type** list, choose **Static IP**.
- In the Destination Address Translation section, in the **Translated Address** list, choose the value from the table (example: **app01-internal-frontend**).
- In the Destination Address Translation section, in the Translated Port box, enter the value from the table (example: **80**).

The screenshot shows the 'NAT Policy Rule' configuration window with the 'Translated Packet' tab selected. The 'Source Address Translation' section has 'Translation Type' set to 'Dynamic IP And Port', 'Address Type' set to 'Interface Address', 'Interface' set to 'ethernet1/2', and 'IP Type' set to 'None'. The 'Destination Address Translation' section has 'Translation Type' set to 'Static IP', 'Translated Address' set to 'app01-internal-frontend', and 'Translated Port' set to '[1 - 65535]'. The 'Enable DNS Rewrite' checkbox is unchecked, and the 'Direction' dropdown is set to 'reverse'. The 'Add' and 'Delete' buttons are visible at the bottom of each list.

- On the Target tab, select only the firewall target from the value from the table (example: **vmseries-3**), and then click **OK**.

11.10 Configure the Inbound Security Policy

In this procedure, you use security Pre Rules. These rules are logically evaluated prior to local rules and cannot be locally overridden on the local device. The security policy example for inbound traffic permits HTTP and HTTPS.

Each firewall needs a unique security policy rule. In Step 8, the destination address must match the IP address of the public interface of each firewall. Table 30 lists only two firewalls as targets. If your firewall layer includes additional firewalls, then you must add a new rule for each additional firewall.

Table 30 Security policy rules for the dedicated inbound firewall option

Name	Source object	Destination address	Target firewall
appgw-example-app-3	appgw-subnet	vmseries-3-public	vmseries-3
appgw-example-app-4	appgw-subnet	vmseries-4-public	vmseries-4

Table 31 Security policy rules for the common firewall option

Name	Source object	Destination address	Target firewall
appgw-example-app-1	appgw-subnet	vmseries-1-public	vmseries-1
appgw-example-app-2	appgw-subnet	vmseries-2-public	vmseries-2

Step 1: Navigate to **Policies**, and then choose a device group:

- If you are using the dedicated inbound firewall option, choose **Transit-Inbound**.
- If you are using the common firewall option, choose **Transit**.

Step 2: If you are deploying the dedicated inbound option, configure the values shown in Table 30.

If you are deploying the common firewall option, configure the values shown in Table 31.

Step 3: Navigate to **Security > Pre Rules**, and then click **Add**.

Step 4: In the **Name** box, enter the value from the table (example: **appgw-example-app-3**).

Step 5: On the Source tab, in the Source Zone pane, click **Add**, and then select **public**.

Step 6: In the Source Address pane, click **Add**, and then select **appgw-subnet**.

Step 7: On the Destination tab, in the Destination Zone pane, click **Add**, and then select **private**.

Step 8: In the Destination Address pane, click **Add**, and then enter the value from the table (example: **vmseries-3-public**).

Step 9: On the Application tab, in the Applications pane, click **Add**.

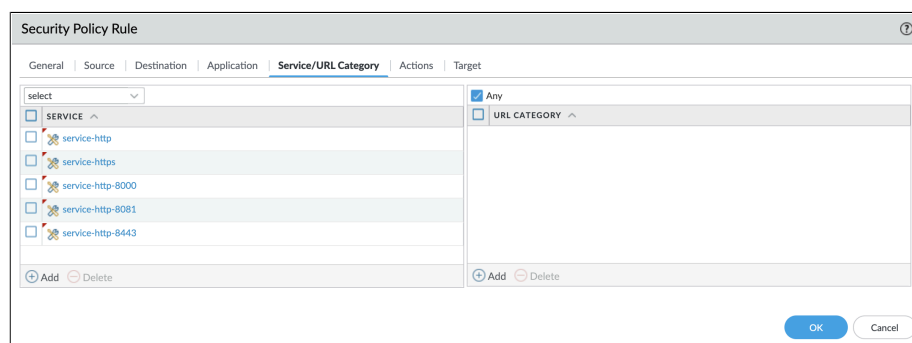
Step 10: In the search box, enter **web-browsing**, and then in the results list, choose **web-browsing**.

Step 11: In the Applications pane, click **Add**.

Step 12: In the search box, enter **ssl**, and then in the results list, choose **ssl**.

Next, you configure the security policy to allow web traffic on multiple non-standard ports as well as the standard ports (80/443). The firewall restricts web traffic on non-standard ports when application-default is configured. To permit web traffic on the non-standard ports, you must explicitly list each service in use.

Step 13: On the Service/URL Category tab, in the Service pane, click **Add**, and then select each service listed in Table 27.

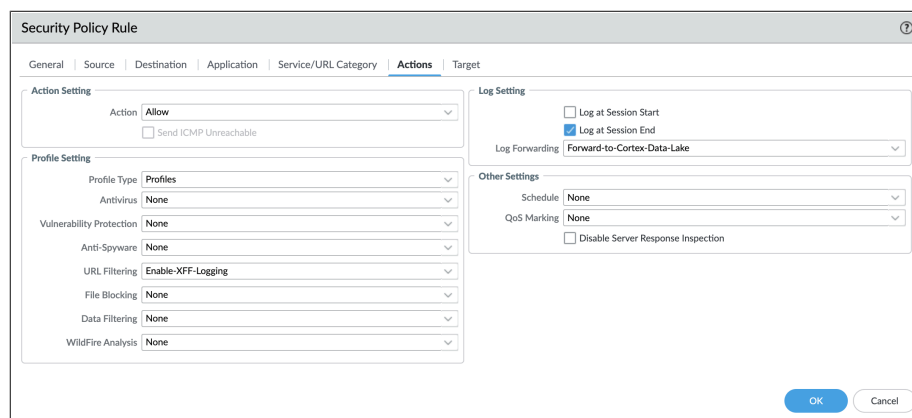


Step 14: On the Actions tab, in the Action Setting section, in the **Action** list, choose **Allow**.

Step 15: In the Profile Setting section, in the **Profile Type** list, choose **Profiles**.

Step 16: In the Profile Setting section, in the **URL Filtering** list, choose **Enable-XFF-Logging**.

Step 17: In the Log Setting section, in the **Log Forwarding** list, choose **Forward-to-Strata-Logging-Service**.



Step 18: On the Target tab, select only the firewall target from the value from the table (example: **vmseries-3**), and then click **OK**.

Step 19: On the **Commit** menu, click **Commit and Push**.



Note

You must modify the default settings for the firewall's URL filtering log in order to view the XFF header information for sessions. Navigate to **Monitor > Logs > URL Filtering**, add the X-Forwarded-For column to the default view, and then rearrange the column order.

Deploying VPN to Prisma Access

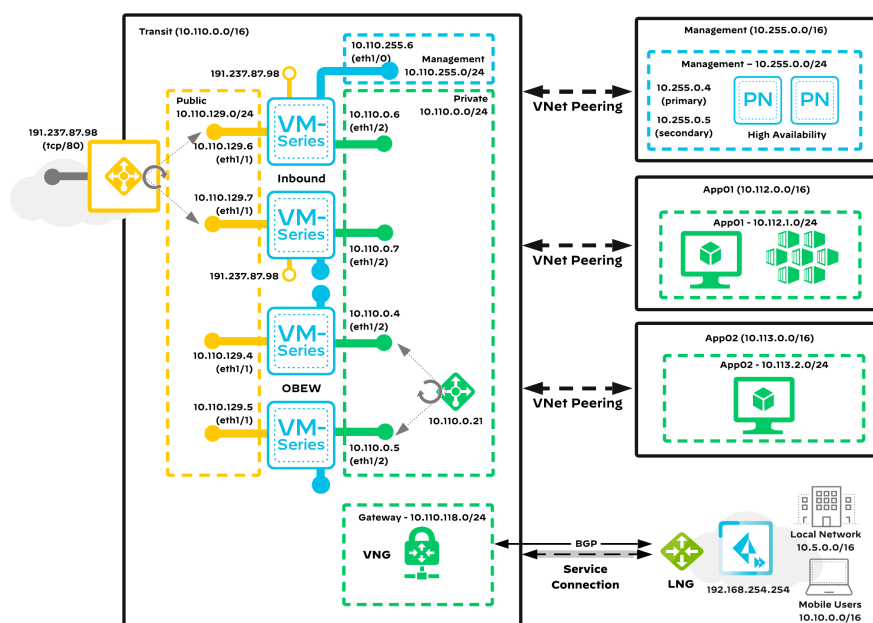
In this section, you configure connectivity from the Azure VNG service to Prisma Access. The connection between the VNG service and Prisma Access uses a service connection with BGP dynamic routing. The VPN endpoints are the Azure VNG and a Prisma Access cloud NGFW referenced by an Azure LNG.



Note

This procedures in this guide assume you have licensed and deployed your Prisma Access instance following the instructions in **SASE for Securing Private Applications: Deployment Guide**. That guide focuses on specific configurations for the service connection, address objects, and security policy in order to include your deployed resources and applications in Azure.

Figure 13 Enterprise network connectivity



Procedures

Creating a Virtual Network Gateway and VPN Attachments

- 12.1 Create and Modify the Route Tables
- 12.2 Configure the User-Defined Routes
- 12.3 Create the VNG Subnet
- 12.4 Create the Virtual Network Gateway Public IP
- 12.5 Create the Virtual Network Gateway

These procedures rely on the following assumptions:

- Prisma Access has a statically assigned public IP address.
- The enterprise network IP address block is **10.5.0.0/16**.
- The reachable management subnets for Panorama and VM-Series are **10.255.0.0/24** and **10.110.255.0/24**.
- The reachable application subnets (app01 and app02) are **10.112.0.0/16** and **10.113.0.0/16**.

12.1 Create and Modify the Route Tables

By default, Azure networking routes traffic originating from all subnets and destined to the enterprise network ranges directly to the VNG. In order to change the default behavior and direct all traffic to the VM-Series you must add UDRs for all connected subnets.



Caution

As a best practice, if you are running BGP, disable VNG route propagation for all subnets except for the Transit Private subnet. This configuration prevents any BGP learned routes from being installed in the active route table for these subnets.

First, you create a route table for the transit VNet public subnet with route propagation disabled.

Step 1: Log in to the Azure Resource Manager at <https://portal.azure.com>.

Step 2: Navigate to **Home > Route tables**, and then click **Create**.

Step 3: In the **Resource group** list, choose **transit**.

Step 4: In the **Region** list, choose **(US) West US** or the appropriate value.

Step 5: In the **Name** box, enter **public**.

Step 6: In **Propagate gateway routes**, select **No**.

Step 7: Click **Review + create**, and then click **Create**.

Next, you associate the route table with the public subnet.

Step 8: Navigate to **Home > Route tables**, and then click **public**.

Step 9: Select **Subnets**, and then click **Associate**. The Associate Subnet page appears.

Step 10: In the **Virtual network** box, select **Transit (Transit)**.

Step 11: In the **Subnet** box, choose **public**, and then click **OK**.

Finally, you remove route propagation from applicable route tables.

Step 12: Navigate to **Home > Route tables > app01**, and then click **Configuration**.

Step 13: In **Propagate gateway routes**, select **No**, and then click **Save**.

Step 14: Repeat Step 12 and Step 13 for the **app02** and **management** route tables, as well as any other applicable VNet route tables.

12.2 Configure the User-Defined Routes

In this procedure, you create the VNG route table and add UDRs for the VNG to reach the application and management subnets by using the transit VNet private load-balancer. You create the route table so that you can associate it when you create the subnet.



Caution

You cannot associate a default route 0.0.0.0/0 to the gateway subnet. This means that you must create UDRs for each connected network or traffic will bypass the VM-Series firewalls.

Table 32 VNG subnet UDRs

Route name	Address prefix	Next-hop type	Next-hop address	Comments
app01	10.112.0.0/16	Virtual appliance	10.110.0.21	Override peering for app01
app02	10.113.0.0/16	Virtual appliance	10.110.0.21	Override peering for app02
Management	10.255.0.0/24	Virtual appliance	10.110.0.21	Override peering for mgmt

First, you create the route table for the VNG.

Step 1: Navigate to **Home > Route tables**, and then click **Create**.

Step 2: In the **Resource group** list, choose **transit**.

Step 3: In the **Name** box, enter **vng**.

Step 4: Click **Review + create**, and then click **Create**.

Next, you add the UDRs for the app01, app02 and management networks with a next-hop of the transit VNet private subnet internal load-balancer.

Step 5: Navigate to **Home > Route tables > vng > Routes**, and then click **Add**.

Step 6: In the **Route name** box, enter **app01**.

Step 7: In the **Address prefix destination** list, choose **IP Addresses**.

Step 8: In the **Destination IP addresses/CIDR ranges** box, enter **10.112.0.0/16**.

Step 9: In the **Next hop type** list, choose **Virtual Appliance**.

Step 10: In the **Next hop address** value, enter **10.110.0.21**, and then click **Add**.

Step 11: Repeat Step 5 through Step 10 for all UDRs in Table 32.

12.3 Create the VNG Subnet

In this procedure, you add a new gateway subnet for the Azure VNG to the existing transit VNet.

Step 1: Navigate to **Home > Virtual networks > transit**, and then click **Subnets**.

Step 2: Click **+ Gateway subnet**.

Step 3: In the **Subnet address range** box, enter **10.110.118.0/24**, and then click **Save**.

Next, you associate the route table with the vng subnet.

Step 4: Navigate to **Home > Route tables**, and then click **vng**.

Step 5: Select **Subnets**, and then click **Associate**. The Associate Subnet page appears.

Step 6: In the Virtual network box, select **Transit (Transit)**.

Step 7: In the **Subnet** box, choose **GatewaySubnet**, and then click **OK**.

Associate subnet	
VNG	
Virtual network *	ⓘ
Transit (Transit)	
Subnet *	ⓘ
GatewaySubnet	

12.4 Create the Virtual Network Gateway Public IP

Step 1: Navigate to **Home > Public IP addresses**, and then click **Create**.

Step 2: In the **Resource group** box, select **transit**.

Step 3: In the **Region** box, select **(US) West US**, or the appropriate value.

Step 4: In the **Name** box, enter **vng-public**.

Caution

Do not enter a value in the **DNS name label** box. Azure does not support dynamic resolution of the FQDN for a VPN gateway.

Step 5: Click **Review + Create**, and then click **Create**.

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription ⓘ *

Resource group ⓘ *

[Create new](#)

Instance details

Region *

Configuration details

Name *

Prisma Access requires a peer IP address for the Azure VNG. Azure does not assign the actual IP address until you create the VNG and associate the public IP address.

12.5 Create the Virtual Network Gateway

In this procedure, you use dynamic routing with BGP.

Note

Every Azure VPN gateway consists of two instances in an active-standby configuration. For any planned maintenance or unplanned disruption that happens to the active instance, the standby instance would take over automatically and resume the VPN connections.

Step 1: Navigate to **Home > Virtual networks gateways**, and then click **Create**.

Step 2: In the **Name** box, enter **vng**.

Step 3: In the **Region** list, choose **(US) West US** or the appropriate value.

Step 4: In the Gateway Type section, select **VPN**.

Step 5: In the VPN Type section, select **Route-based**.

Step 6: In the **SKU** list, choose **VpnGw1**. The basic SKU does not support BGP or IKEv2.

Step 7: In the **Virtual network** list, select **transit**.

Step 8: In **Public IP address**, select **Use existing**.

Step 9: In the **Choose public IP address** list, select **vng-public**.

Step 10: For **Configure BGP**, select **Enabled**.

Step 11: In the **Autonomous system number (ASN)** box, accept the proposed default value of **65515**.

 **Caution**

If you want to use an Azure Route Server (ARS) in your environment, it can only be deployed with ASN 65515. The ARS also requires the VNGs use the same ASN for iBGP peering, so Palo Alto Networks recommends deploying your VNGs now with an ASN of 65515.

Step 12: Click **Review + create**, and then click **Create**.

Instance details	
Name *	vng ✓
Region *	West US
Gateway type *	☒ VPN ☐ ExpressRoute
VPN type *	☒ Route-based ☐ Policy-based
SKU *	VpnGw1
Generation	Generation1
Virtual network *	Transit Create virtual network
Subnet	GatewaySubnet (10.110.118.0/24)
Public IP Address Type	☒ Only virtual networks in the currently selected subscription and region are listed.
Public IP address	☐ Basic ☒ Standard
Public IP address *	☐ Create new ☒ Use existing
Public IP address name *	vng-public ✓
Public IP address SKU	Standard
Assignment	☐ Dynamic ☒ Static
Enable active-active mode *	☐ Enabled ☒ Disabled
Configure BGP *	☒ Enabled ☐ Disabled
Autonomous system number (ASN) *	65515

 **Note**

The VNG deployment typically takes 15 minutes or more to complete.

Step 13: Navigate to **Home > Virtual network gateways > vng**, and then click **Configuration**.

Step 14: Record the public IP address (example: **13.91.7.42**).

Step 15: Record the default Azure BGP peer IP address (example: **10.110.118.5**).

SKU * ⓘ
VpnGw1

Active-active mode
Enabled Disabled

Gateway Private IPs
Enabled **Disabled**

☒ Configure BGP

Autonomous system number (ASN) * ⓘ
65515

Public IP Address
13.91.7.42

Default Azure BGP peer IP address
10.110.118.5

Custom Azure APIPA BGP IP address ⓘ
Peer Address

Procedures

Configuring Prisma Access Service Connection VPN Tunnels and Routing

- 13.1 Configure Prisma Access Service-Connection VPN Tunnels
- 13.2 Configure Prisma Access Service-Connection Routing
- 13.3 Deploy Prisma Access Service Connection

In this group of procedures, you configure and deploy the Prisma Access service connection VPN tunnels and BGP routing.

13.1 Configure Prisma Access Service-Connection VPN Tunnels

In this procedure, you configure IPsec VPN tunnels on Prisma Access in order to support service connections to your Azure VNG. In this procedure, you configure a primary IPsec tunnel for the service connection.

Table 33 Prisma Access service-connection parameters

Parameter	Primary IPsec tunnel
Service connection name	Azure-US-West
Location	US West

(Table 33 continues on the following page)

Table 33 (continued)

Parameter	Primary IPSec tunnel
Tunnel name	Azure-US-West-1
IKE peer ID (VNG Public IP)	13.91.7.42
Static IP (VNG Public IP)	13.91.7.42

Step 1: Log in to Strata **Cloud Manager**.

Step 2: Navigate to **Workflows > Prisma Access Setup > Service Connections**, and then in the Service Connections pane, click **Add Service Connection**.

Step 3: In the General pane, in the **Name** box, enter **Azure-US-West**.

Step 4: In the **Region** list, choose **North America**.

Step 5: In the **Location** list, choose **US West**.

Step 6: In the Primary Tunnel pane, click **Set Up**.

Step 7: In the Create IPSec Tunnel dialog box, click **Create New**.

Step 8: In the Create IPSec Tunnel dialog box, in the **Tunnel Name** box, enter **Azure-US-West-1**.

Step 9: In the **Branch Device Type** list, choose **Palo Alto Networks**.

Step 10: In the **Pre-shared Key** box, enter a complex key string.

Step 11: In the **Confirm Pre-shared Key** box, re-enter the complex key string.

Step 12: In the **IKE Peer Identification** list, choose **IP address**, and in the box, enter **13.91.7.42**. This is the Public IP address of the Azure VNG.

Step 13: In the Branch Device IP Address section, select **Static IP**, and in the box enter **13.91.7.42**. This is the Public IP address of the Azure VNG.

Step 14: Continuing in the Create IPSec Tunnel dialog box near the bottom, click **IKE Advanced Options**.

Step 15: In the IKE Advanced Options dialog box, in the **IKE Protocol Version** list, choose **IKEv2 only mode**.

Step 16: In the **IKEv2 Crypto Profile** list, choose **PaloAlto-Networks-IKE-Crypto**, and then click **Save**.

Step 17: Continuing in the Create IPSec Tunnel dialog box, click **IPSec Advanced Options**.

Step 18: In the **IPSec Crypto Profile** list, choose **PaloAlto-Networks-IPSec-Crypto**.

Step 19: Select **Copy ToS**, and then click **Save**.

Step 20: In the Create IPsec Tunnel dialog box, click **Save**.

13.2 Configure Prisma Access Service-Connection Routing

After you have configured the service-connection VPN tunnels, you configure dynamic routing (using the BGP routing protocol) for the service connections.

Table 34 Prisma Access service-connection BGP parameters

Parameter	Primary IPsec tunnel
Location	US West
Name	Azure-US-West
BGP peer IP address (VNG)	10.110.118.5
BGP peer AS (VNG)	65515
Local IP address (Prisma Access)	192.168.254.254

Step 1: Continuing in the Add Service Connection dialog box, in the Routing pane, click **Set Up**.

Step 2: In the Set Up Routing dialog box, in Dynamic Routing, select **Enable BGP for Dynamic Routing**.

Step 3: On the Primary WAN tab, in the **Peer IP Address** box, enter the BGP peer IP address for the virtual network gateway (example: **10.110.118.5**).

Step 4: In the **Peer AS** box, enter the BGP AS number for the virtual network gateway (example: **65515**).

Step 5: In the **Local IP Address** box, enter the BGP local IP address for Prisma Access (example: **192.168.254.254**), and then click **Save**.

Caution

Do not add a BGP Secret. The Azure VNG configuration does not support BGP secrets in its peering relationships.

Dynamic Routing

☒ Enable BGP for Dynamic Routing

☐ Summarize Mobile User Routes before advertising

☐ Advertise Default Route

☐ Use different BGP Peer for Secondary Tunnel

Add no-export Community

disabled

Primary WAN
Secondary WAN

☐ Do Not Export Routes

Peer IP Address

10.110.118.5

Peer AS *

65515

Local IP Address

192.168.254.254

Step 6: In the Add Service Connection dialog box, click **Save**.

13.3 Deploy Prisma Access Service Connection

In this procedure, you push the configuration to Prisma Access and deploy the service connection.

Step 1: In Service Connections Setup, in the upper-right corner, click **Push Config**, and then click **Push**.

Step 2: In the **Description** box, enter a description.

Step 3: Under Container, select **Service Connections**, and then click **Push**.

The screenshot shows a configuration window titled 'Latest Config Version' with a 'Target' button in the top right. The main area is a tree view under the 'Container' column. The 'Global' container is expanded, showing 'Prisma Access' as a sub-container. Under 'Prisma Access', the 'Mobile Users Container' is expanded, showing 'GlobalProtect', 'Explicit Proxy', 'Remote Networks', and 'Service Connections'. The 'Service Connections' item is selected, indicated by a blue checkmark in the first column. At the bottom right, there are 'Cancel' and 'Push' buttons.

Step 4: In the Jobs dialog box, when the push job result changes to OK, click **Done**.

Prisma Access should complete the service-connection deployment in approximately 15 to 30 minutes. To complete the Azure VNG configurations, you need the IP address for the Service IP.

Step 5: In **Workflows > Prisma Access Setup > Service Connections**, make note of the Prisma Access Service IP address.

When your newly created service-connection Config Status is **In sync**, you should see the Service IP address and the EBGP Router IP Address. Make note of these two values for the next set of procedures.

Service Connections (3)							
Site		Status		Prisma Access			
☐	Name	Subnets	Tunnel	Config	Location	Service IP	EBGP Router
☐	Azure-US-West			In sync	US West	130.41.249.19	192.168.254.254



Note

The IPsec tunnels to Prisma Access will not be operational until you finish the configuration of the Azure VNG service. The output in the tunnel column displays an error status until the configuration is complete and the tunnels are established.

Procedures

Configuring Azure VPN Tunnels and Routing

- 14.1 Create the Local Network Gateway
- 14.2 Create the VPN Connection from the VNG to the LNG
- 14.3 Configure the VNet Peers to Use the Virtual Network Gateway

In this group of procedures, you finish the configuration of the Azure VPN tunnels to advertise the application and management networks to Prisma Access through BGP.

14.1 Create the Local Network Gateway

In this procedure, you create the local network gateway, which is a configuration object that corresponds to the IP address information from the Prisma Access firewall.

Step 1: Log in to the Azure Resource Manager at <https://portal.azure.com>.

Step 2: Navigate to **Home > Local network gateways**, and then click **Create**.

Step 3: In the **Resource group** list, choose **transit**.

Step 4: In the **Region** list, choose **(US) West US** or the appropriate value.

Step 5: In the **Name** box, enter **prisma-access**.

Step 6: In the **IP address** box, enter the public IP address of the Prisma Access peer (example: **130.41.249.19**). This is the Service IP from the previous procedure.

Step 7: In the **Address space(s)** box, enter the IP prefix for the BGP peer address from the Prisma Access firewall (example: **192.168.254.254/32**). This is the EBGP router IP from the previous procedure.

Instance details	
Region *	West US
Name *	prisma-access
Endpoint ⓘ	IP address FQDN
IP address *	130.41.249.19
Address Space(s) ⓘ	192.168.254.254/32

Step 8: Click **Next : Advanced**.

Step 9: In **Configure BGP settings**, select **Yes**.

Step 10: In the **Autonomous system number (ASN)** box, enter the AS number for Prisma Access (example: **65534**).

Step 11: In the **BGP peer IP address** box, enter **192.168.254.254**.

Step 12: Click **Review + create**, and then click **Create**.

14.2 Create the VPN Connection from the VNG to the LNG

In this procedure, you create the connection that is the site-to-site IPsec tunnel between the VNG and Prisma Access.

Step 1: Navigate to **Home > Connections**, and then click **Create**.

Step 2: In the **Resource group** list, choose **transit**.

Step 3: In the **Connection type** list, choose **Site-to-site (IPsec)**.

Step 4: In the **Name** box, enter **transit-to-prisma**.

Step 5: In the **Region** list, choose **(US) West US** or the appropriate value.

Instance details	
Connection type *	Site-to-site (IPsec) 
Name *	transit-to-prisma 
Region *	West US 

Step 6: Click **Next : Settings**.

Step 7: In the **Virtual network gateway** list, select **vng**.

Step 8: Click the **Local Network Gateway** section, and then select **prisma-access**.

Step 9: In the **Shared key (PSK)** box, enter the value for the pre-shared key (complex password).

Step 10: Select **Enable BGP**.

Step 11: Click **Review + create**, and then click **Create**.

14.3 Configure the VNet Peers to Use the Virtual Network Gateway

Azure networking allows you to share a VNG across multiple VNets. In this procedure, you enable each peer connection separately.

Table 35 Transit VNet peer connections

Source VNet	Peering name	Destination VNet	VNet peer setting
transit	transit-to-management	management	Use this virtual network's gateway or route server
transit	transit-to-app01	app01	Use this virtual network's gateway or route server
transit	transit-to-app02	app01	Use this virtual network's gateway or route server

Table 36 Spoke VNet peer connections

Source VNet	Peering name	Destination VNet	VNet peer setting
management	management-to-transit	transit	Use the remote virtual network's gateway or route server
app01	app01-to-transit	transit	Use the remote virtual network's gateway or route server
app01	app02-to-transit	transit	Use the remote virtual network's gateway or route server

First, you configure the transit VNet side of the peering connection.

Step 1: Navigate to **Home > Virtual networks > transit**, and then click **Peerings**.

Step 2: Select **transit-to-management**.

Step 3: In the Virtual Network Gateway or Route Server section, select **Use this virtual network's gateway or route server**, and then click **Save**.

Virtual network gateway or Route Server ⓘ

☒ Use this virtual network's gateway or Route Server
☐ Use the remote virtual network's gateway or Route Server
☐ None (default)

Step 4: Repeat Step 2 and Step 3 for all remaining connections in Table 35.

Step 5: Navigate to **Home > Virtual networks > transit - Peerings**, and then in the Gateway transit column, verify that gateway transit has been enabled for all peered VNets.

Next, you configure the spoke side of the peering connection.

Step 6: Navigate to **Home > Virtual networks > management**, and then click **Peerings**.

Step 7: Select **management-to-transit**.

Step 8: In the Virtual Network Gateway or Route Server section, select **Use the remote virtual network's gateway or route server**, and then click **Save**.

Virtual network gateway or Route Server ⓘ

☐ Use this virtual network's gateway or Route Server
☒ Use the remote virtual network's gateway or Route Server
☐ None (default)

Step 9: Repeat Step 6 through Step 8 for all remaining source VNet and connections in Table 36.

Procedures

Configuring Prisma Access Security Policy

- 15.1 Create Address Objects
- 15.2 Configure Security Policy for Azure Application Access

In this group of procedures, you configure the security policy in Prisma Access for users connecting to applications and resources in Azure.

15.1 Create Address Objects

In this procedure, you configure IP address objects to use in security policies. The IP addresses apply to the example design. You configure the address objects to be shared by all services.

Table 37 IP address objects

Address object name	IP address
Azure-US-West-App01	10.112.0.0/16
Azure-US-West-App02	10.113.0.0/16
Azure-US-West-Mgmt	10.255.0.0/24

Step 1: Log in to **Cloud Manager**.

Step 2: Navigate to **Manage > Configuration > NGFW and Prisma Access > Objects > Address > Addresses**.

Step 3: In the Configuration Scope pane, select **Prisma Access**.

Step 4: In the Addresses pane, click **Add Address**.

Step 5: In the **Name** box, enter [Azure-US-West-App01](#).

Step 6: In the **Description** box, enter a valid description.

Step 7: In the **IP Netmask** box, enter [10.112.0.0/16](#), and then click **Save**.

Step 8: Repeat Step 4 through Step 7 for all remaining objects in Table 37.

15.2 Configure Security Policy for Azure Application Access

In this procedure, you configure security policy to allow mobile users and remote sites access to enterprise applications and resources hosted in Azure.



Note

These security policies use the Prisma Access app's best-practice-based security protection and logging settings.

Table 38 Internet-access applications

Application	Description
web-browsing	Unencrypted web browsing (http)
ssl	Encrypted web browsing (https)
ssh	Secure shell
ms-rdp	Microsoft Remote Desktop Connection

This example policy allows mobile and remote-site users access to applications and management VNets in Azure. If you want to enforce a more restrictive security policy, you can modify or replace the following rule.

Step 1: Continuing in Cloud Manager, navigate to **Manage > Configuration > NGFW and Prisma Access > Security Services > Security Policy**.

Step 2: In the Configuration Scope pane, select **Prisma Access**.

Step 3: In the Security Policy Rules pane, click **Add Rule** and then select **Pre Rules**.

Step 4: In the **Name** box, enter **Access to enterprise applications**.

Step 5: In the **Description** box, enter **Rule to allow users to connect to enterprise applications**.

Step 6: In the Match Criteria section, in the Destination pane, next to Zones, click **Select**.

Step 7: In the resulting list, choose **trust**, and then click outside the pane.

Step 8: In the Destination pane, next to Addresses, click **Select**.

Step 9: In the resulting list, choose **Address**, and then select **Azure-US-West-app01**.

Step 10: Still in the **Address** list, select **Azure-US-West-app02** and **Azure-US-West-mgmt**.

Step 11: In the Applications/Service pane, next to Application, click **Select**.

Step 12: In the resulting list, in the search box, type **web-browsing**.

Step 13: From the search results, select **web-browsing**.

Step 14: Still in the list, in the search box, type **ssl**, and then from the search results, select **ssl**.

Step 15: In the search box, type **ssh**, and from the search results, select **ssh**.

Step 16: In the search box, type **ms-rdp**, and then from the search results, select **ms-rdp**.

Step 17: Click outside the pane.

Step 18: In the Actions pane, ensure that the action is set to **Allow** and that the profile group is set to **best-practice**, and then click **Save**.

Step 19: Click **Push Config**, and then click **Push**.

Step 20: In the **Description** box, enter a description.

Step 21: Under Container, select **GlobalProtect** and **Remote Networks**, and then click **Push**.

Procedures

Configuring Azure VM-Series Security Policy

16.1 Configure the Enterprise Security Policy

To allow users to connect to applications and resources in Azure, you now configure the security policy for the outbound east-west VM-Series NGFWs in Azure.

16.1 Configure the Enterprise Security Policy

In this procedure, you configure an enterprise-to-application security policy rule (in Panorama) that allows traffic from Prisma Access to reach applications in the app01 to app02 VNets. You use security Pre Rules, which are logically evaluated prior to local rules and cannot be overridden on the local device.

The security policy example for the enterprise network-connection traffic profile permits these applications. As required, you can add the following applications to your policy:

- SSH (ssh)
- RDP (ms-rdp)
- Web browsing (web-browsing)
- SSL (ssl)

Step 1: In the primary Panorama appliance, navigate to **Policies**, and then in the **Device Group** list, choose **Transit**.

Step 2: Navigate to **Security > Pre Rules**, and then click **Add**.

Step 3: In the **Name** box, enter **enterprise-to-applications**.

Step 4: On the **Source** tab, under **Source Zone**, click **Add**.

Step 5: In the **Source Zone** list, choose **private**.

Step 6: On the **Destination** tab, under **Destination Zone**, click **Add**.

Step 7: In the **Destination Zone** list, choose **private**.

Step 8: In the **Destination Address** list, choose **add**, then select **app01**.

Step 9: In the **Destination Address** list, choose **add**, then select **app02**.

Step 10: In the **Destination Address** list, choose **New Address**.

Step 11: In the **Name** box, enter **management**.

Step 12: In the **Type** list, choose **IP Netmask**.

- Step 13:** In the **IP** box, enter the management subnet **10.255.0.0/16**, and then click **OK**.
- Step 14:** On the Application tab, in the Applications pane, click **Add**.
- Step 15:** In the search box, enter **ssh**, and then in the results list, choose **ssh**.
- Step 16:** In the Applications pane, click **Add**.
- Step 17:** In the search box, enter **ms-rdp**, and then in the results list, choose **ms-rdp**.
- Step 18:** In the Applications pane, click **Add**.
- Step 19:** In the search box, enter **web-browsing**, and then in the results list, choose **web-browsing**.
- Step 20:** In the Applications pane, click **Add**.
- Step 21:** In the search box, enter **ssl**, and then in the results list, choose **ssl**.
- Step 22:** On the Service/URL Category tab, in the **Service** list, choose **application-default**.
- Step 23:** On the Actions tab, in the **Action** list, choose **Allow**.
- Step 24:** In the **Log Forwarding** list, choose **Forward-to-Strata-Logging-Service**.
- Step 25:** On the Target tab, select **Any (target to all devices)**, and then click **OK**.
- Step 26:** On the **Commit** menu, click **Commit and Push**, and then click **Commit and Push** again.

Feedback

You can use the **feedback form** to send comments about this guide.

HEADQUARTERS

Palo Alto Networks
3000 Tannery Way
Santa Clara, CA 95054, USA

Phone: +1 (408) 753-4000
Sales: +1 (866) 320-4788
Fax: +1 (408) 753-4001
<https://www.paloaltonetworks.com>
info@paloaltonetworks.com

©2024 Palo Alto Networks, Inc. Palo Alto Networks is a registered trademark of Palo Alto Networks. A list of our trademarks can be found at <https://www.paloaltonetworks.com/company/trademarks.html>. All other marks mentioned herein may be trademarks of their respective companies. Palo Alto Networks reserves the right to change, modify, transfer, or otherwise revise this publication without notice.



You can use the [feedback form](#) to send comments about this guide.